



Artificial Intelligence and Machine Learning

Block-1 Unit-1

Introduction to Artificial Intelligence

Topics to be Covered

- Introduction
- Basics of Artificial Intelligence (AI)?
- Brief history of Artificial Intelligence
- Components of Intelligence
- Approaches to Artificial Intelligence



Topics to be Covered

- Comparison between Artificial Intelligence (AI), Machine Learning (ML) and DeepLearning (DL).
- Application Areas of Artificial Intelligence Systems
- Intelligent Agents
- Important Topics
- Summary



Introduction

- AI is everywhere in today's world, and people are gradually becoming accustomed to its presence. It is utilized in systems that recognise both voices and faces.
- In addition to this, it can provide you with shopping recommendations that are tailored to your own purchasing preferences.
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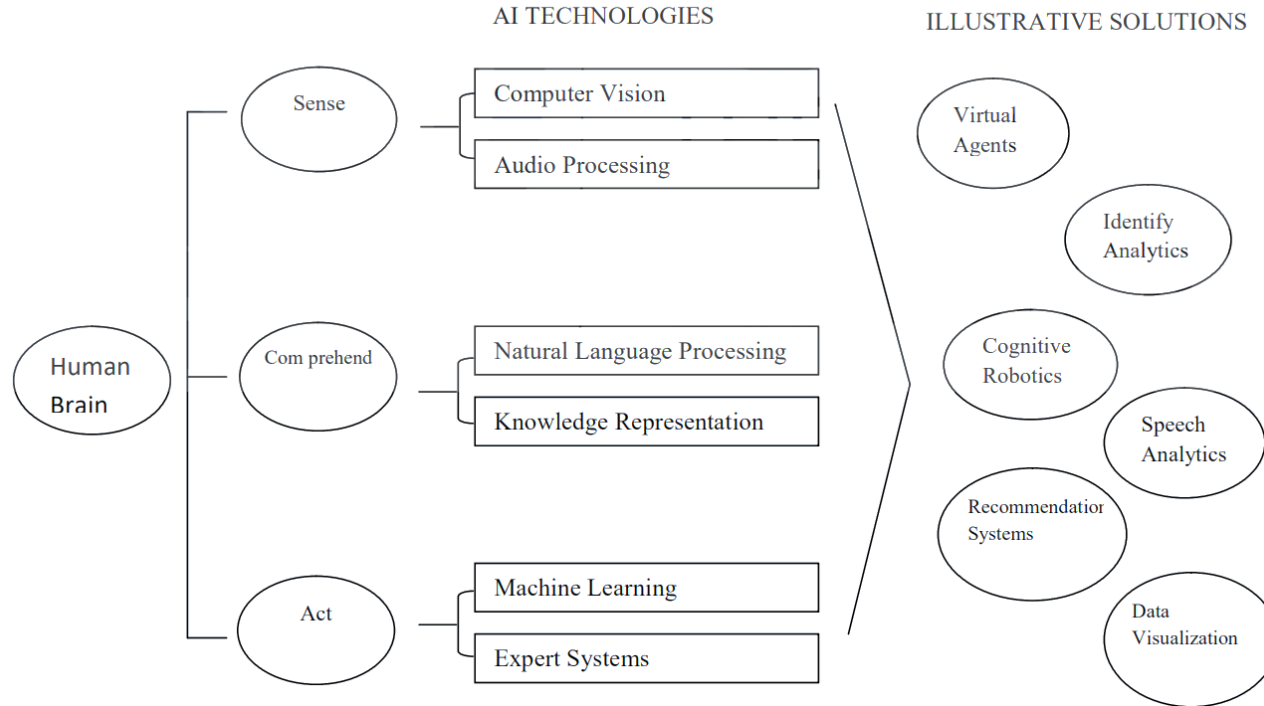
Introduction

- There is a well-known proverb that says "marks are not the measure of intelligence." This is due to the fact that intelligence is not a measurement of how much information one possesses. In fact, it is the measure of how much we comprehend and put into practice.
- People who are knowledgeable may gain a lot of information, but an intelligent person understands how to comprehend, analyze, and use the information.
- You could have a lot of knowledge but still be the least intelligent person in the room. Knowledge and intelligence are inextricably linked, and each contributes to the other's development.
- Knowledge enables one to learn the understandings that others have of things, whereas intelligence is the foundation for one's ability to grasp the things themselves.

Basics of Artificial Intelligence ?

- The following is a list of definitions of artificial intelligence that have been provided by well-known authors of artificial intelligence textbooks.
 - "The Exciting New Effort to Make Computers Think... Machines with Minds, in the Full and Literal Sense.
 - “The automation of behaviors that we connect with human thinking, activities such as decision-making, problem-solving, and learning.
 - The art of building machines that execute functions that demand intellect when performed by people.

Basics of Artificial Intelligence ?



What is Artificial Intelligence? – Emerging AI Technologies

+91 – 9319133134, query@cftedu.in

Basics of Artificial Intelligence ?

- The various categories of Artificial Intelligence are :
- **Artificial Narrow Intelligence (ANI)**, also called Weak AI or Narrow AI:
 - Weak AI is a term for thinking that is "simulated." Such systems seem to act intelligently, but they don't have any awareness of what they are doing.
 - For example, a chatbot might talk to you in a way that seems natural, but it doesn't know who it is or why it's talking to you.
 - Artificial intelligence is a system that was built to do a certain job.

Basics of Artificial Intelligence ?

- **Artificial General Intelligence (AGI):** Strong or General Artificial Intelligence, also called "actual" thinking. That is, acting like a smart human and thinking like one with a conscious, subjective mind.
- For instance, when two humans talk, they probably both know who they are, what they're doing, and why. Systems with strong or general artificial intelligence can do things that people can do.
- These systems tend to be harder to understand and more complicated. They are set up to handle situations where they might need to solve problems on their own without help from a person.
- Uses for these kinds of systems include self-driving cars and operating rooms in hospitals.

Basics of Artificial Intelligence ?

- **Artificial Super Intelligence (ASI)** - Super intelligence: The term "super intelligence" usually refers to a level of general and strong AI that is smarter than humans, if that's even possible.
- The ASI is seen as the logical next step after the AGI because it can do more than humans can.
- This includes making decisions, making rational decisions, and even things like building emotional relationships.
- There is a marginal difference between AGI and ASI.

Brief History of Artificial Intelligence

- In the late 19th century and early 20th century, mathematicians and philosophers like Gottlob Frege, Bertram Russell, Alfred North Whitehead, and Kurt Godel built on Boole's first ideas about logic to make mathematical representations of logic problems.
- In 1950, Turing wrote a paper called "Computing Machines and Intelligence." This article gave a good overall picture of AI. To learn more about Alan Turing, go to <http://www.turing.org.uk/turing>.
- Turing's paper talked about a lot of things, one of which was how to solve problems by using heuristics as guides to look through the space of possible solutions. He used the game of chess to explain how his ideas about how machines can think work. He even said that the machine could change its own instructions so that machines could learn from what they do.

Brief History of Artificial Intelligence

- In the 1990s, AI made a lot of progress, especially in machine learning, data mining, intelligent tutoring, case-based reasoning, multi-agent planning and scheduling, uncertain reasoning, understanding and translating natural language, vision, virtual reality, games, and other areas.
- In 2000, The Nomad robot goes to remote parts of Antarctica to look for meteorite samples. Space probes that are made of robots can work on their own to learn more about space. They keep an eye on what's going on around them, make decisions, and take action to get where they want to go.
- In April 2004, the first three-month missions of NASA's Mars rovers went well. The Spirit rover was looking at a group of hills on Mars that took two months to reach. It is finding strangely eroded rocks that might be new pieces of the puzzle that is the history of the area. Spirit's twin sister, Opportunity, was looking at the layers of rock in a crater

Components of Intelligence

- According to the dominant school of thought in psychology, human intelligence should not be viewed as a singular talent or cognitive process but rather as a collection of distinct components. The majority of attention in the field of artificial intelligence research has been paid to the following aspects of intelligence: learning, reasoning, problem-solving, perception, and language comprehension.
- **Learning:** There are numerous approaches to develop a learning system. Making mistakes is the simplest way to learn. A basic software that solves "mate in one" chess issues, for example, might test different moves until it finds one that answers the problem. The programme remembers which move worked so that the next time the computer is given the identical situation, it can provide an immediate response. The simple act of memorising things like answers to problems, words in a vocabulary list, and so on is known as "rote learning" or memorization.

Components of Intelligence

- According to this system, there are five ways to learn:
- Rote Learning or memorizing.
- Learning by Instructions
- Learning by analogy.
- Learning by Induction
- Learning by deduction.

Components of Intelligence

- **Reasoning:** To reason is to draw conclusions that are appropriate for the situation. Both deductive and inductive reasoning can be used to make conclusions. An example of a deductive inference is, "Fred is either in the museum or in the cafe. He isn't in the cafe, so he must be in the museum."
- An example of inductive inference is, "In the past, accidents just like this one have been caused by instrument failure." The difference between the two is that in the deductive case, the truth of the premises guarantees the truth of the conclusion, while in the inductive case, the truth of the premises supports the conclusion that instrument failure caused the accident, but more research could show that the conclusion is actually false, even though the premises are true.

Components of Intelligence

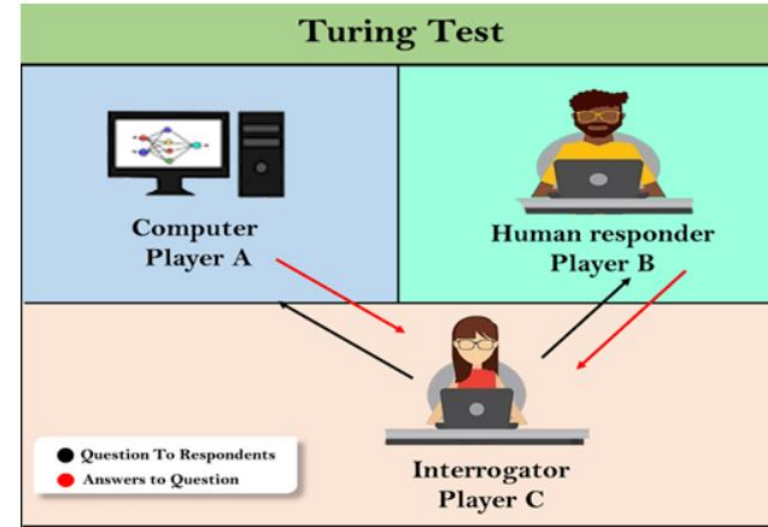
- **Problem-solving:** Problems usually go like this: given these data, find x. AI is used to solve a very wide range of problems. Some examples are finding the best way to win a board game, figuring out who someone is from a picture, and planning a series of steps that will allow a robot to do a certain task.
- **Perception:** Perception involves scanning the surroundings with numerous sense organs, genuine or artificial, and internal processes for analyzing the scene into objects, their features, and relationships. The fact that one and the same item can have various appearances depending on the angle from which it is viewed, whether or not parts of it are projecting shadows, and so on, complicates analysis.

Approaches to Artificial Intelligence

- **Turing Test** : The Turing Test is a method of inquiry in AI for determining whether or not a computer is capable of thinking like a human being. The test is named after Alan Turing, the founder of the Turing Test and an English computer scientist, cryptanalyst, mathematician and theoretical biologist.
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- Consider, Player A is a computer, Player B is human, and Player C is an interrogator. Interrogator is aware that one of them is machine, but he needs to identify this on the basis of questions and their responses.

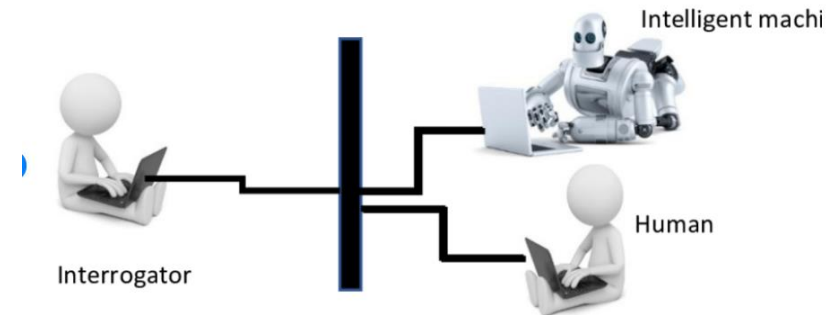
Approaches to Artificial Intelligence

- The conversation between all players is via keyboard and screen so the result would not depend on the machine's ability to convert words as speech.
- The test result does not depend on each correct answer, but only how closely its responses like a human answer. The computer is permitted to do everything possible to force a wrong identification by the interrogator.
- The questions and answers can be like:
- **Interrogator:** Are you a computer?



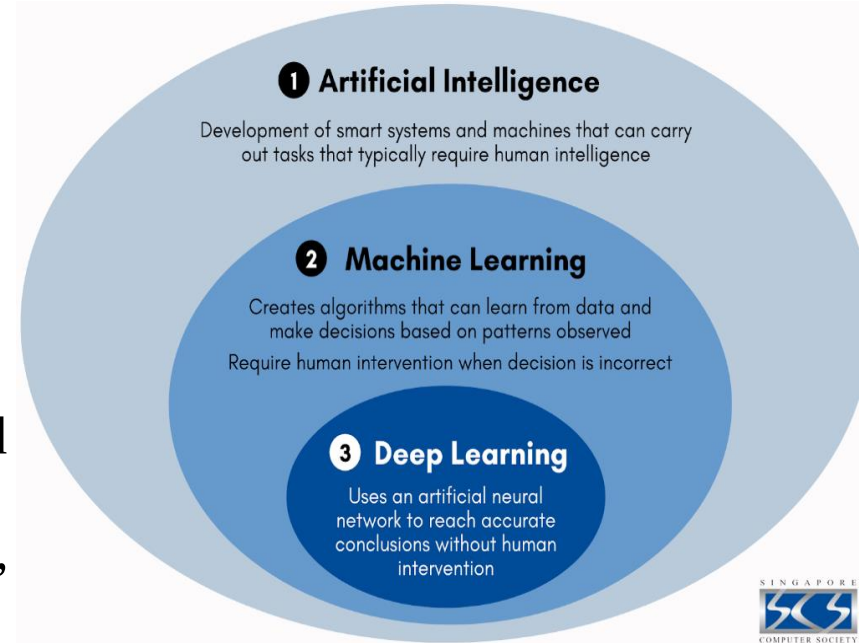
Approaches to Artificial Intelligence

- **PlayerA (Computer):** No
- **Interrogator:** Multiply two large numbers such as $(256896489 * 456725896)$
- **Player A:** Long pause and give the wrong answer.
- In this game, if an interrogator would not be able to identify which is a machine and which is human, then the computer passes the test successfully, and the machine is said to be intelligent and can think like a human.



Comparison - Artificial Intelligence, Machine Learning and Deep Learning

- Artificial intelligence is a big field that includes a lot of different ways of doing things, from top down (knowledge representation) to bottom-up (machine learning). In recent years, people have often talked about three related ideas: artificial intelligence (AI), machine learning (ML), and deep learning (DL) (DL). AI is the most general term, machine learning is a part of AI, and deep learning is a type of machine learning.



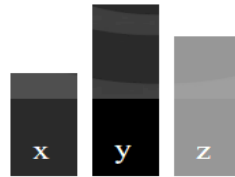
Comparison - Artificial Intelligence, Machine Learning and Deep Learning

- **Introduction to Machine Learning (ML):** Machine learning is a branch of artificial intelligence (AI). It explains one of the most important ideas in AI, which has to do with learning through experience and not through being taught.
- One of the most recent advances in AI, this way of learning is made possible by applying machine learning to very large data sets.
- Machine learning algorithms find patterns and learn how to make predictions and recommendations by using data and experiences instead of explicit programming instructions.
- The algorithms also change as they get new data and learn from their experiences. This makes them more effective over time.

Comparison - Artificial Intelligence, Machine Learning and Deep Learning

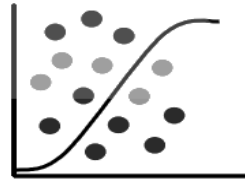
Machine learning provides predictions and prescriptions
Types of analytics (in order of increasing complexity)

Descriptive



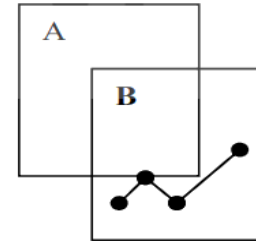
- Describe **what happened**
- Employed heavily across all industries

Predictive



- Anticipate **what will happen** (inherently probabilistic)
- Employed in data-driven organizations as a key source of insight

Prescriptive



- Provide recommendations on **what to do** to achieve goals
- Employed heavily by leading data and internet companies

Focus of machine learning

Comparison - Artificial Intelligence, Machine Learning and Deep Learning

- **Introduction to Deep Learning (DL)** : Deep Learning is a subfield of machine learning that focuses on algorithms called "Artificial Neural Networks" (ANN) that are based on how the brain is built and how it works. Deep Learning is a type of machine learning that can handle a wider range of data sources, needs less pre-processing of data, and often gives more accurate results than traditional machine learning methods.
- A neural network is made up of layers of software-based calculators called "neurons" that are linked together. This neural network can take in a huge amount of data and process it through many layers. At each layer, the network learns more complex features of the data. The network can then decide what to do with the data, find out if it was right, and use what it has learned to decide what to do with new data.

Comparison - Artificial Intelligence, Machine Learning and Deep Learning

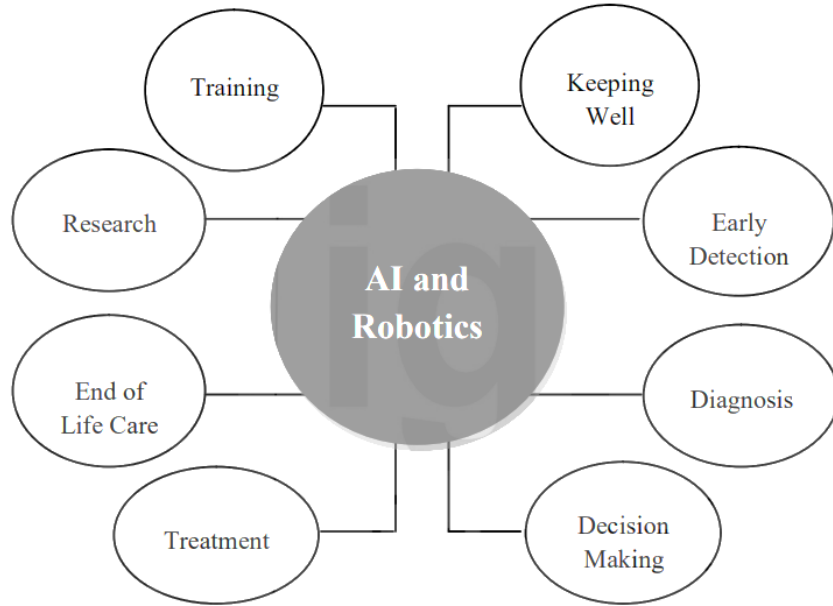
- Deep learning is a way of programming computers that uses the way neural networks work to teach computers to do things that humans do naturally. So, Deep Learning is a way to teach a computer model to run classification algorithms based on an image, text, or sound. Once a neural network knows what an object looks like, it can spot that object in a new picture.
- Deep learning is becoming more popular because its models can get better results. It uses large sets of labeled data and neural network architectures to train the models..

Application Areas of Artificial Intelligence systems

- Artificial intelligence is the most important factor in the transformation of economies straight from the ground up, and it is contributing as an efficient alternative. It has a lot of potential to perform optimization in any industry, whether it smart cities or the health sector or agriculture or any other prospective sector of relevance, and below we have included a few of the systems in which AI is functioning as the major source of competitive advantage:



Application Areas of Artificial Intelligence systems



Potential Use of AI in Health Care

- Healthcare: The application of AI in healthcare can help address issues of high barriers to access to healthcare facilities, particularly in rural areas that suffer from poor connectivity and a limited supply of healthcare professionals.
- This is especially true in areas where the supply of healthcare professionals is limited. The deployment of use cases like as AI-driven diagnostics, personalized treatment, early diagnosis of potential pandemics, and imaging diagnostics, amongst others, is one way to accomplish this goal.

Application Areas of Artificial Intelligence systems

- **Agriculture:** AI has the potential to bring in a food revolution while simultaneously satisfying the ever-increasing need for food (global need to produce 50 percent more food and cater to an additional 2 billion people by 2050 as compared to today). It also has the ability to resolve issues such as inadequate demand prediction, a lack of secure irrigation, and the abuse or misuse of pesticides and fertilizers.
- **Smart Mobility, including Transports and Logistics:** Autonomous fleets for ride sharing, semiautonomous features such as driver assistance, and predictive engine monitoring and maintenance are all possible use cases for smart mobility, which includes transportation and logistics.
- **Retail:** The retail industry was one of the first to use AI solutions. For example, personalised suggestions, browsing based on user preferences, and image-based product search have all been used to improve the user experience.

Application Areas of Artificial Intelligence systems

- **Top Used Applications of Artificial Intelligence**
 - Tools and checkers for plagiarism
 - Recognizing faces
 - Putting an AI autopilot on commercial planes
 - Applications for sharing rides (E.g.: Uber, Lyft)
 - E-mail spam filters; voice-to-text features; search suggestions
 - Google's predictions based on AI (E.g.: Google Maps)
 - Protecting against and stopping fraud.
 - Smart personal assistants (E.g.: Siri, Alexa)

Intelligent Agents

- A software agent is a system which, when given a goal to be achieved, could carry out the details of the appropriate (computer) operations and further, in case it gets stuck, it can ask for advice and can receive it from humans, may even evaluate the appropriateness of the advice and then act suitably. Essentially, a computer agent is a computer software that additionally has the following attributes:
- It has autonomous control i.e., it operates under its own control
- it is perceptive, i.e., it is capable of perceiving its own environment
- it persists over a long period of time
- it is adaptive to changes in the environment and
- it is capable of taking over others' goals.

Intelligent Agents

- An agent is supposed to possess some or all of the following properties:
- **Reactivity:** The ability of sensing the environment and then acting accordingly.
- **Autonomy:** The ability of moving towards its goal, changing its moves or strategy, if required, without much human intervention.
- **Communicating ability:** The ability to communicate with other agents and humans.
- **Ability to coexist by cooperating:** The ability to work in a multi-agent environment to achieve a common goal.

Intelligent Agents

- **Ability to adapt to a new situation:** Ability to learn, change and adapt to the situations in the world around it.
- **Ability to draw inferences:** The ability to infer or conclude facts, which may be useful, but are not available directly.
- **Temporal continuity:** The ability to work over long periods of time.
- **Personality:** Ability to impersonate or simulate someone, on whose behalf the agent is acting.
- **Mobility:** Ability to move from one environment to another.

Intelligent Agents

- **PEAS (Performance, Environment, Actuators, Sensors)**
- For designing an agent, the first requirement is to specify the task environment to the maximum extent possible.
- The task environment for an agent to solve one type of problems, may be described by the four major parameters namely, performance (which is actually the expected performance), environment (i.e., the world around the agent), actuators (which include entities through which the agent may perform actions) and sensors (which describes the different entities through which the agent will gather information about the environment).

Intelligent Agents

- Example (An Automated Public Road Transport Driver Agent)
- We describe the task environment of the agent on the basis of PEAS.
- Performance Measures: Some of the performance measures which can easily be perceived of an automated public road transport driver would be:
 - Maximizing safety of passengers
 - Maximizing comfort of passengers
 - Ability to reach correct destination
 - Ability to minimize the time to reach the destination Obeying traffic rules
 - Causing minimum discomfort or disturbance to other agents □ Minimizing costs, etc.

Intelligent Agents

- Environment (or the world around the agent) We must remember that the environment or the world around the agent is extremely uncertain or open ended. There are unlimited combinations of possibilities of the environment situations, which such an agent could face.
- Let us enumerate some of the possibilities or circumstances which an agent might face:
- Variety of roads e.g., from 12-lane express-ways, freeways to dusty rural bumpy roads; different road rules including the ones requiring left-hand drive in some parts of the world and right-hand drive-in other parts.
- The degree of knowledge of various places through which and to which driving is to be done.

Intelligent Agents

- Various kinds of passengers, including high cultured to almost ruffians etc. □ All kind of other traffic possibly including heavy vehicles, ultra-modern cars, three-wheelers and even bullock carts.
- Actuators: These include the following:
- Handling steering wheel, brakes, gears and accelerator
- Understanding the display screen
- A device or devices for all communication required

Intelligent Agents

- **Sensors:** The agent acting as automated public road transport driver must have some way of sensing the world around it i.e., the traffic around it, the distance between the automobile and the automobiles ahead of it and its speed, the speeds of neighboring vehicles, the condition of the road, any turn ahead etc. It may use sensors like odometer, speedometer, sensors telling the different parameters of the engine, Global Positioning System (GPS) to understand its current location and the path ahead. Also, there should be some sort of sensors to calculate its distance from other vehicles etc
- There are two parts of an agent or its structure:
- A (hardware) device with sensors and actuators in which that agent will reside, called the architecture of the agent.
- An agent program that will convert or map the percepts in to actions

Intelligent Agents

- **Goal Based Agents** : In order to design appropriate agent for a particular type of task, we know the nature of the task environment plays an important role. Also, it is desirable that the complexity of the agent should be minimum and just sufficient to handle the task in a particular environment. In this regard, first we discussed the simplest type of agents, viz., Simple Reflex Agents. The action of this type of agent is decided by the current precept only.
- **Utility Based Agents** :Goal based agent's success or failure is judged in terms of its capability for achieving or not achieving its goal. A goal-based agent, for a given pair of environment state and possible input, only knows whether the pair will lead to the goal state or not. Such an agent will not be able to decide in which direction to proceed when there are more than one conflicting goals.

Intelligent Agents

- **Stimulus-Response Agents** : A stimulus response agent (or a reactive agent) take input from the world through sensors, and then take action based on those inputs through actuators. Between the stimulus and response, there is a processing unit that can be arbitrarily complex.
- An example of such an agent is one that controls a vehicle in a racing game: the agent “looks” at the road and nearby vehicles, and then decides how much to turn and break. Such Agents (Stimulus-Response Agents are the Reactive agents) represents a special category of agents, which do not possess internal, symbolic models of their environments; instead, they act/respond in a stimulus-response manner to the present state of the environment in which they are embedded. These agents are relatively simple and they interact with other agents in basic ways. Nevertheless, complex patterns of behavior emerged from the interactions when the ensemble of agents is viewed globally

Important Topics

- Components of Intelligence
- Approaches to Artificial Intelligence
- Comparison between Artificial Intelligence (AI), Machine Learning (ML) and DeepLearning (DL).
- Application Areas of Artificial Intelligence Systems
- Intelligent Agents

Summary

- In this session we learned about the difference between knowledge and intelligence and also pointed out the meaning of Artificial Intelligence (AI), along with the application of AI systems in various fields.
- The session also covers the historical development of the field of AI systems. Along with the development of the AI as a discipline, the need of classification of AI systems was felt, and hence the unit discussed the classification of the AI systems in detail. Further, the unit discussed about the concepts of Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL).
- Finally, the session discussed the components of Intelligence, which was extended for the understanding of the concepts of Intelligent Agents, with special emphasis on Stimulus - Response Agents

*Thank
You*



Artificial Intelligence and Machine Learning

Block-1 Unit-2

Problem Solving Using Search

Topics to be Covered

- Introduction
- Introduction to State Space Search
- Formulation of 8 puzzle problem from AI perspective
- N-queen's problem- Formulation and Solution
- Two agent search: Adversarial search
- Minimax search strategy
- Alpha-Beta Pruning algorithm
- Important Topics
- Summary



Introduction

- Many AI-based applications need to figure out how to solve problems. In the world, there are two types of problems. First, the problem which can be solved by using deterministic procedure and the success is guaranteed. But most real-world problems can be solved only by searching a solution. AI is concerned with these second types of problems solving. To build a system to solve a problem, we need to
- Define the problem precisely-find initial and final configuration for acceptable solution to the problem.
- Analyze the problem-find few important features that may have impact on the appropriateness of various possible techniques for solving the problem

Introduction

- Isolate and represent task knowledge necessary to solve the problem
- Choose the best problem-solving technique(s) and apply it to the particular problem.
- To provide a formal description of a problem, we need to do the following:
- Define a state space that contains all the possible configurations of the relevant objects.
- Specify one or more states that describe possible situations, from which the problem solving process may start. These states are called initial states.
- Specify one or more than one goal states.
- Defining a set of rules for the actions (operators) that can be taken.

Introduction to State Space Search

- It is necessary to represent an AI problem in the form of a state space before it can be solved. A state space is the set of all states reachable from the initial state. A state space forms a graph in which the nodes are states and the arcs between nodes are actions. In state space, a path is a sequence of states connected by a sequence of actions.
- **Structure of a state space:** The structures of state space are trees and graphs. A tree has one and only one path from any point to any other point. Graph consists of a set of nodes (vertices) and a set of edges (arcs). Arcs establish relationship (connections) between the nodes, i.e., a graph has several paths to a given node. Operators are directed arcs between nodes. The method of solving problem through AI involves the process of defining the search space, deciding start and goal states and then finding the path from start state to goal state through search space. Search process explores the state space. In the worst case, the search explores all possible paths between the initial state and the goal state. +91 - 9319133134 , query@cftedu.in

Introduction to State Space Search

- **Problem Solution:** In a state space, a solution is a path from the initial state to a goal state or sometime just a goal state. A numeric cost is assigned to each path. It also gives the cost of applying the operators to the states. A path cost function is used to measure the quality of solution and out of all possible solutions, an optimal solution has the lowest path cost. The importance of cost depends on the problem and the type of solution asked.
- **Problem formulation:** Many problems can be represented as state space. The state space of a problem includes: an initial state, one or more goal state, set of state transition operator (or a set of production rules), used to change the current state to another state. This is also known as actions. A control strategy is used that specifies the order in which the rules will be applied. For example, Depthfirst search (DFS), Breath-first search (BFS) etc. It helps to find the goal state or a path to the goal state.

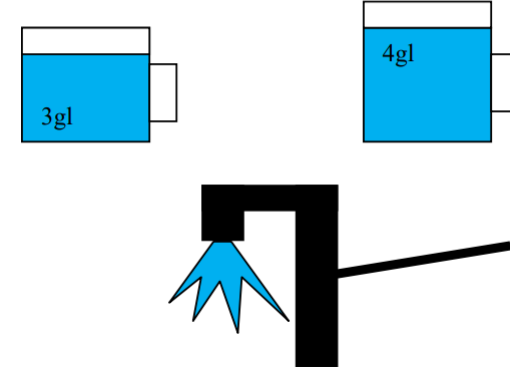
Introduction to State Space Search

- **SEARCHING FOR SOLUTIONS** After formulating our problem, we are ready to solve it, this can be done by searching through the state space for a solution, this search will be applied On a search tree or generally a graph that is generated using the initial state and the successor function.
- Example : State space representation of Water-Jug problem (WJP):
 - Given two jugs, a 4-gallon and a 3-gallon, both of which do not have measuring indicators on them. The jugs can be filled with water with the help of a pump that is available (as shown in figure 4) The question is “how can you get exactly 2 gallons of water into 4-gallon jug”

Introduction to State Space Search

- The state space or production rule of this problem can be defined as a collection of ordered pairs of integers (x,y) where $x=0,1,2,3,4$ and $y=0,1,2,3$.
- In the order pair (x,y) , x is the amount of water in four-gallon jug and y is the amount of water in the three-gallon jug.
- State space: all possible combination of (x,y)
- The start state is $(0,0)$ and
- **Goalstate** is $(2, n)$, where $n = 0,1,2,3$
- The following table 1 shows the set of production rules (actions) that can be used to change one state to another.

The water Jug Problem



Introduction to State Space Search

- Table 1: Production rules for Water Jug problem

Production Rules		
Rule No	Production	Meaning
R1	$(x, y \mid x < 4) \rightarrow (4, y)$	Fill 4-gallon jug
R2	$(x, y \mid y < 3) \rightarrow (x, 3)$	Fill 3-gallon jug
R3	$(x, y \mid x > 0) \rightarrow (0, y)$	Empty 4-gallon jug
R4	$(x, y \mid y > 0) \rightarrow (x, 0)$	Empty 3-gallon jug
R5	$(x, y \mid x + y \geq 4 \text{ and } y > 0) \rightarrow (4, y - (4 - x))$	Pour water from 3-gallon jug into 4-gallon jug until 4-gallon jug is full
R6	$(x, y \mid x + y \geq 3 \text{ and } x > 0) \rightarrow (x - (3 - y), 3)$	Pour water from 4-gallon jug into 3-gallon jug until 3-gallon jug is full
R7	$(x, y \mid x + y \leq 4 \text{ and } y > 0) \rightarrow (x + y, 0)$	Pour all water from 3-gallon jug into 4-gallon jug.
R8	$(x, y \mid x + y \leq 3 \text{ and } x > 0) \rightarrow (0, x + y)$	Pour all water from 4-gallon jug into 3-gallon jug.
R9	$(x, y \mid x > 0) \rightarrow (x - d, y)$	Pure some water d out from 4-gallon jug
R10	$(x, y \mid y > 0) \rightarrow (x, y - d)$	Pure some water d out from 3-gallon jug

Introduction to State Space Search

- The following 2 solutions are found for the problem “how can you get exactly 2 gallons of water into 4-gallon jug”, as shown in Table-2 and in Table-3.
- Solution-1: Table-2 Getting exactly 2 gallons of water into 4-gallon jug (solution1)

4-gal jug	3-gal jug	Rule applied
0	0	Initial state
4	0	R1 {fill 4-gal jug}
1	3	R6 {Pure all water from 4-gal jug to 3-gal jug}
1	0	R4 {empty 3-gal jug}
0	1	R8 {Pour water from 4 to 3-gal jug}
4	1	R1 {Fill 4-gal jug}
2	3	R6 {Pour water from 4-gal jug to 3-gal jug until it is full}
2	0	R4 {empty 3-gal jug}

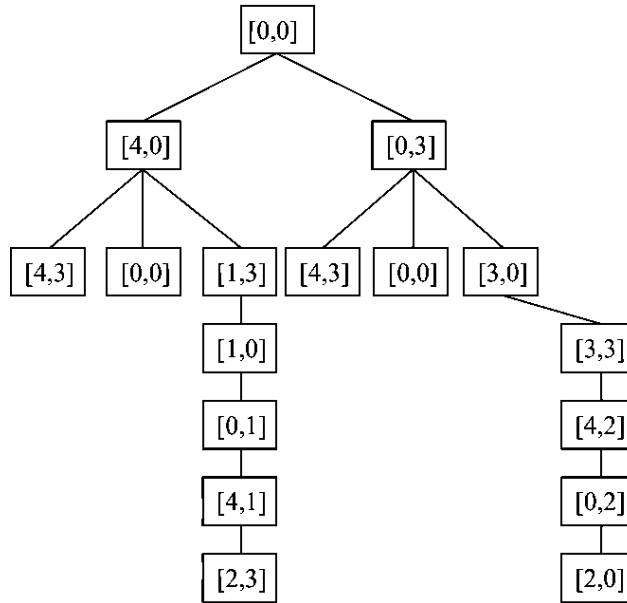
Introduction to State Space Search

- Solution-2 Table-3 Getting exactly 2 gallons of water into 4-gallon jug (solution2)

4-gal jug	3-gal jug	Rule applied
0	0	Initial state
0	3	R2 {fill 3-gal jug}
3	0	R7 {Pour all water from 3-gal jug to 4-gal jug}
3	3	R2 {fill 3-gal jug}
4	2	R5 {Pour from 3 to 4-gal jug until it is full}
0	2	R3 {Empty 4-gal jug}
2	0	R7 {Pour all water from 3-gal jug to 4-gal jug}

Introduction to State Space Search

- A state space tree for WJP with all possible solution is shown in the given figure .



All possible Solution of
WJP using state space tree

Formulation of 8 Puzzle Problem

- The eight-tile puzzle consists of a 3-by-3 (3×3) square frame board which holds eight (8) movable tiles numbered as 1 to 8. One square is empty, allowing the adjacent tiles to be shifted. The objective of the puzzle is to find a sequence of tile movements that leads from a starting configuration to a goal configuration
- The Eight Puzzle Problem formulation: Given a 3×3 grid with 8 sliding tiles and one “blank” Initial state: some other configuration of the tiles.
- **Operator:** Slide tiles (Move Blank) to reach the goal (as shown below). There are 4 operators that is, “Moving the blank”:
- Move the blank UP, Move the blank DOWN, Move the blank LEFT and Move the blank RIGHT.

Formulation of 8 Puzzle Problem

- Path Cost: Sum of the cost of each path from initial state to goal state. Here cost of each action (blank move) = 1, so cost of a sequence of actions = the number of actions. A optimal solution is one which has a lowest cost path.
- Performing State-Space Search: Basic idea:
- If the initial state is a goal state, return it. If not, apply the operators to generate all states that are one step from the initial state (its successors)

3	1	2
4		5
6	7	8

initial state



3		2
4	1	5
6	7	8

3	1	2
4	7	5
6		8

3	1	2
	4	5
6	7	8

3	1	2
4	5	
6	7	8

Its successors

Formulation of 8 Puzzle Problem

- Find the minimum cost path for the 8-puzzle problem, where the start and goal state are given as follows:

- Initial State: $\{(1,2,3),(4,8,-),(7,6,5)\}$
- Successor State-
- $\{(1,2,3),(4,8,5),(7,6,-)\}$; Move 5 up or – to down
- $\{(1,2,3),(4,8,5),(7,-,6)\}$; Move 6 right or – to left
- $\{(1,2,3),(4,-,5),(7,8,6)\}$; Move 8 down or – to up
- $\{(1,2,3),(4,5,-),(7,8,6)\}$; Move 5 to left or – to right
- Goal State:**
- $\{(1,2,3),(4,5,6),(7,8,-)\}$; Move 6 to up or – to down
- PATH COST=5**

1	2	3
4	8	-
7	6	5

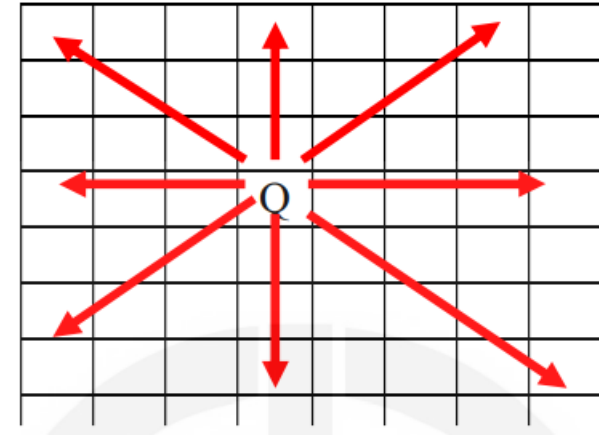
Start State

1	2	3
4	5	6
7	8	-

Goal State

N Queen's Problem-Formulation and Solution

- The N-Queen problem is the problem of placing N Queen's ($Q_1, Q_2, Q_3, \dots, Q_n$) on an $N \times N$ chessboard so that no two queens attack each other. The colour of the queens is meaningless in this puzzle, and any queen is assumed to be attack any other. So, a solution requires that no two queens share the same row, column, or diagonal. The N-queen problem must follow the following rules:
- There is at most one queen in each column.
- There is at most one queen in each row.
- There is at most one queen in each diagonal.



N Queen's Problem-Formulation and Solution

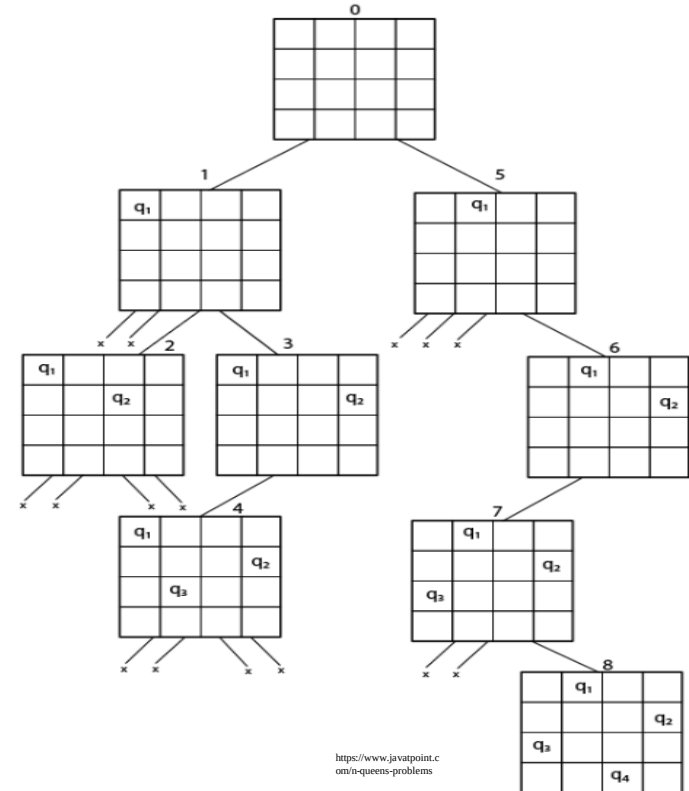
- Given a 4 x 4 chessboard and number the rows and column of the chessboard 1 through 4.
- Since, we have to place 4 queens such as q_1 q_2 q_3 and q_4 on the chessboard, such that no two queens attack each other. In such a conditional each queen must be placed on a different row, i.e., we put queen "i" on row "i."

	1	2	3	4
1				
2				
3				
4				

4x4 chessboard

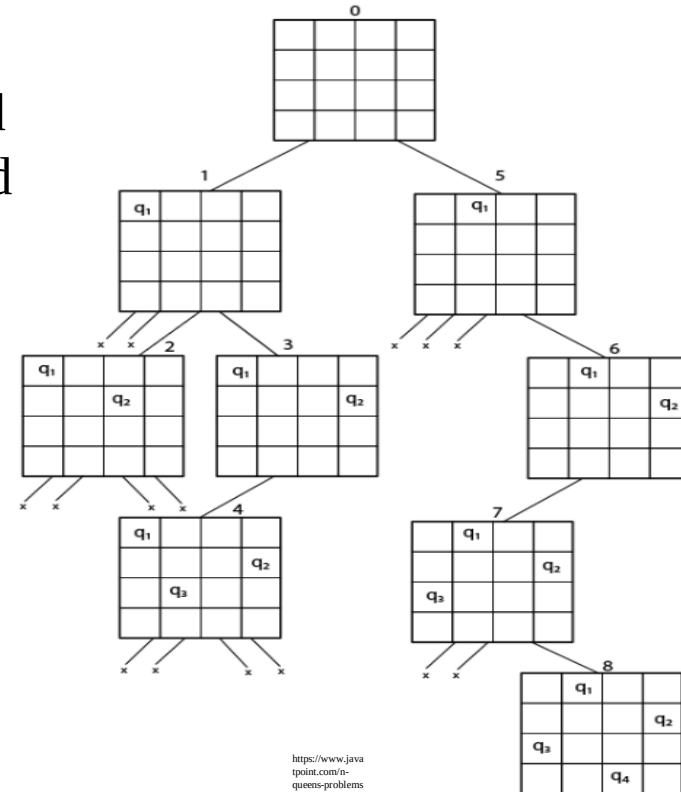
N Queen's Problem-Formulation and Solution

- Now, we place queen q_1 in the very first acceptable position (1, 1). Next, we put queen q_2 so that both these queens do not attack each other. We find that if we place q_2 in column 1 and 2, then the dead end is encountered.
- Thus the first acceptable position for q_2 in column 3, i.e. (2, 3) but then no position is left for placing queen ' q_3 ' safely. So we backtrack one step and place the queen ' q_2 ' in (2, 4), the next best possible solution.



N Queen's Problem-Formulation and Solution

- Then we obtain the position for placing ' q_3 ' which is (3, 2). But later this position also leads to a dead end, and no place is found where ' q_4 ' can be placed safely. Then we have to backtrack till ' q_1 ' and place it to (1, 2) and then all other queens are placed safely by moving q_2 to (2, 4), q_3 to (3, 1) and q_4 to (4, 3).
- That is, we get the solution (2, 4, 1, 3). This is one possible solution for the 4-queens problem. For another possible solution, the whole method is repeated for all partial solutions. The other solutions for 4 - queens problems is (3, 1, 4, 2) i.e.

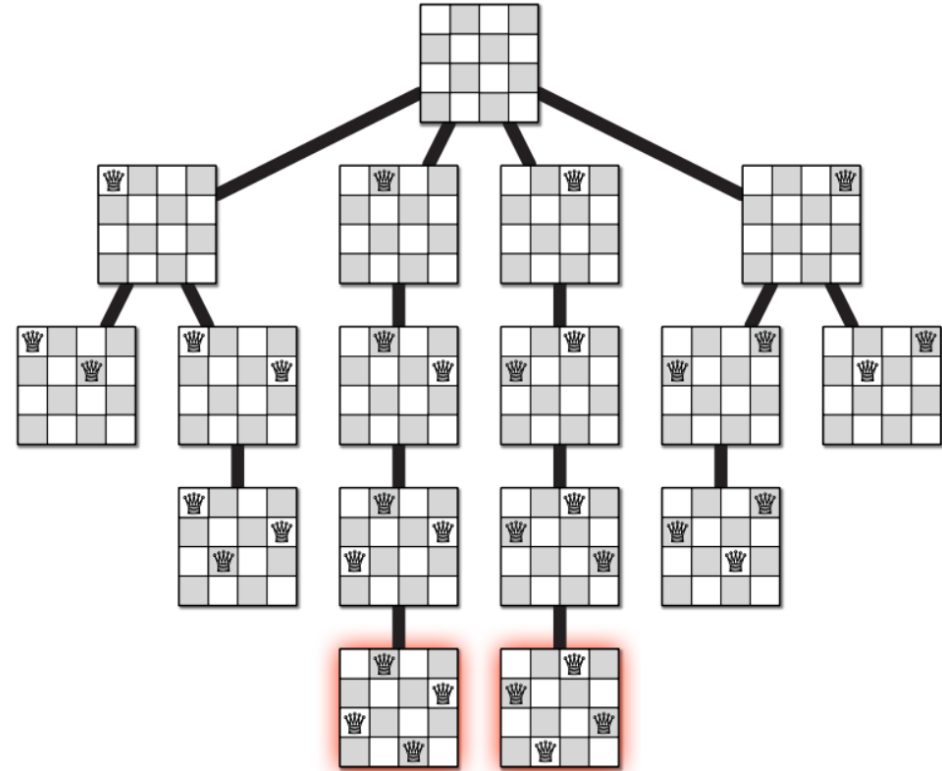


N Queen's Problem-Formulation and Solution

- **Following is the backtracking algorithm for solving the N-Queen problem**
- Initialize an empty chessboard of size $N \times N$.
- Start with the leftmost column and place a queen in the first row of that column.
- Move to the next column and place a queen in the first row of that column.
- Repeat step 3 until either all N queens have been placed or it is impossible to place a queen in the current column without violating the rules of the problem.
- If all N queens have been placed, print the solution.
- If it is not possible to place a queen in the current column without violating the rules of the problem, backtrack to the previous column.
- Remove the queen from the previous column and move it down one row.
- Repeat steps 4-7 until all possible configurations have been tried.

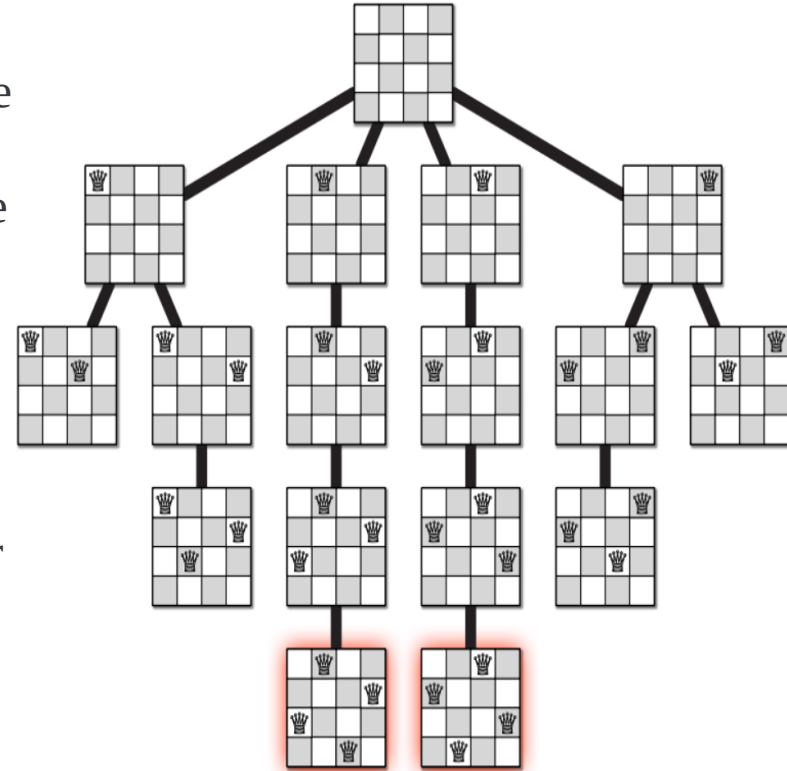
N Queen's Problem-Formulation and Solution

- Backtracking algorithms are commonly used to make a *sequence of decisions*, with the goal of building a recursively defined structure satisfying certain constraints.
- Often (but not always) this goal structure is itself a sequence. Here, for the n-queens problem, the goal is a sequence of queen positions, one in each row, such that no two queens attack each other. For each row, the algorithm *decides* where to place the queen.



N Queen's Problem-Formulation and Solution

- In each recursive call to the backtracking algorithm, we need to make **exactly one** decision, and our choice must be consistent with all previous decisions. Thus, each recursive call requires not only the portion of the input data we have not yet processed, but also a suitable summary of the decisions we have already made.
- For the sake of efficiency, the summary of past decisions should be as small as possible. For the n-queens problem, we must pass in not only the number of empty rows, but the positions of all previously placed queens. Here, unfortunately, we must remember our past decisions in complete detail.



Adversarial Search-Two Agent Search

- Adversarial search is a method applied to a situation where you are planning while another actor prepares against you. Your plans, therefore, could be affected by your opponent's actions. The term “search” in the context of adversarial search refers to games, at least in the realm of artificial intelligence (AI).
- Techniques required to get the best optimal solution : There is always a need to choose right algorithms which provide the best optimal solution in a limited time.
- So we use the following technique which could fulfil our requirements :
- Pruning: A technique which allows ignoring the unwanted portions of a search tree which make no difference in its final result.
- Heuristic Evaluation Function: It allows to approximate the cost value at each level of the search tree, before reaching the goal node.

Adversarial Search-Two Agent Search

- Elements of Game Playing search To play a game, we use a game tree to know all the possible choices and to pick the best one out. There are following elements of a game-playing:
- S_0 : It is the initial state from where a game begins.
- PLAYER (s): It defines which player is having the current turn to make a move in the state.
- ACTIONS (s): It defines the set of legal moves that a player can make to change the state.
- RESULT (s, a): It is a transition model which defines the result of a move.

Adversarial Search-Two Agent Search

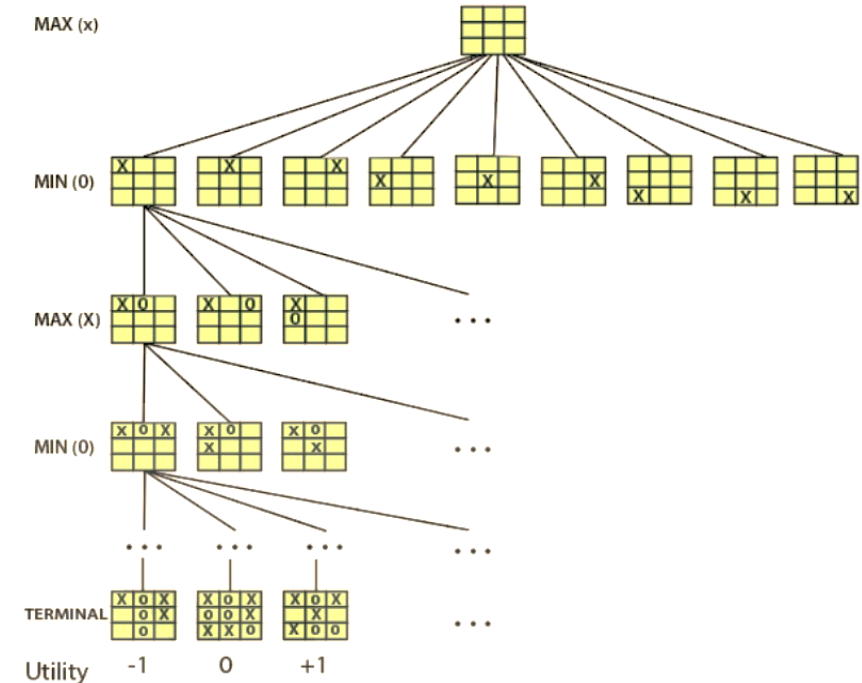
- **TERMINAL-TEST (s):** It defines that the game has ended (or over) and returns true. States where the game has ended are called terminal states.
- **UTILITY (s,p):** It defines the final value with which the game has ended. This function is also known as Objective function or Payoff function. This utility function gives a numeric value for the outcome of a game i.e.
- For example, in chess, tic-tac-toe, we have two or three possible outcomes. Either win or lose, or draw the match, which we can represent by the values +1,-1 or 0. In other word we can say that (-1): if the PLAYER loses, (+1), if the PLAYER wins and (0): If there is a draw between the PLAYERS.

Adversarial Search-Two Agent Search

- Let's understand the working of the elements with the help of a game tree designed for tic-tactoe. Here, the node represents the game state and edges represent the moves taken by the players. The root of the tree is the initial state. Next level is all of MAX's moves, then next level is all of MIN's moves and so on. Note that root has 9 blank square (MAX), level 1 has 8 blank squares (MIN), level 2 has 7 blank square (MAX) and so on.
- **Objective:**
- **Player1:** Maximize outcome and
- **Player2:** Minimize outcome Terminal (goal) state:
- **utility:** -1, 0, +1 (that is win for X is +1 and win for O is -1 and 0 for draw. The number on each leaf node indicates the utility value of the terminal state from the point of view of MAX.

Adversarial Search-Two Agent Search

- High values are assumed to be good for MAX and bad for MIN (which is how the players get their names). It is MAX's job to use the search tree to determine the best move. Note that if MAX win then Utility value is +1, if MIN wins then utility value is -1 and if DRAW then utility value is 0.



Adversarial Search-Two Agent Search

- In a tic-tac-toe game playing, as shown in figure above we have the following elements:
- INITIAL STATE (S₀): The top node in the game-tree represents the initial state in the tree and shows all the possible choice to pick out one.
- PLAYER (s): There are two players, MAX and MIN. MAX begins the game by picking one best move and place X in the empty square box.
- ACTIONS (s): Both the players can make moves in the empty boxes chance by chance.

Adversarial Search-Two Agent Search

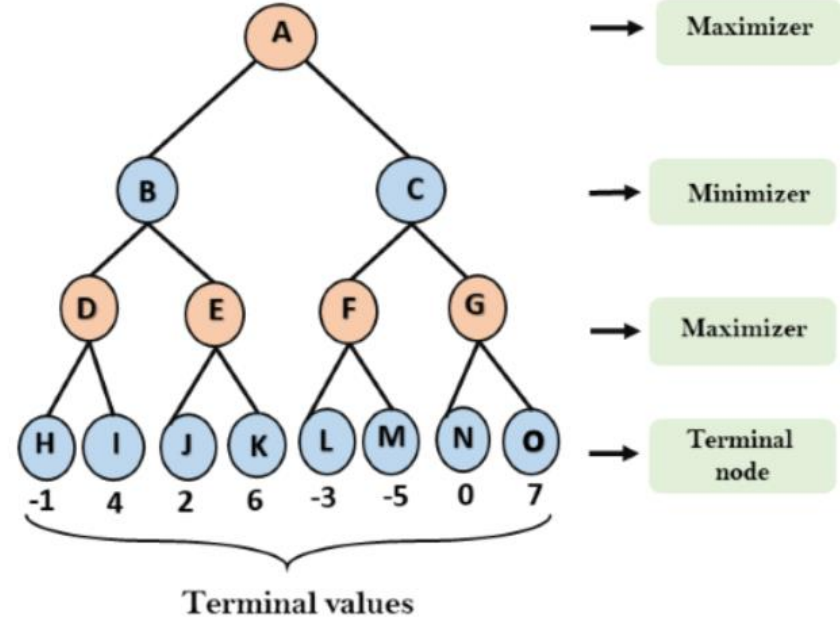
- **RESULT (s, a):** The moves made by MIN and MAX will decide the outcome of the game.
- **TERMINAL-TEST(s):** When all the empty boxes will be filled, it will be the terminating state of the game.
- **UTILITY:** At the end, we will get to know who wins: MAX or MIN, and accordingly, the price will be given to them. If MAX win then Utility value is +1, if MIN wins then utility value is -1 and if DRAW then utility value is 0.

Min-Max Search Strategy

- **MINIMAX algorithm** is a backtracking algorithm where it backtracks to pick the best move out of several choices. MINIMAX strategy follows the DFS (Depth-first search) concept.
- Here, we have two players MIN and MAX, and the game is played alternatively between them, i.e., when MAX made a move, then the next turn is of MIN. It means the move made by MAX is fixed and, he cannot change it.
- The same concept is followed in DFS strategy, i.e., we follow the same path and cannot change in the middle. That's why in MINIMAX algorithm, instead of BFS, we follow DFS. The following steps are used in MINMAX algorithm:

Min-Max Search Strategy

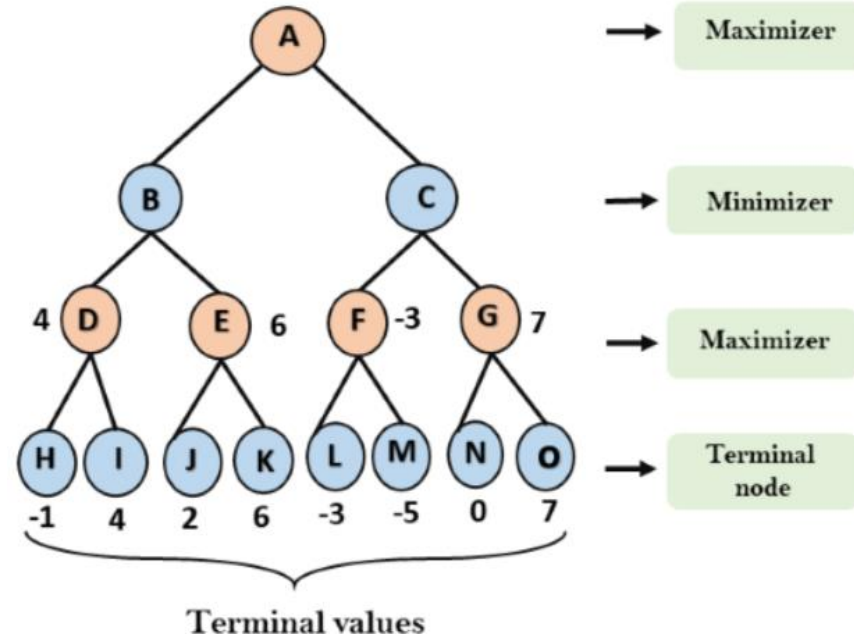
- Step-1:** In the first step, the algorithm generates the entire game-tree and apply the utility function to get the utility values for the terminal states. In the below tree diagram, let's take A is the initial state of the tree. Suppose maximizer takes first turn which has worst-case initial value = -infinity, and minimizer will take next turn which has worst-case initial value = +infinity.



<https://www.javatpoint.com/mini-max-algorithm-in-ai>

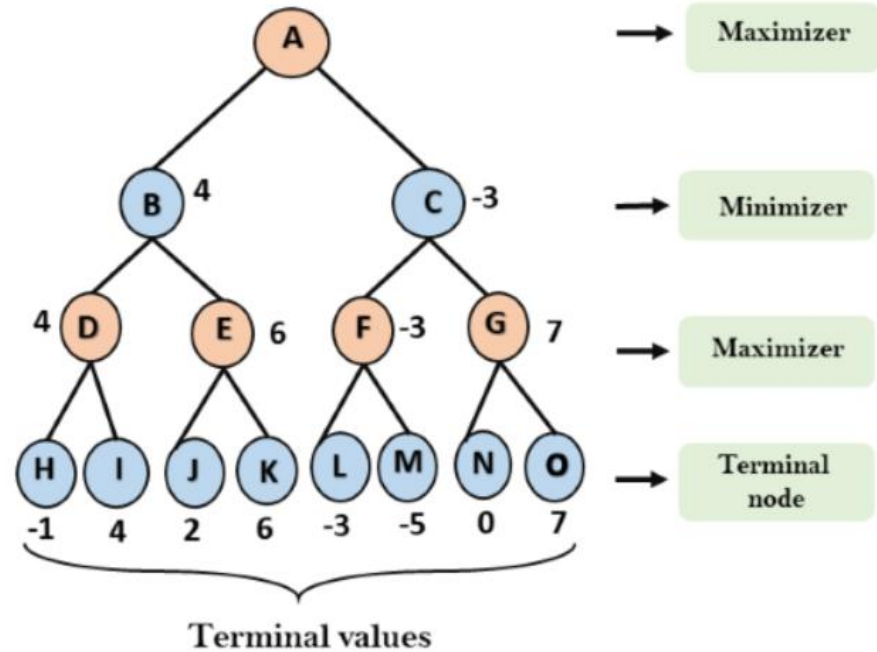
Min-Max Search Strategy

- **Step 2:** Now, first we find the utilities value for the Maximizer, its initial value is $-\infty$, so we will compare each value in terminal state with initial value of Maximizer and determines the higher nodes values. It will find the maximum among the all.
- For node D $\max(-1, -\infty) \Rightarrow \max(-1, 4) = 4$
- For Node E $\max(2, -\infty) \Rightarrow \max(2, 6) = 6$
- For Node F $\max(-3, -\infty) \Rightarrow \max(-3, -5) = -3$
- For node G $\max(0, -\infty) = \max(0, 7) = 7$



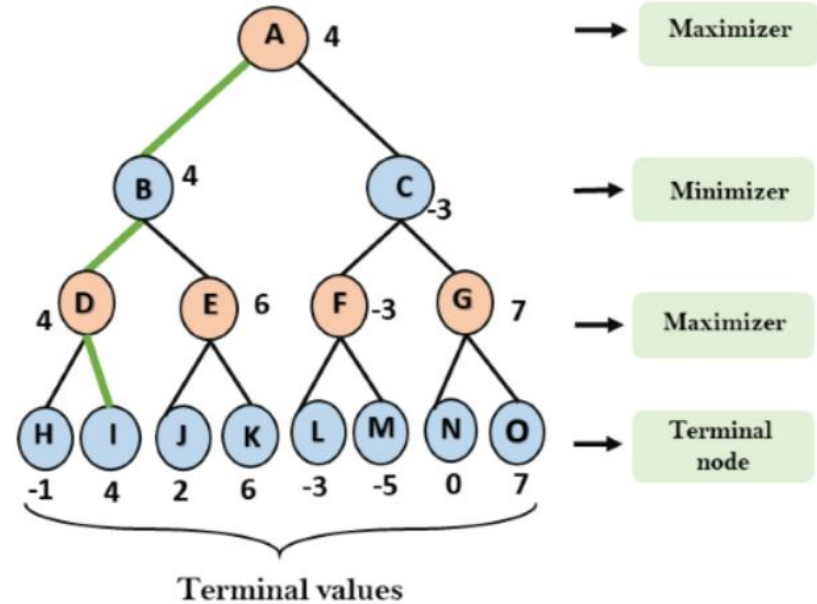
Min-Max Search Strategy

- **Step 3:** In the next step, it's a turn for minimizer, so it will compare all nodes value with $+\infty$, and will find the 3rd layer node values.
- For node B = $\min(4, 6) = 4$
- For node C = $\min(-3, 7) = -3$



Min-Max search strategy

- **Step 4:** Now it's a turn for Maximizer, and it will again choose the maximum of all nodes value and find the maximum value for the root node. In this game tree, there are only 4 layers, hence we reach immediately to the root node, but in real games, there will be more than 4 layers.
- For node A $\max(4, -3) = 4$

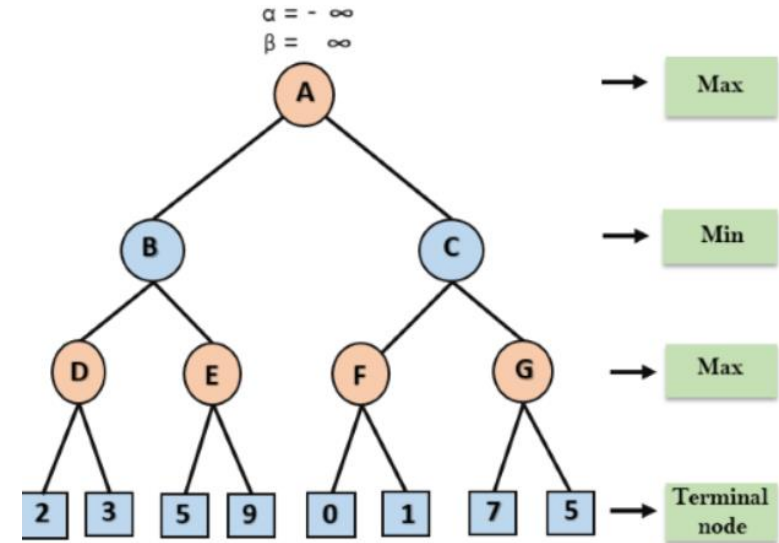


Alpha-Beta Pruning

- Alpha-beta pruning is a modified version of the minimax algorithm. It is an optimization technique for the minimax algorithm.
- We have seen in the minimax search algorithm that the number of game states it has to examine are exponential in depth of the tree. As we cannot eliminate the exponent, but we can cut it to half. Hence there is a technique by which without checking each node of the game tree we can compute the correct minimax decision, and this technique is called **pruning**. This involves two threshold parameter Alpha and beta for future expansion, so it is called **alpha-beta pruning**. It is also called as **Alpha-Beta Algorithm**.

Alpha-Beta Pruning

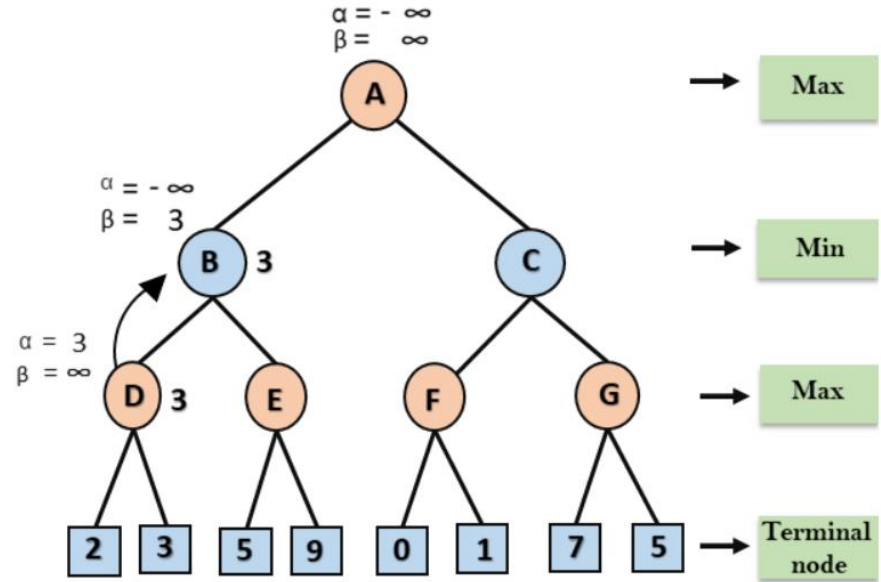
- **Working of Alpha-beta Pruning.** Let's take an example of two-player search tree to understand the working of Alpha-beta pruning
- **Step 1:** At the first step the, Max player will start first move from node A where $\alpha = -\infty$ and $\beta = +\infty$, these value of alpha and beta passed down to node B where again $\alpha = -\infty$ and $\beta = +\infty$, and Node B passes the same value to its child D.



<https://www.javatpoint.com/ai-alpha-beta-pruning>

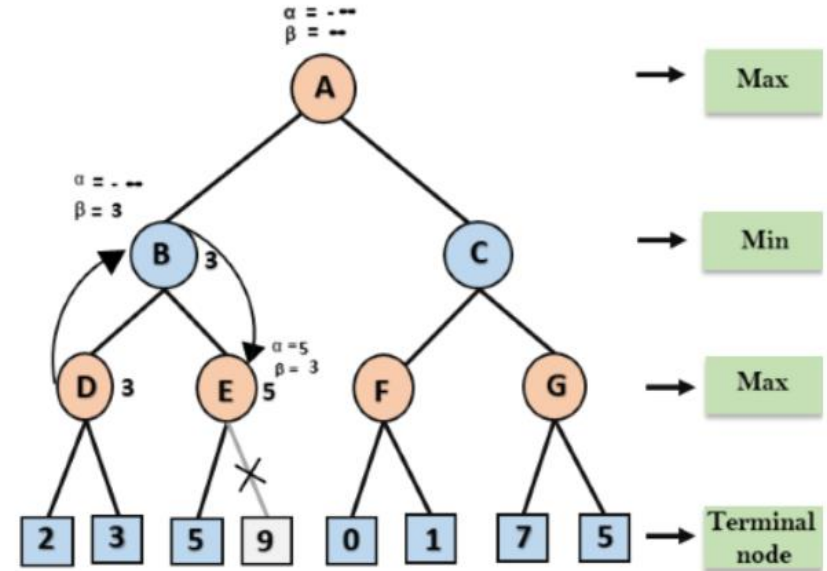
Alpha-Beta Pruning

- Step 2:** At Node D, the value of α will be calculated as its turn for Max. The value of α is compared with firstly 2 and then 3, and the max (2, 3) = 3 will be the value of α at node D and node value will also 3.
- Step 3:** Now algorithm backtrack to node B, where the value of β will change as this is a turn of Min, Now $\beta = +\infty$, will compare with the available subsequent nodes value, i.e. min (∞ , 3) = 3, hence at node B now $\alpha = -\infty$, and $\beta = 3$.



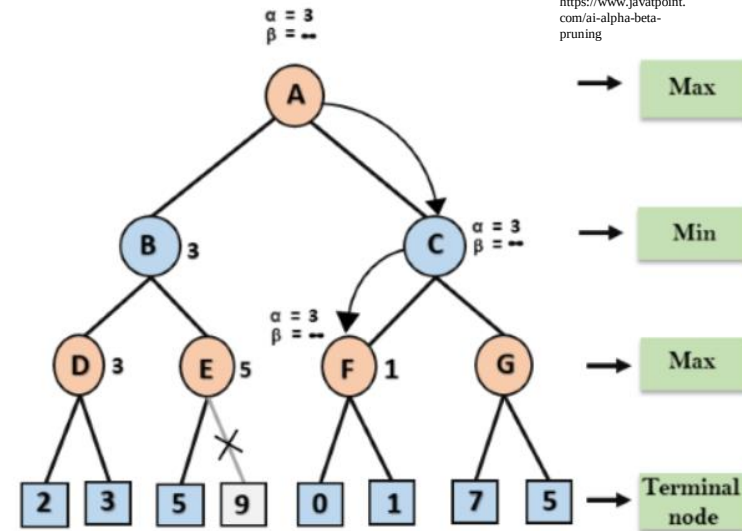
Alpha-Beta Pruning

- In the next step, algorithm traverse the next successor of Node B which is node E, and the values of $\alpha = -\infty$, and $\beta = 3$ will also be passed.
- Step 4:** At node E, Max will take its turn, and the value of alpha will change. The current value of alpha will be compared with 5, so $\max(-\infty, 5) = 5$, hence at node E $\alpha = 5$ and $\beta = 3$, where $\alpha \geq \beta$, so the right successor of E will be pruned, and algorithm will not traverse it, and the value at node E will be 5.



Alpha-Beta Pruning

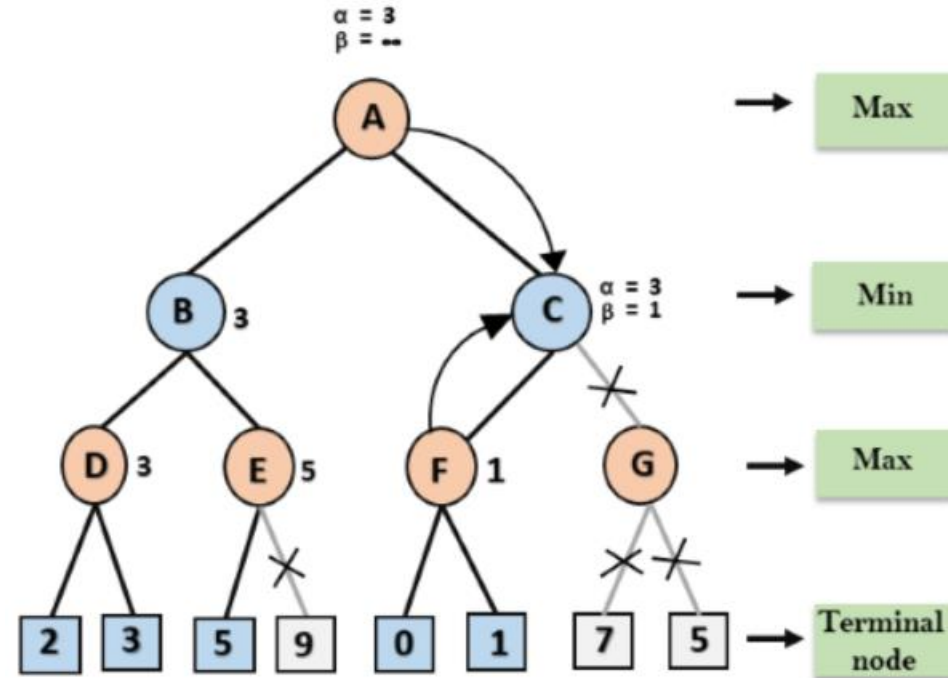
- Step 5:** At next step, algorithm again backtrack the tree, from node B to node A. At node A, the value of alpha will be changed the maximum available value is 3 as $\max(-\infty, 3) = 3$, and $\beta = +\infty$, these two values now passes to right successor of A which is Node C.
- At node C, $\alpha = 3$ and $\beta = +\infty$, and the same values will be passed on to node F.
- Step 6:** At node F, again the value of α will be compared with left child which is 0, and $\max(3, 0) = 3$, and then compared with right child which is 1, and $\max(3, 1) = 3$ still α remains 3, but the node value of F will become 1.



Alpha-Beta Pruning

<https://www.javatpoint.com/m/ai-alpha-beta-pruning>

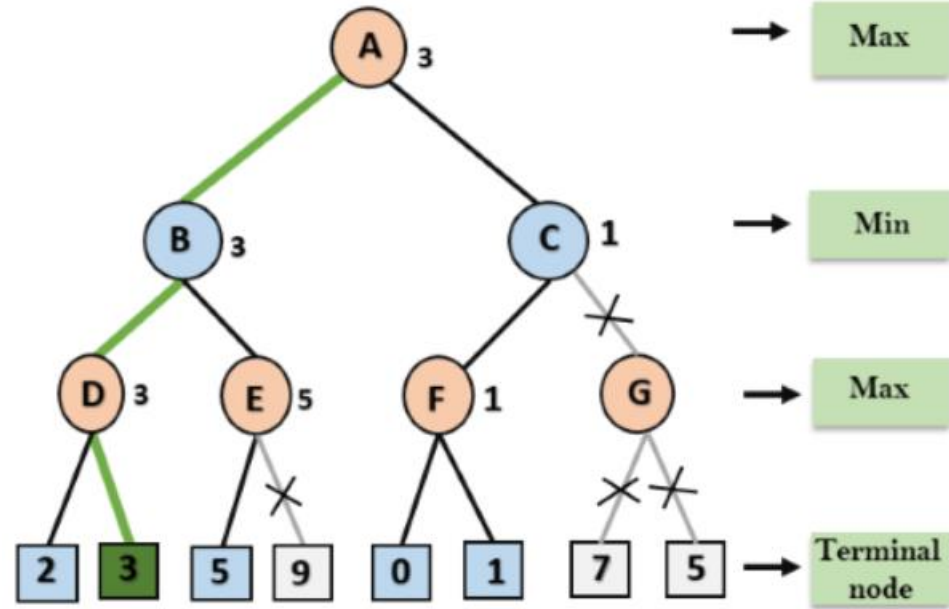
- Step 7:** Node F returns the node value 1 to node C, at C $\alpha = 3$ and $\beta = +\infty$, here the value of beta will be changed, it will compare with 1 so $\min(\infty, 1) = 1$. Now at C, $\alpha = 3$ and $\beta = 1$, and again it satisfies the condition $\alpha \geq \beta$, so the next child of C which is G will be pruned, and the algorithm will not compute the entire sub-tree G.



Alpha-Beta Pruning

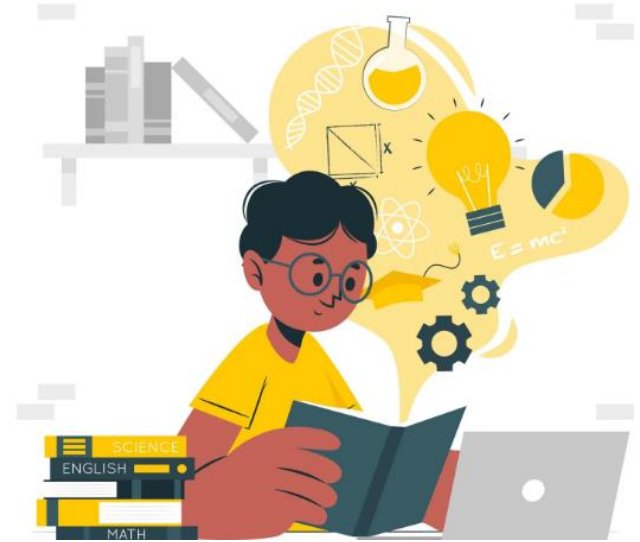
<https://www.javatpoint.com/ai-alpha-beta-pruning>

- Step 8:** C now returns the value of 1 to A here the best value for A is $\max(3, 1) = 3$. Following is the final game tree which is showing the nodes which are computed and nodes which has never computed. Hence the optimal value for the maximizer is 3 for this example.



Important Topics

- State Space Search
- Formulation of 8 Queen's problem
- State space tree for 4-Queen's problem
- Backtracking approach to solve N Queen's problem
- Two agent search: Adversarial search
- Minimax search strategy
- Alpha-Beta Pruning algorithm



<https://www.gyanarth.com/web-stories/bose-12th-chemistry-exam-paper-analysis/>

Summary

- **In this session we have seen that -**
- Before an AI problem can be solved it must be represented as a state space. Among all possible states, there are two special states called initial state (the start point) and final state (the goal state).
- A successor function (a set of operators) is used to change the state. It is used to move from one state to another.
- A state space is set of all possible states of a problem.
- A state space essentially consists of a set of nodes representing each state of the problem, arcs between nodes representing the legal moves from one state to another, an initial state, and a goal state. Each state space takes the form of a tree or a graph.

Summary

- The process of searching means a sequence of action that take you from an initial state to a goal state.
- search is fundamental to the problem-solving process. Search means the problem is solved by using the rules, in combination with an appropriate control strategy, to move through the problem space until a path from an initial state to a goal state is found.
- In a normal search, we follow a sequence of actions to reach the goal or to finish the game optimally. But in an adversarial search, the result depends on the players which will decide the result of the game. It is also obvious that the solution for the goal state will be an optimal solution because the player will try to win the game with the shortest path and under limited time.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-1 Unit-3 Part-2

Uninformed and Informed Search

Topics to be Covered

- Introduction
- A* Algorithm
- Problem reduction search
- Memory Bound heuristic search
- Recursive Best First search (RBFS)
- Important Topics
- Summary



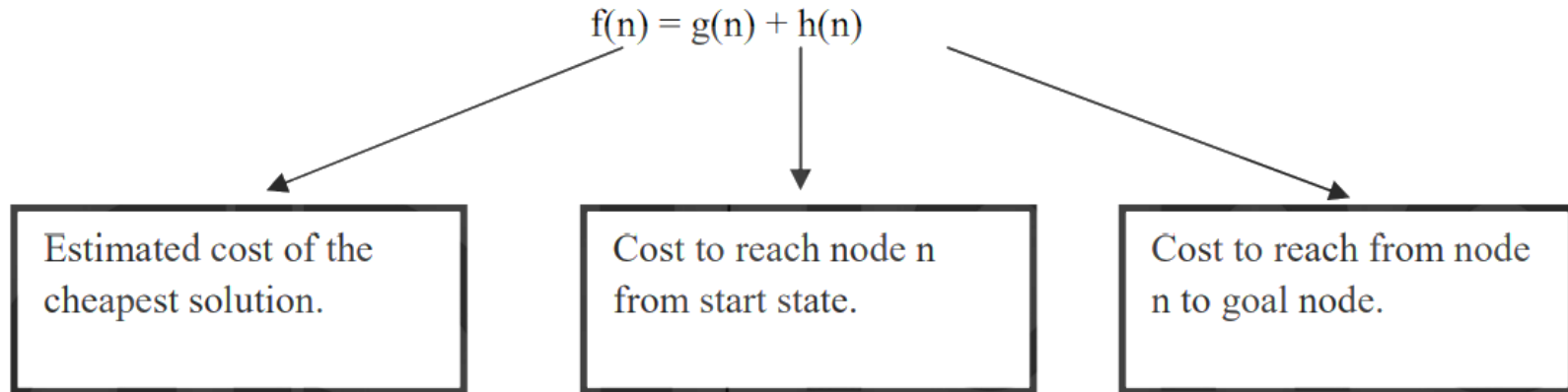
<http://ijess.org/index.php/topics/>

Introduction

- In Artificial Intelligence, Search techniques are universal problem-solving methods. Rational agents or Problem-solving agents in AI mostly used these search strategies or algorithms to solve a specific problem and provide the best result.
- Problem-solving agents are the goal-based agents and use atomic representation. One disadvantage of state space representation is that it is not possible to visualize all states for a given problem. Also, the resources of the computer system are limited to handle huge state space representation. But many problems in AI take the form of state-space search.

A* Algorithm

- A* search is the most commonly known form of best-first search. It uses heuristic function $h(n)$, and cost to reach the node n from the start state $g(n)$. It has combined features of uniform cost search (UCS) and greedy best-first search, by which it solves the problem efficiently. A* search algorithm finds the shortest path through the search space using the heuristic function. This search algorithm expands less search tree and provides optimal result faster.



A* Algorithm

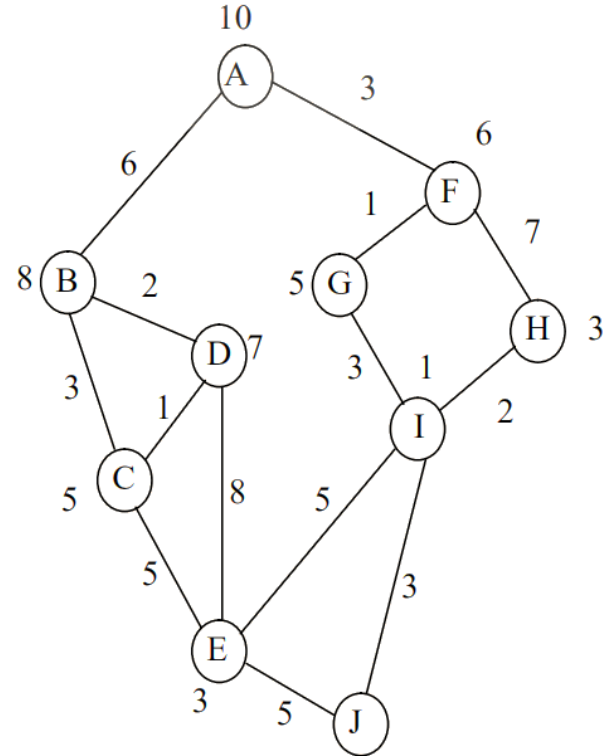
Algorithm A*

1. **Initialize:** Set $OPEN = (s)$, $CLOSED = \{\}$, $g(s) = 0$, $f(s) = h(s)$
2. **Fail:** If $OPEN = \{\}$, Terminate & fail
3. **Select :** Select the minimum cost state, n , from $OPEN$, Save n in $CLOSED$.
4. **Terminate:** If $n \in G$, terminate with success, and return $f(n)$
5. **Expand:** For each successor, m , of n
If $m \notin [OPEN \cup CLOSED]$
Set $g(m) = g(n) + C(n,m)$
Set $f(m) = g(m) + h(m)$
Insert m in $OPEN$
If $m \in [OPEN \cup CLOSED]$
Set $g(m) = \min \{ \min \{g(m), g(n) + C(n,m)\} \}$
Set $f(m) = g(m) + h(m)$
If $f(m)$ has decreased and $m \in CLOSED$,
move m to $OPEN$

A* Algorithm

Working of A* algorithm Example:

Let's us consider the following graph to understand the working of A* algorithm. The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic value. Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.



A* Algorithm

- **Step-1:** We start with node A. Node B and Node F can be reached from node A. A* Algorithm calculates $f(B)$ and $f(F)$. Estimated Cost $f(n) = g(n) + h(n)$ for Node B and Node F is: $f(B) = 6+8=14$ $f(F) = 3+6=9$
- Since $f(F) < f(B)$, so it decides to go to node F. → Closed list (F) Path- A → F
- **Step-2:** Node G and Node H can be reached from node F. A* Algorithm calculates $f(G)$ and $f(H)$. $f(G) = (3+1)+5=9$ $f(H) = (3+7)+5=13$ Since $f(G) < f(H)$, so it decides to go to node G. → Closed list (G) Path- A → F → G
- **Step-3:** Node I can be reached from node G. A* Algorithm calculates $f(I)$. $f(I) = (3+1+3)+1=8$; It decides to go to node I. → Closed list (I). Path- A → F → G → I

A* Algorithm

- Step-4: Node E, Node H and Node J can be reached from node I.
- A* Algorithm calculates $f(E)$, $f(H)$, $f(J)$.
- $f(E) = (3+1+3+5) + 3 = 15$
- $f(H) = (3+1+3+2) + 3 = 12$
- $f(J) = (3+1+3+3) + 0 = 10$
- Since $f(J)$ is least, so it decides to go to node J.
- $\rightarrow$ Closed list (J)
- Shortest Path - $A \rightarrow F \rightarrow G \rightarrow I \rightarrow J$
- Path Cost is $3+1+3+3=10$

A* Algorithm

- **Advantages:**
- A* search algorithm is the best algorithm than other search algorithms.
- A* search algorithm is optimal and complete.
- This algorithm can solve very complex problems.
- **Disadvantages:**
- It does not always produce the shortest path as it mostly based on heuristics and approximation.
- A* search algorithm has some complexity issues.
- The main drawback of A* is memory requirement as it keeps all generated nodes in the memory, so it is not practical for various large-scale problems.

A* Algorithm

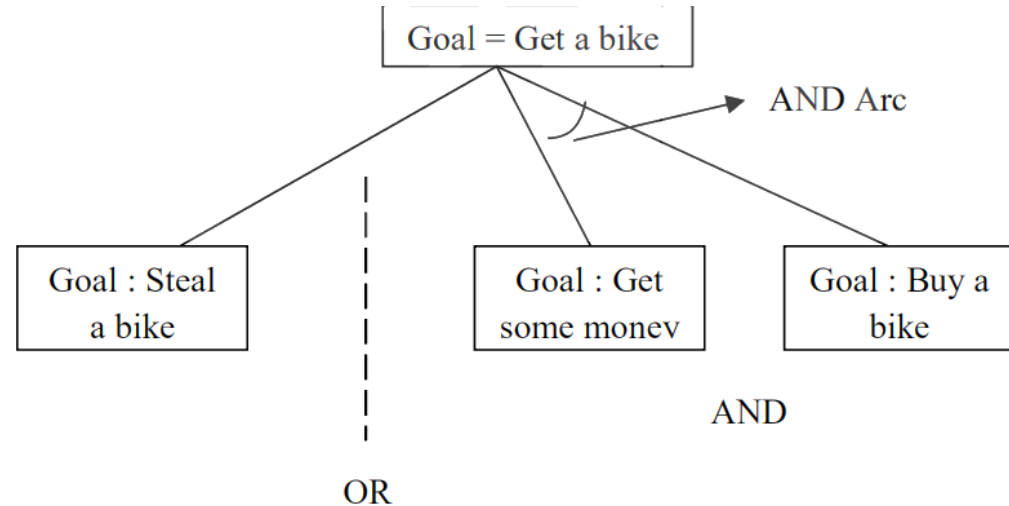
- **Properties of heuristic Algorithm:**
- **Admissibility condition:** Algorithm A is admissible if it guarantees to return an optimal solution when one exists. A heuristic function h is called admissible if many general estimates., we always have $h(n) \leq h^*(n)$
- **Completeness condition:** Algorithm A is complete if it always terminates with a solution when one exists.
- **Dominance property:** If A1 and A2 are two Admissible versions of Algorithm A such that A is more informed than A2 $h_1(n) > h_2(n)$.
- **Results on A* Algorithm**
- **A* is admissible:** Algorithm A* is admissible , that is, if there is a path from S to goal state, A* terminates by finding an optimal solution.
- **A* is complete:** If there is a path from S to goal state, A terminates (Even when the state space is ∞).

Problem Reduction Search

- Problem reduction search is broadly defined as a planning how best to solve a problem that can be recursively decomposed into subproblems in multiple ways. There are many ways to decompose a problem, we have to find the best decomposition, which gives the quality of searching or cost is minimum.
- When a problem can be divided into a set of sub problems, where each sub problem can be solved separately and a combination of these will be a solution, AND-OR graphs or AND - OR trees are used for representing the solution. The decomposition of the problem or problem reduction generates AND arcs. Consider the following example to understand the AND-OR graph given in figure.

Problem Reduction Search

- The figure shows an AND-OR graph. In an AND-OR graph, OR node represents a choice between possible decompositions, and an AND node represents given decomposition. For example, to Get a bike, we have two options, either:
 - 1.(Steal a bike) **OR**
 - 2. Get some money AND Buy a Bike



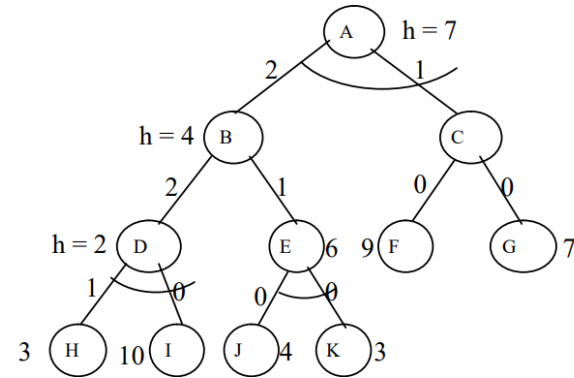
Problem Reduction Search

- In this graph we are given two choices, first Steal a bike or get some money AND Buy a Bike. When we have more than one choice and we have to pick one, we apply OR condition to choose one.(That's what we did here). Basically, the ARC here denotes AND condition.
- Here we have replicated the arc between the Get some money and buy a bike because by getting some money possibility of buying a bike is more than stealing.
- AO* search algorithm is based on AND-OR graph, so it is called AO* search algorithm. AO* Algorithm basically based on problem decomposition (Breakdown problem into small pieces).
- The main difference between the A*(A star) and AO*(AO star) algorithms is that A* algorithm represents an OR graph algorithm that is used to find a single solution (either this or that).

Problem Reduction Search

- Example2: Consider the following AND-OR Graph with estimated heuristic cost at every node. Note that A,D,E are AND node and B, C are OR node. Edge cost (i.e., $g(n)$ value) is also given. Apply AO* algorithm and find the optimal cost path using AO* algorithm.

- Heuristic (Estimated) cost at every Node is given. For example, heuristic cost at node A is $h=7$, which means at least 7-unit cost required to find a solution. Since A is AND Node, so we have to solve Both of its successor Node B and C. Cost of Node A i.e., $f(A-B-C) = (2+4) + (1+3) = 10$

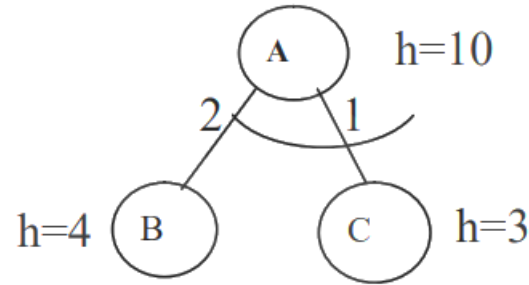


A,D,E → AND Node
B,C → OR Node

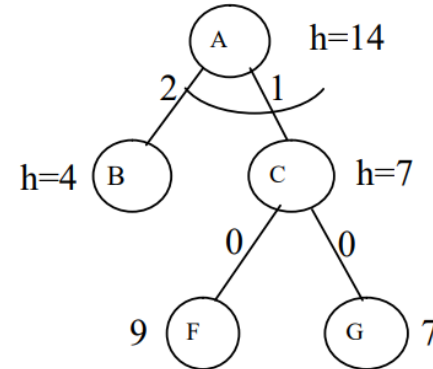
AND-OR graph with heuristic function value

Problem Reduction Search

- We perform cost revision in Bottom-up fashion. i.e., heuristic value of Node A will be 10.
- Now, Let's see R.H.S first, Node C is an OR Node. So, cost $f(C-F) = 0+9=9$ & $f(C-G) = 0+7=7$
- So Best successor of C in G, so we perform cost revision in Bottom-up fashion as follows:



Cost revision



Cost revision in Bottom-up.

Problem Reduction Search

- Now revision cost of Node A is 14; So, till now, we can say that the best path to solve the problem is: G-C-A. Now expend left node of root, i.e., Node B.
- Since B is an OR Node; So, we have to See, which successor is best; i.e., D or E.
- The revised cost of Node D is $= (3+1) + (10+0) = 14$
- And the revised cost of Node E $= (40+0) + (3+0) = 7$
- So, the Best promising successor
- Node is E. Now, perform cost Revision in Bottom-up fashion.

Problem Reduction Search

- **AO* algorithm** Our real-life situations cannot be exactly decomposed into either AND tree or OR tree but is always combination of both. So, we need an AO* algorithm where O stands for 'ordered'. Instead of two lists OPEN and CLOSED of A* algorithm, we use a single structure GRAPH in AO* algorithm.

Algorithm AO*

1. Initialize: Set $G^* = \{s\}$, $f(s) = h(s)$
If $s \in T$, label s as SOLVED
2. Terminate: If s is SOLVED, then Terminate
3. Select: Select a non-terminal leaf node n from the marked sub-free
4. Expand: Make explicit the successors of n for each new successors, m :
Set $f(m) = h(m)$
If m is terminal, label m SOLVED
5. Cost Revision: Call **Cost-Revise (n)**
6. Loop: Go to Step 2.

Cost Revision in AO*: Cost-Revise(n)

Problem Reduction Search

- **Advantage of AO*** algorithm
- Note that AO* will always find a minimum cost solution if one exists if $h'(n) < h(n)$ and that all arc costs are positive. The efficiency of this algorithm will depend on how closely $h'(n)$ approximates $h(n)$. Also note that AO* is guaranteed to terminate even on graphs that have cycles.
- **Note:** When the graph has only OR node then AO* algorithm works just like A* algorithm.

Memory Bound Heuristic Search

- The following are the commonly used memory bound heuristics search:
- Iterative deepening A^* (IDA*)
- Recursive Best-First search (RBFS)
- Memory bound A^* (MBA*)

Memory Bound Heuristic Search

- **Iterative Deepening A* (IDA*)** IDA* is a variant of the A* search algorithm which uses iterative deepening to keep the memory usage lower than in A*. It is an informed search based on the idea of the uniformed iterative deepening search. Iterative deepening A* or IDA* is similar to iterative-deepening depth-first, but with the following modifications: The depth bound modified to be an f-limit
- Start with limit = $h(\text{start})$
- Prune any node if $f(\text{node}) > \text{f-limit}$
- Next f-limit = minimum cost of any node pruned
- Iterative Deepening is a kind of uniformed search strategy. It combines the benefits of depth-first and breadth- first search.
- **Advantage of IDA* is:** It is optimal and complete like breadth first search and modest memory requirement like depth-first search.

Memory Bound Heuristic Search

- **IDA* algorithm**
 1. Set $C = f(s)$
 2. Perform DFBB with cut-off C
 3. Update C to the minimum f -value which exceeded C among states which were examined and Go TO Step 2.
- Expand a state, n , only if its f -value is less than or equal to C
- If a goal is selected for expansion, then return C and terminate
- **Working of IDA*** : Perform depth-first search LIMITED to some f -bound.
- If goal found then ok.
- Else: increase the f -bound and restart.

Memory Bound Heuristic Search

How to establish the f-bound?

Initially: $f(S)$

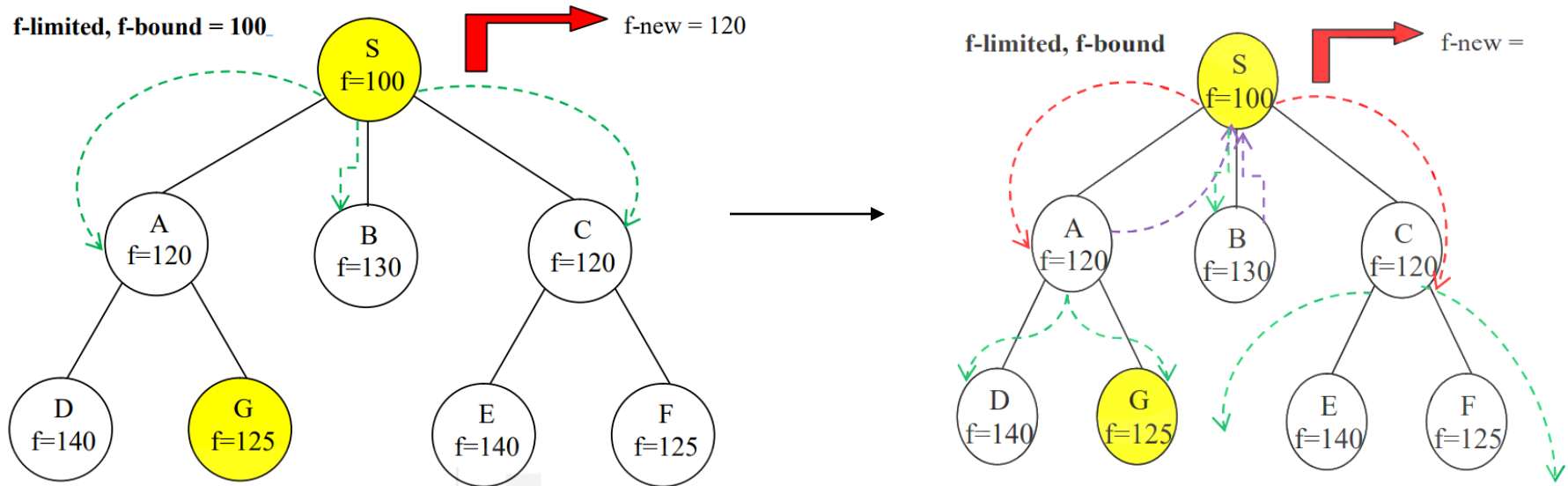
Generate all successors

Record the minimal $f(\text{succ}) > f(S)$

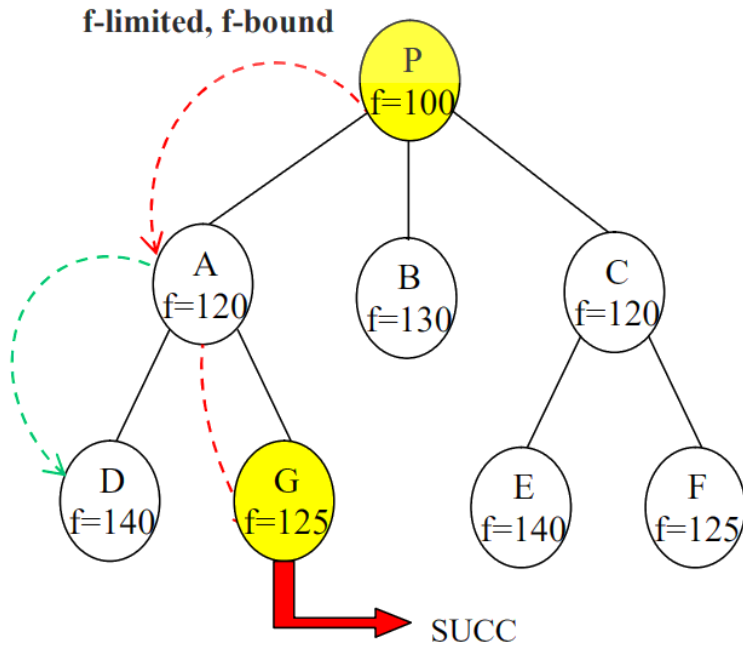
Continue with minimal $f(\text{succ})$ instead of $f(S)$

Memory Bound Heuristic Search

- Consider the following example to understand the IDA*

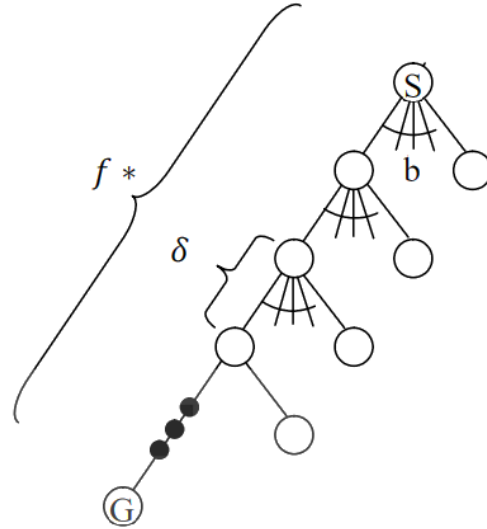


Memory Bound Heuristic Search



Memory Bound Heuristic Search

- **Analysis of IDA***
- IDA* is complete, optimal, and optimally efficient (assuming a consistent, admissible heuristic), and requires only a polynomial amount of storage in the worst case:
- F^* = optimal path cost to a goal
- B = branching factor
- δ = minimum operator step cost
- Bf^* / δ nodes of storage required



Memory Bound Heuristic Search

- Comparison of A* and IDA* algorithm:
- Iterative Deepening Search (IDS) is nothing but BFS plus DFS for tree search.
- IDA* algorithm is “complete and Optimal” algorithm.
- BFA and A* is good for optimality, but not memory.
- DFS: good for memory $O(bd)$, but not optimality
- In the worst case, only one new state is expanded in each iteration
- (IDA*). If A* expands N states, then IDA* can expand : $1+2+3+\dots+N=O(N^2)$

Recursive Best First Search (RBFS)

- The idea of recursive best first search is to simulate A* search with $O(bd)$ memory, where b is the branching factor and d is the solution depth.
- It is a memory bound, simple recursive algorithm that works like a standard best first search but only takes up linear space. There are some things that make it different from recursive DFS. It keeps track of f , the value of the best alternative path that can be found from any ancestor of the current node, instead of continuing indefinitely down the current path.
- RBFS mimic the operation of standard Best-First search algorithm. IBFS keep track of the f value of the best alternative path available from any ancestor of the current node.

Recursive Best First Search (RBFS)

- Advantages and Disadvantages of RBFS:
- More efficient than IDA* and still optimal.
- Best-first Search based on next best f-contour; fewer regeneration of nodes.
- Exploit results of search at a specific f-contour by saving next f-contour associated with a node whose successors have been explored.
- Like IDA* still suffers from excessive node regeneration.
- IDA* and RBFS not good for graphs.
- Can't check for repeated states other than those on current path.
- Both are hard to characterize in terms of expected time complexity.

Important Topics

- A* Algorithm
- Problem reduction search
- Memory Bound heuristic search
- Working of IDA*
- Recursive Best First search (RBFS)



<https://gyaanarth.com/web-stories/cbse-12th-chemistry-exam-paper-analysis/>

Summary

- **In this session we have seen that –**
- Best-first search: Order agenda based on some measure of how ‘good’ each state is.
- Uniform-cost search: Cost of getting to current state from initial state = $g(n)$
- Greedy search: Estimated cost of reaching goal from current state – Heuristic evaluation functions, $h(n)$
- A* search: $f(n) = g(n) + h(n)$
- Admissibility: $h(n)$ never overestimates the actual cost of getting to the goal state.

Summary

- Informed Search: A search strategy which searches less of the states in order to find a goal state is more informed.
- A* algorithm avoids expanding paths that are already expensive.
- In A*, Evaluation function $f(n) = g(n) + h(n)$, where $g(n)$ = cost so far to reach n , $h(n)$ = estimated cost to goal from n , $f(n)$ = estimated total cost of path through n to goal.
- A* search uses an admissible heuristic function, i.e., $h(n) \leq h^*(n)$ where $h^*(n)$ is the true cost of cheapest solution from n .
- A* search has a very good property: A* search is optimal! So, if there is any solution, A* search is guaranteed to find a least cost solution. Remember, this needs an admissible heuristic.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-1 Unit-4 Part - 2

Predicate and Propositional Logic

Topics to be Covered

- Introduction
- Inferencing in Predicate Logic
- Proof Systems
- Natural Deduction
- Propositional Resolution
- Important Topics
- Summary



Introduction

- Logic is the study and analysis of the nature of the valid argument, the reasoning tool by which valid inferences can be drawn from a given set of facts and premises.
- It is the basis on which all the sciences are built, and this mathematical theory of logic is called symbolic logic.
- The reason why the subject-matter of the study is called Symbolic Logic is that symbols are used to denote facts about objects of the domain and relationships between these objects.

Inferencing in Predicate Logic

- We are given a set of arguments in predicate logic. Now, using the rules of inference, we can deduce other arguments (or predicates) based on the given arguments (or predicates). With the help of the following example we will learn how the rules of inference, discussed above, can be used to solve the problems. **Example :**
- There is a village that consists of two types of people – those who always tell the truth, and those who always lie. Suppose that you visit the village and two villagers A and B come up to you. Further, suppose
- A says, “B always tells the truth,” and
- B says, “A and I are of opposite types”.
- What types are A and B ?

Inferencing in Predicate Logic

- Solution: Let us start by assuming A is a truth-teller.
- What A says is true.
- B is a truth-teller.
- What B says is true.
- A and B are of opposite types.

Inferencing in Predicate Logic

- This is a contradiction, because our premises say that A and B are both truth-tellers.
- The assumption we started with is false.
- A always tells lies.
- What A has told you is lie.
- B always tells lies.
- A and B are of the same type, i.e., both of them always lie.

Proof Systems

- A sequence of statements that follows logically from the previous set of statements or observations are the mathematical proofs in propositional logic. The last statements in the proof becomes the theorem. This proof system symbolizes the science of valid inference. There are more than two main styles of proof system for propositional logic, but the two main styles are:
- **Axiomatic Proof Systems** : This is a system where conclusion is derived from either the given hypothesis or using assumed premise, which are considered as truth. Such hypothesis or assumed premise is an axiom.
- Example : Show that $(\alpha_1 \vee \alpha_2)$ is a logical consequence of $(\alpha_2 \wedge \alpha_1)$ using proof systems.

Proof Systems

- **Proof:**
- Step 1: $(\alpha_2 \wedge \alpha_1)$ (Premise)
- Step 2: α_1 (Simplification, Step 1)
- Step 3: α_2 (Simplification, Step 1)
- Step 4: $\alpha_1 \vee \alpha_2$ (Addition on either Step 2 or Step 3)
- We observe here that each step is either a truth or a logical consequence of previously established truths. In general, there can be more than one proof for establishing a given conclusion. Thus, the proof system in propositional logic is quite similar to the proofs in mathematics, which follows a step-wise derivation of consequent from the given hypothesis. However, in propositional logic, we use well-formed formulas obtained from literals and connectors, rather than using English statements.

Natural Deductions

- Till now, we have discussed methods, of solving problems requiring reasoning of propositional logic, that were based on
 - i) Truth-table construction
 - ii) Use of inference rules,
- and follow, directly or indirectly, natural deduction approach.

Natural Deductions

- Example 2: Show that $\sim P$ is concluded from RV S, $S \rightarrow \sim Q$, $P \rightarrow Q$ and $R \rightarrow \sim Q$.
- **Proof :**
- Step 1: $\sim(\sim P)$ (Negated conclusion as Premise)
- Step 2: P (Step 1, Double negation)
- Step 3: $P \rightarrow Q$ (Premise)
- Step 4: Q (Step 2, 3, Modus Ponens)
- Step 5: $S \rightarrow \sim Q$ (Premise)
- Step 6: $\sim S$ (Step 4, 5, Modus Tollens)

Natural Deductions

- Step 7: $R \vee S$ (Premise)
- Step 8: R (Step 1, 3, 5, 7, Disjunctive Syllogism)
- Step 9: $R \rightarrow \sim Q$ (Premise)
- Step 10: $\sim Q$ (Step 1, 3, 5, 7, 9 and Modus Ponens)
- We obtain $\sim Q$ in Step 10 and Q in step 4, which is a contradiction. Thus, our assumption is not valid. Hence, the conclusion follows from the set of premises.

Propositional Resolution

- For the most part, there are two distinct strategies that can be implemented in order to demonstrate the correctness of a theorem or derive a valid conclusion from a given collection of axioms:
- **Natural Deduction** : In the method known as natural deduction, one begins with a given set of axioms, applies various rules of inference, and ultimately arrives at a conclusion. This method is strikingly similar to the intuitive reasoning that is characteristic of humans

Propositional Resolution

- **Method of Refutation** : When using a refutation approach, one begins with the denial of the conclusion that is to be drawn and then proceeds to deduce a contradiction or the word "false." We are able to deduce a contradiction as a result of having presupposed that the conclusion is incorrect; hence, the premise that the conclusion is incorrect is itself incorrect.
- Therefore, the argument concerning the technique of resolution leads to the correctness of the conclusion.

Propositional Resolution

- **Resolution Rule:** The rule states given two statements as $\{\alpha_1, \alpha_2, \alpha_3 \dots \alpha_m\}$ and $\{\gamma_1, \gamma_2, \gamma_3 \dots \gamma_n\}$, then the conclusion is $\{\alpha_1, \alpha_2, \alpha_3 \dots \alpha_m, \gamma_1, \gamma_2, \gamma_3, \dots \gamma_n\}$. For example, the statements $\{A, B\}$ and $\{C, \sim B\}$ leads to the conclusion $\{A, C\}$.
- **Resolution Refutation:**
 - a) Convert all the given statements to Conjunctive Normal Form (CNF). It is also described as AND of ORs'. For eg., $(A \vee B) \wedge (\sim A \vee B) \wedge (\sim B \vee A)$ is a CNF.
 - b) Obtain the negation of the given conclusion
 - c) Apply the resolution rule until either the contradiction is obtained or the resolution rule cannot be applied anymore.

Propositional Resolution

- **Clausal Form** Any atomic sentence or its negation is called as a literal. A literal or the disjunction of at least two literals is called as the clausal form or clause expression. Next, a clause is defined as the set of literals in the clause form or clause expression. For example, consider two atomic statements represented using the literals X and Y . Their clausal expressions are X , $\sim X$ and $(X \vee Y)$ and the clauses for these expressions are $\{X\}$, $\{\sim X\}$ and $\{X, Y\}$. It is noteworthy that the empty set $\{\}$ is always a clause as it represents an empty disjunction and hence, is unsatisfiable. Kindly note, conjunctive normal form (CNF) of an expression represents the corresponding clausal form.

Propositional Resolution

- Now, we shall first understand certain rules for converting the statements to the clause form as given below.
- Operator: $(\beta_1 \vee \beta_2 \vee \beta_3 \vee \dots \vee \beta_m) \Rightarrow \{\beta_1, \beta_2, \beta_3, \dots, \beta_m\}$
- $(\beta_1 \wedge \beta_2 \wedge \beta_3 \wedge \dots \wedge \beta_m) \Rightarrow \{\beta_1\}, \{\beta_2\}, \{\beta_3\}, \dots, \{\beta_m\}$
- ii) Negation: same as Double Negation and De Morgan's Law
- iii) Distribution: as in sub-section
- iv) Implications: as Material implication and Material Equivalence

Propositional Resolution

- Derive the Clausal form or Conjunctive normal form of $X \leftrightarrow Y$.
- Step 1: Replace bi-condition using Material equivalence rule:
 - $(X \rightarrow Y) \wedge (Y \rightarrow X)$
- Step 2: Use Material Implication replacement rule to replace the implication:
 - $(\neg X \vee Y) \wedge (\neg Y \vee X)$

Propositional Resolution

- **Determining Unsatisfiability** : If the obtained set of clauses is not satisfiable, then one can derive an empty clause using resolution principle as described above. In other words, to determine whether a set of propositions or premises $\{P\}$ logically entails a conclusion C , write $P \vee \{\neg C\}$ in clausal form and try to derive the empty clause as explained in examples below.
- Example 1: Given a set of propositions : $X \rightarrow Y$, $Y \rightarrow Z$. Prove $X \rightarrow Z$
- Proof: To prove the conclusion, we add the negation of conclusion i.e $\sim(X \rightarrow Z)$ to the set of premises, and derive an empty clause.

Propositional Resolution

- Step 1: $X \rightarrow Y$ (Premise)
- Step 2: $\sim X \vee Y$ (Premise, Material Implication)
- Step 3: $Y \rightarrow Z$ (Premise)
- Step 4: $\sim Y \vee Z$ (Premise, Material Implication)
- Step 5: $\sim(X \rightarrow Z)$ (Premise, Negated Conclusion)
- Step 6: $\sim(\sim X \vee Z)$ (Premise, Material Implication)
- Step 7: $X \wedge \sim Z$ (Premise, DeMorgans)

Propositional Resolution

- Step 8: X (Clausal form, Operator)
- Step 9: $\sim Z$ (Clausal form, Operator)
- Step 10: Y (Resolution rule on Premises in Step 2 and Step 8)
- Step 11: Z (Resolution rule on Step 10 and Step 4)
- Step 12: $\{ \}$ (Conjunction on Step 11 and Step 9)
- Thus, the given set of premises entail the conclusion.

Propositional Resolution

- Example 2: Use propositional resolution to derive the goal from the given knowledge base.
- a) Either it is a head, or Lisa wins.
- b) If Lisa wins, then Mary will go.
- c) If it is a head, then the game is over.
- d) The game is not over.
- Conclusion: Mary will go.
- Proof: First consider propositions to represent each of the statement in knowledge base.

Propositional Resolution

- Let H: It is a head
- L: Lisa wins
- M: Mary will go
- G: Game is over.
- Re-writing the given knowledge base using the propositions defined.
- a) $H \vee L$
- b) $L \rightarrow M$
- c) $H \rightarrow G$
- d) $\sim G$

Propositional Resolution

- Conclusion: M
- Step 1: HVL (Premise)
- Step 2: $L \rightarrow M$ (Premise)
- Step 3: $\sim L \vee M$ (Step 2, Material Implication)
- Step 4: $H \rightarrow G$ (Premise)
- Step 5: $\sim H \vee G$ (Step 4, Material Implication)
- Step 6: $\sim G$ (Premise)
- Step 7: $\sim M$ (Negated conclusion as Premise)

Propositional Resolution

- Step 8: $H \vee M$ (Resolution principle on Step 1 and 3)
- Step 9: $M \vee G$ (Resolution principle on Step 8 and 5)
- Step 10: M (Resolution principle on Step 9 and 6)
- Step 11: $\{\}$ (Sep 10 and 7)
- After applying Proof by Refutation i.e., contradicting the conclusion, the problem is terminated with an empty clause ($\{\}$). Hence, the conclusion is derived.

Important Topics

- Inference in Predicate Logic
- Proof Systems
- Natural Deduction
- Propositional Resolution



<https://cetking.com/important-topics-tips-cat-preparation/>

Summary

- In this session we also learned about the propositions that contain the quantifiers ‘for All’ and ‘there exists’. In symbolic logic, the goal was to determine which propositions are true and which are false.
- Using symbolic logic, we can formalize our arguments and logical reasoning in a manner that can easily show if the reasoning is valid, or is a fallacy. How we symbolize the reasoning is what is presented in this session.
- Known different types of connectives, their associated semantics and corresponding truth tables.
- Differentiate between valid and satisfiable arguments.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-2 Unit-1

First Order Predicate Logic

Topics to be Covered

- Introduction
- Syntax of First Order Predicate Logic(FOPL)
- Interpretations in FOPL
- Semantics of Quantifiers
- Inference & Entailment in FOPL
- Conversion to clausal form
- Resolution & Unification
- Important Topics
- Summary

https://en.wikipedia.org/wiki/Wikipeak_artificial_intelligence



Introduction

- The symbolic system of FOPL treats an atomic statement not as an indivisible unit. Rather, FOPL not only treats an atomic statement divisible into subject and predicate but even further deeper structures of an atomic statement are considered in order to handle larger class of arguments.
- First Order Predicate Logic (FOPL) is the most well-known symbolic system for the purpose.
- FOPL symbolizes and establishes validity/invalidity and consistency/inconsistency of arguments is the subject matter of this unit.

Syntax of First Order Predicate Logic

- A propositional function, or a predicate, in a variable x is a sentence $p(x)$ involving x that becomes a proposition when we give x a definite value from the set of values it can take. We usually denote such functions by $p(x)$, $q(x)$, etc. The set of values x can take is called the universe of discourse.
- Propositional function, in logic, a statement expressed in a form that would take on a value of false or true were it not for the appearance within it of a variable x (or of several variables), which leaves the statement undetermined as long as no definite values are specified for the variables.
- So, if $p(x)$ is ' $x > 5$ ', then $p(x)$ is not a proposition. But when we give x particular values, say $x = 6$ or $x = 0$, then we get propositions. Here, $p(6)$ is a true proposition and $p(0)$ is a false proposition.

Interpretations in FOPL

- FOPL stands for First Order Predicate Logic. Let us understand from an example :
- Every man is lucky. Raghav is a man.
- Hence, he is mortal.
- In order to derive the validity of above simple argument, instead of looking at an atomic statement as indivisible, to begin with, we divide each statement into subject and predicate. The two predicates which occur in the above argument are:
- ‘is lucky’ and ‘is man’.

Interpretations in FOPL

- Let us use the notation
- IL: is_lucky and
- IN: is_man.
- In view of the notation, the argument on para-phrasing becomes:
- For all x, if IN (x) then IL (x).
- IN (Raghav).
- Hence, IL (RAMAN)

Interpretations in FOPL

- Function symbols can be used in the first-order logic. For example, we can use product (x, y) to denote $x * y$ and father (x) to mean the 'father of x '. The statement: Mohan's father loves Mohan can be symbolised as $\text{LOVE}(\text{father}(\text{Mohan}), \text{Mohan})$. Thus, we need not know name of father of Mohan and still we can talk about him. A function serves such a role.
- We may note that $\text{LIKE}(\text{Ram}, \text{Mohan})$ and $\text{LOVE}(\text{father}(\text{Mohan}), \text{Mohan})$ are atoms or atomic statements of PL, in the sense that, one can associate a truth-value True or False with each of these, and each of these does not involve a logical operator like $\sim, \wedge, \rightarrow, \leftrightarrow$ or $\vee$.

Interpretations in FOPL

- From the above discussion we learned the following concepts of symbols.
- **Individual symbols or constant symbols:** These are usually names of objects, such as Ram, Mohan, numbers like 3, 5 etc.
- **Variable symbols:** These are usually lowercase unsubscripted or subscripted letters, like x , y , z , x_3 .
- **Function symbols:** These are usually lowercase letters like f , g , h ,....or strings of lowercase letters such as father and product.
- **Predicate symbols:** These are usually uppercase letters like P , Q , R ,....or strings of lowercase letters such as greater-than, is_tall etc.

Semantics of Quantifiers

- A propositional function, or a predicate, in a variable x is a sentence $p(x)$ involving x that becomes a proposition when we give x a definite value from the set of values it can take. We usually denote such functions by $p(x)$, $q(x)$, etc. The set of values x can take is called the universe of discourse.
- So, if $p(x)$ is ' $x > 5$ ', then $p(x)$ is not a proposition. But when we give x particular values, say $x = 6$ or $x = 0$, then we get propositions. Here, $p(6)$ is a true proposition and $p(0)$ is a false proposition.
- Examples: Prove for the following closed formula
- **Ques :** $(\forall x) P(x) \wedge (\exists y) \sim P(y)$ is inconsistent.
- **Solution:** Consider

Semantics of Quantifiers

- $(\forall x) P(x) \wedge (\forall y) \sim P(y) = (\forall x) P(x) \wedge \sim (\forall y) P(y)$ (taking negation out)
- But we know for each bound occurrence, a variable is dummy, and can be replaced in the whole scope of the variable uniformly by another free variable. Hence,
- $R = (\forall x) P(x) \wedge \sim (\forall x) P(x)$
- Each conjunct of the formula is either
- True or False and, hence, can be thought of as a formula of PL, in stead of formula of FOPL, Let us replace $(\forall x) (P(x))$ by Q , a formula of PL.
- $R = Q \wedge \sim Q = \text{False}$
- Hence, the proof.

Inferencing & Entailment in FOPL

- Inference in First-Order Logic is used to deduce new facts. It can also be used to deduce sentences from existing sentences. As propositional logic we also have inference rules in first-order logic, so following are some basic inference rules in FOL:
- **Universal Generalization:** is a valid inference rule which states that if premise $P(c)$ is true for any arbitrary element c in the universe of discourse, then we can have a conclusion as $\forall x P(x)$.
- It can be represented as:
$$\frac{P(c)}{\forall x P(x)}$$
- **Example:** Let's represent, $P(c)$: "A byte contains 8 bits", so for $\forall x P(x)$ "All bytes contain 8 bits.", it will also be true.

Inferencing & Entailment in FOPL

- **Universal instantiation** is also called as universal elimination or UI is a valid inference rule. It can be applied multiple times to add new sentences.
- It is represented as:
$$\frac{\forall x P(x)}{P(c)}$$
- **Example** : "All kings who are greedy are Evil." So let our knowledge base contains this detail as in the form of FOL:
- $\forall x \text{ king}(x) \wedge \text{greedy}(x) \rightarrow \text{Evil}(x),$
- $\text{King}(\text{John}) \wedge \text{Greedy}(\text{John}) \rightarrow \text{Evil}(\text{John}),$
- $\text{King}(\text{Richard}) \wedge \text{Greedy}(\text{Richard}) \rightarrow \text{Evil}(\text{Richard}),$
- $\text{King}(\text{Father}(\text{John})) \wedge \text{Greedy}(\text{Father}(\text{John})) \rightarrow \text{Evil}(\text{Father}(\text{John}))$
- So from this information, have inferred the above statements using Universal Instantiation

Inferencing & Entailment in FOPL

- **Existential instantiation** is also called as Existential Elimination, which is a valid inference rule in first-order logic. It can be applied only once to replace the existential sentence.
- It can be represented as:
$$\frac{\exists x P(x)}{P(c)}$$
- Example:
- From the given sentence: $\exists x \text{Crown}(x) \wedge \text{OnHead}(x, \text{John})$,
- So we can infer: $\text{Crown}(K) \wedge \text{OnHead}(K, \text{John})$, as long as K does not appear in the knowledge base.
- The above used K is a constant symbol, which is called Skolem constant.
- The Existential instantiation is a special case of Skolemization process.

Inferencing & Entailment in FOPL

- **Existential Generalization Rule (E.G)** The Universal Generalization (U.G) and Existential Instantiation rules should be applied with utmost care, however, other two rules may be applied, whenever, it appears to be appropriate.
- It is represented as :
$$\frac{P(a)}{(\exists x)P(x)}$$
- Example: Symbolize the following and then construct a proof for the argument:
- (i) Anyone who repairs his own car is highly skilled and saves a lot of money on repairs
- (ii) Some people who repair their own cars have menial jobs. Therefore,
- (iii) Some people with menial jobs are highly skilled.

Inferencing & Entailment in FOPL

- Solution: Let us use the notation:
- $P(x)$: x is a person
- $S(x)$: x saves money on repairs
- $M(x)$: x has a menial job
- $R(x)$: x repairs his own car
- $H(x)$: x is highly skilled
- Therefore, (i), (ii) and (iii) can be symbolized as:
- (i) $(\forall x) (R(x) \rightarrow (H(x) \wedge S(x)))$
- (ii) $\exists(x) (R(x) \wedge M(x))$
- (iii) $(\exists x) (M(x) \wedge H(x))$ (to be concluded)

Conversion to Clausal Form

- The first step towards the Clausal form is to begin with Prenex Normal Form (PNF), and the second step is skolemization, which will be discussed after PNF.
- **Prenex Normal Form (PNF)** : is relates to re-alignment of the quantifiers, i.e. to bring all the quantifiers in the beginning of the expression and then replacement the existential and universal quantifiers with constants and the functions is performed for skolemization i.e. to bring the statement in the clausal form.
- **Steps for Transforming an FOPL Formula into Prenex Normal Form**
- Step 1 : Remove the connectives ' $\rightarrow$ ' and ' $\leftrightarrow$ ' using the equivalences.
- Step 2 Use the equivalence to remove even number of $\sim$'s

Conversion to Clausal Form

- Step 3 Apply De Morgan's laws in order to bring the negation signs immediately before atoms.
- Step 4 rename bound variables if necessary
- Step 5 Bring quantifiers to the left before any predicate symbol appears in the formula. This is achieved by using (i) to (vi) discussed above.

Conversion to Clausal Form

- **Example:** Transform the following formulas into prenex normal forms:

- $(\forall x) (Q(x) \rightarrow (\exists x) R(x, y))$

- **Solution** *Step 1: By removing ' $\rightarrow$ ', we get*

$$(\forall x) (\sim Q(x) \vee (\exists x) R(x, y))$$

Step 2: By renaming x as z in $(\exists x) R(x, y)$ the formula becomes

$$(\forall x) (\sim Q(x) \vee (\exists z) R(z, y))$$

Step 3: As $\sim Q(x)$ does not involve z , we get

$$(\forall x) (\exists z) (\sim Q(x) \vee R(z, y))$$

Resolution & Unification

- Resolution is used, if there are various statements are given, and we need to prove a conclusion of those statements. Unification is a key concept in proofs by resolutions. Resolution is a single inference rule which can efficiently operate on the conjunctive normal form or clausal form.
- Steps for Resolution:
- Conversion of facts into first-order logic.
- Convert FOL statements into CNF
- Negate the statement which needs to prove (proof by contradiction)
- Draw resolution graph (unification).

Important Topics

- Interpretations in FOPL
- Semantics of Quantifiers
- Inference & Entailment in FOPL
- Conversion to clausal form
- Resolution & Unification



<https://gyaanarth.com/web-stories/cbse-12th-chemistry-exam-paper-analysis/>

Summary

- In this session , initially, we discuss how PL is inadequate to solve even simple problems, requires some extension of PL or some other formal inferencing system so as to compensate for the inadequacy. First Order Predicate Logic (FOPL), is such an extension of PL that is discussed in the unit. Next, syntax of proper structure of a formula of FOPL is discussed. In this respect, a number of new
- concepts including those of quantifier, variable, constant, term, free and bound occurrences of variables; closed and open wff, consistency/validity of wffs etc. are introduced.
- Next, two normal forms viz. Prenex Normal Form (PNF) and Skolem Standard Normal Form are introduced. Finally, tools and techniques developed in the unit, are used to solve problems involving logical reasoning.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-2 Unit-2

Rule Based Systems and Other Formalism

Topics to be Covered

- Introduction
- Rule Based Systems
- Semantic nets
- Frames
- Scripts
- Important Topics
- Summary



<https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence>

Introduction

- Building a expert system is known as knowledge engineering and its practitioners are called knowledge engineers. It is the job of the knowledge engineer to ensure to make sure that the computer has all the knowledge needed to solve a problem.
- The knowledge engineer must choose one or more forms in which to represent the required knowledge i.e., s/he must choose one or more knowledge representation schemes.

Rule Based Systems

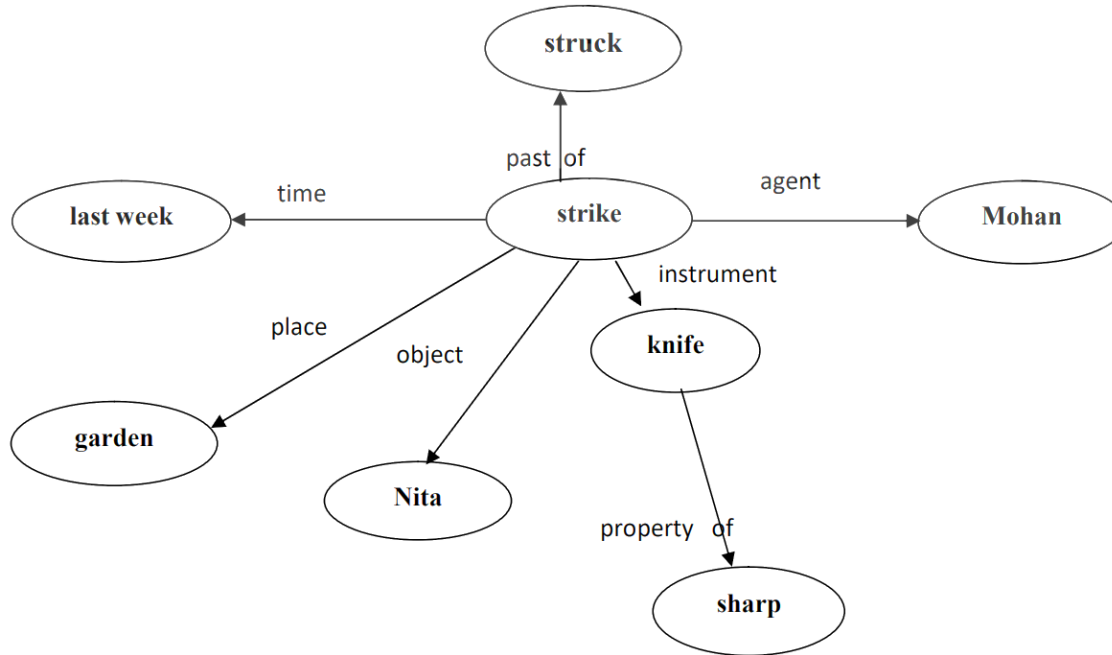
- Planning is the process that exploits the structure of the problem under consideration for designing a sequence of actions in order to solve the problem under consideration. There are two broad kinds of rule-based systems:
- In a **forward chaining** system we start with the initial facts, and keep using the rules to draw new intermediate conclusions (or take certain actions) given those facts. The process terminates when the final conclusion is established.
- In a **backward chaining** system, we start with some goal statements, which are intended to be established and keep looking for rules that would allow us to conclude, setting new subgoals in the process of reaching the ultimate goal. In the next round, the subgoals become the new goals to be established. The process terminates when in this process all the subgoals are given fact.
- Forward chaining systems are primarily data-driven, while backward chaining systems are goal-driven.

Semantic Nets

- Semantic Network representations provide a structured knowledge representation. In such a network, parts of knowledge are clustered into semantic groups.
- In semantic networks, the concepts and entities/objects of the problem domain are represented by nodes and relationships between these entities are shown by arrows, generally, by directed arrows.
- In view of the fact that semantic network representation is a pictorial depiction of objects, their attributes and the relationships that exist between these objects and other entities.
- A semantic net is just a graph, where the nodes in the graph represent concepts, and the arcs are labeled and represent binary relationships between concepts.

Semantic Nets

- For example, the fact (a piece of knowledge): **Mohan struck Nita in the garden with a sharp knife last week**, is represented by the semantic network as :



Frames

- Frames are a variant of semantic networks that are one of the popular ways of representing nonprocedural knowledge in an expert system. In a frame, all the information relevant to a particular concept is stored in a single complex entity, called a frame. Frames look like the data structure, record. Frames support inheritance.
- We may represent some knowledge about a lion in frames as follows:
- Mammal :
- Subclass : Animal
- warm blooded : yes

Frames

- Lion :
 - subclass : Mammal
 - eating-habit : carnivorous
 - size : medium
 - Raja :
 - instance : Lion
 - colour : dull-Yellow
 - owner : Amar Circus
- A particular frame (such as Lion) has a number of attributes or slots such as eating-habit and size. Each of these slots may be filled with particular values, such as the eating-habit for lion may be filled up as carnivorous.
 - Frame systems are pretty complex and sophisticated knowledge representation tools. This representation has become so popular that special high level frame based representation languages have been developed.

Scripts

- A script is a structured representation describing a stereotyped sequence of events in a particular context.
- Scripts are used in natural language understanding systems to organize a knowledge base in terms of the situations that the system should understand. Scripts use a frame-like structure to represent the commonly occurring experience like going to the movies eating in a restaurant, shopping in a supermarket.
- A script is a structure that prescribes a set of circumstances that could be expected to follow on from one another.

Scripts

Scripts are beneficial because:

Events tend to occur in known runs or patterns.

A casual relationship between events exist.

An entry condition exists which allows an event to take place.

Prerequisites exist upon events taking place.

Scripts

- The components of a script include:
- Entry condition: These are basic condition which must be fulfilled before events in the script can occur.
- Results: Condition that will be true after events in script occurred.
- Props: Slots representing objects involved in events
- Roles: These are the actions that the individual participants perform.
- Track: Variations on the script. Different tracks may share components of the same scripts.
- Scenes: The sequence of events that occur.

Important Topics

- Forward chaining
- Backward chaining
- Conflict resolution
- Semantic nets
- Frames
- Scripts



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Summary

- This session majorly discussed the various knowledge representation mechanisms, used in Artificial Intelligence.
- The session begins with the discussion on Rule Based Systems, and discussed the related concept of Forward chaining and Backward chaining, later the concept of Conflict resolution is discussed.
- The session also discussed the other techniques of knowledge representation like Semantic nets, Frames and Scripts.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-2 Unit-3

Probabilistic Reasoning

Topics to be Covered

- Introduction
- Reasoning with uncertain information
- Review of Probability Theory
- Introduction to Bayesian Theory
- Baye's Networks
- Probabilistic Inference
- Basic idea of Inferencing with Bayes Networks
- Other Paradigm of Uncertain Reasoning
- Dempster Scheffer Theory
- Important Topics
- Summary

Introduction

- This session is dedicated to probability theory and its usage in decision making for various problems. Contrary to the classical decision making of True and False propositions, the probability of the truth value with a certain probability is used for making decisions.
- The inclusion of such a probabilistic approach is quite relevant since uncertainties are quite obvious in the real world.



Reasoning with Uncertain Information

- Reasoning is an important step for various decision making. The amount of information and its correctness plays a crucial role in reasoning.
- Decision making is easier when we have certain information i.e. the correctness of information can be ascertained.
- In the other situation when the certainty of information can not be ascertained, the decision-making process is likely to be erroneous or may not be correct.

Review Of Probability Theory

- **Trials, Sample Space, Events** : You must have often observed that a random experiment may comprise a series of smaller sub-experiments. These are called trials. Consider for instance the following situations.
- Example 1: Suppose the experiment consists of observing the results of three successive tosses of a coin. Each toss is a trial and the experiment consist of three trials so that it is completed only after the third toss (trial) is over.
- Example 2: Consider the experiment of throwing a coin. Suppose we repeat the process of throwing a coin 5 times and suppose the frequencies of occurrence of head is tabulated below in Table:

Review Of Probability Theory

No. of repetitions (n)	Frequency of head ($f_n(H)$)	Relative frequency of head $r_n(H)$
1	0	0
2	1	$1/2$
3	2	$2/3$
4	3	$3/4$
5	3	$3/5$

- Notice that the third column in Table gives the relative frequencies $r_n(H)$ of heads. We can keep on increasing the number of repetitions n and continue calculating the values of $r_n(H)$ in Table. Merely to fix ideas regarding the concept of probability of an event, we present below a very naive approach which in no way is rigorous, but it helps to see things better at this stage.

Review of Probability Theory

- **Conditional probability and independent events :** Let ζ be the sample space corresponding to an experiment and E and F are two events of ζ Suppose the
- experiment is performed and the outcome is known only partially to the effect that the event F has taken place. Thus there still remains a scope for speculation about the occurrence of the other event E. Keeping this additional piece of information confirming the occurrence of F in view, it would be appropriate to modify the probability of occurrence of E suitably. That such modifications would be necessary can be readily appreciated through two simple instances as follows:
- **Example :** Suppose, E and F are such that F E so that occurrence of F would automatically imply the occurrence of E. Thus with the information that the event F has taken place in view, it is plausible to assign probability 1 to the occurrence of E irrespective of its original probability.

Introduction to Bayesian Theory

- Bayes' theorem is widely used to calculate the conditional probabilities of events without a joint probability. It is also used to calculate the conditional probability where intuition fails. In simple terms, probability of a given hypothesis H conditional on E can be defined as $P_E(H) = P(H \& E) / P(E)$, where $P(E) > 0$, and the term $P(H \& E)$ also exists. Here P_E is referred to as a probability function. To simply understand the Bayes' theorem, have a look at the following definitions.
- **Joint Probability:** This refers to the probability of two or more events simultaneously occurring, e.g. $P(A \text{ and } B)$ or $P(A, B)$.
- **Marginal Probability:** It is the probability of an event occurring irrespective of outcome of the other random variables e.g. $P(A)$.
- **Conditional probability:** A conditional probability is defined as the probability of occurrence of an event provided that another event has occurred. e.g. $P(A | B)$.

Baye's Networks

- Bayesian belief network is key computer technology for dealing with probabilistic events and to solve a problem which has uncertainty. We can define a Bayesian network as:
- "A Bayesian network is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph."

Probabilistic Inference

- The probabilistic inference is very much dependent on the conditional probability of the specified events provided the information of occurrence of other events is available. For example, two events E and F such that $P(F) > 0$, the conditional probability of event E when F has occurred can be written as :

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

- When an experiment is repeated a large number of times (say n), the above expression can be given a frequency interpretation. Let the number of occurrences of an event F is represented as No. (F) and the probability of a joint event of E and F as No. (E F). The relative frequencies of both these events can be computed as f_r :

$$f_r(F) = \frac{\text{No. (F)}}{n}$$

Basic idea of Inferencing with Baye's Networks

- A network represents the degree of belief of proposition and their causal interdependence. The inference in a network can be done by propagating the given probabilities of related information through the network giving the output to one of the conclusion nodes. The network representation also reduces the time and space requirements for huge computations involving the probabilities of uncertain knowledge of propositional variables.
- When the outcome of an inference chain results in one or more proposed conclusions, the alternatives must be compared and one should be chosen on the basis of likelihood.

Other Paradigm of Uncertain Reasoning

- The other ways of dealing with uncertainty are the ones with no theoretical proof. These are mostly based on intuition. These are selected over formal methods as a pragmatic solution to a particular problem, when the formal methods impose difficult or impossible conditions.
- One such ad hoc procedure is used to diagnose meningitis and infectious blood disease, the system is called MYCIN. The MYCIN uses If and then rules to assess various forms of patient evidence. It also measures both belief and disbelief to represent degree of confirmation and disconfirmation respectively in a given hypothesis. The ad hoc methods have been used in a larger number of knowledge-based systems than formal methods.
- This is due to the difficulties encountered in acquiring a large number of reliable probabilities related to the given domain and to the complexities to the ensuing calculations.

Dempster Scheffer Theory

- The D-S theory is a generalization of Bayesian probability theory where multiple possible events are assigned probabilities opposed to mutually exclusive singletons. The D-S theory assumes the existence of ignorance in knowledge creating uncertainty which in turn induces belief. Here, uncertainty of the hypothesis is represented by the belief function. The main characteristic of the theory is:
- Multiple possible events are permitted to assign probabilities.
- These events should be exhaustive and exclusive.
- Let us now define a few terms used in D-S theory which will be useful for us.

Dempster Scheffer Theory

- **Evidence** These are events related to one hypothesis or set of hypotheses. Here, a relation is not permitted between various pieces of evidence or set of hypotheses. Also, the relation between the set of hypotheses and the piece of evidence is only quantified by a source of data. In context of D-S theory, we have four types of evidences as following:
- **Consonant Evidence:** These are basically appearing included into the next bigger subset and so on. Here with each increasing subset size, the information refines the evidentiary set over the time.
- **Consistent Evidence:** This assures the presence of at least one
- **Arbitrary Evidence:** A situation where there is not a common element occurring in the subsets though some of the subsets may have a few common element(s).
- **Disjoint Evidence:**

Important Topics

- Reasoning with uncertain information
- Bayesian Theory
- Baye's Networks
- Probabilistic Inference
- Dempster Scheffer Theory



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Summary

- This session relates to the discussion over Reasoning with uncertain information, which involves Review of Probability Theory, and Introduction to Bayesian Theory. Unit also covers the concept of Baye's Networks, which is later used for the purpose of inferencing.
- Finally, the session discussed about the Other Paradigm of Uncertain Reasoning, including the Dempster Scheffer Theory

*Thank
You*



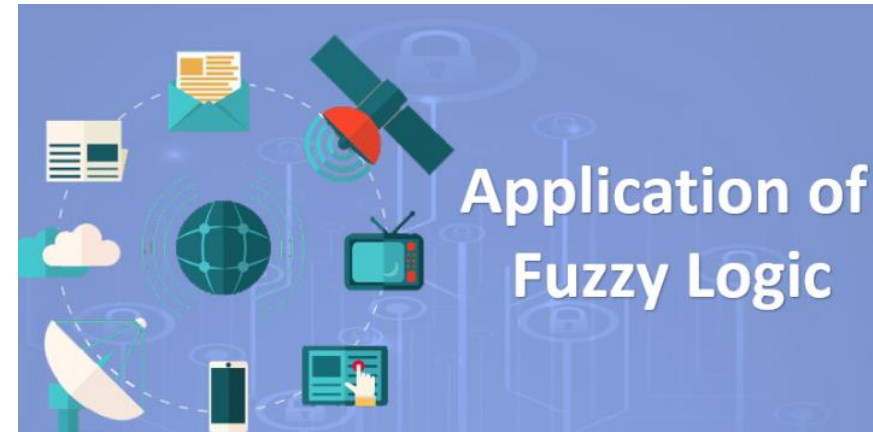
Artificial Intelligence and Machine Learning

Block-2 Unit-4

Fuzzy and Rough Sets

Topics to be Covered

- Introduction
- Fuzzy Systems
- Introduction to Fuzzy Sets
- Fuzzy Set Representation
- Fuzzy Reasoning
- Fuzzy Inference
- Rough Set Theory
- Important Topics
- Summary



<https://www.educba.com/applications-of-fuzzy-logic/>

Introduction

- In this session, we discuss a number of techniques and formal systems that attempt to handle some of these blemishes. To begin with we discuss the fuzzy systems that attempt to handle imprecision in knowledge bases, specially, due to use of natural language words like hot, good, tall etc.
- We will discuss two formal systems that attempt to handle incompleteness of the available information. These systems are called Default Reasoning Systems and Closed World Assumption Systems.

Fuzzy Systems

- In the symbolic Logic systems like, PL and FOPL, that we have studied so far, any (closed) formula has a truth-value which must be binary, viz., True or False.
- However, in our everyday experience, we encounter problems, the descriptions of which involve some words, because of which, to statements of situations, it is not possible to assign a truth value: True or False. For example, consider the statement:
- If the water is too hot, add normal water to make it comfortable for taking a bath. In the above statement, for a number of words/phrases including ‘too hot’ ‘add’, ‘comfortable’ etc., it is not possible to tell when exactly water is too hot, when water is (at) normal (temperature), when exactly water is comfortable for taking a bath.

Fuzzy Systems

- For example, we cannot tell the temperature T such that for water at temperature T or less, truth value False can be associated with the statement 'Water is too hot' and at the same time truth-value True can also be associated to the same statement 'Water is too hot' when the temperature of the water is, say, at degree $T + 1$, $T + 2$etc.
- **Some other cases of Fuzziness in a Natural Language**
- Healthy Person: we cannot even enumerate all the parameters that determine health. Further, it is even more difficult to tell for what value of a particular parameter, one is healthy or otherwise.
- Old/young person: It is not possible to tell exactly upto exactly what age, one is
- young and, by just addition of one day to the age, one becomes old. We age gradually. Aging is a continuous process.

Fuzzy Systems

- **Achieving Human Capability** : In order that computers achieve human capability in solving such problems, computers must be able to solve problems for which only incomplete and/or imprecise information/knowledge is available.
- **Modelling of Solutions and Data/Information/Knowledge** We know that for any problem, the plan of the proposed solution and the relevant information is fed in the computer in a form acceptable to the computer. However, the problems to be solved with the help of computers are, in the first place, felt by the human beings. And then, the plan of the solution is also prepared by human beings.

Introduction to Fuzzy Sets

- In the case of Crisp sets, we have the concepts of Equality of sets, Subset of a set, and Member of a set, as illustrated by the following examples:

- **Equality of two sets** : Set A is equal to set B denoted by $A = B$. But A is not equal to C, denoted by $A \neq C$.

$$\text{Let } A = \{1, 4, 3, 5\}$$

$$B = \{4, 1, 3, 5\}$$

$$C = \{1, 4, 2, 5\}$$

- **Subset** : Consider sets, Then B is a subset of A, denoted by $B \subset A$. Also C is not a subset of A, denoted by $C \not\subset A$.

$$A = \{1, 2, 3, 4, 5, 6, 7\}$$

$$B = \{4, 1, 3, 5\}$$

$$C = \{4, 8\}$$

- **Belongs to/is a member of** : If $A = \{1, 4, 3, 5\}$
- Then each of 1, 4, 3 and 5 is called an element or member of A and the fact that 1 is a member of A is denoted by $1 \in A$.

Introduction to Fuzzy Sets

- **Corresponding Definitions/ concepts for Fuzzy Sets** : In order to define for fuzzy sets, the concepts corresponding to the concepts of Equality of Sets, Subset and Membership of a Set considered so far only for crisp sets, first we illustrate the concepts through an example:
- Let X be the set on which fuzzy sets are to be defined, e.g.,
- $X = \{\text{Mohan, Sohan, John, Abdul, Abrahm}\}$.
- Then X is called the Universal Set.

Introduction to Fuzzy Sets

- **Degree of Membership:** In respect of fuzzy sets, we do not speak of just ‘membership’, but speak of ‘degree of membership’. In the set $A = \{\text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7; \text{Abrahm}/.4\}$,
Degree (Mohan) = .2, degree (John) = .4
- **Equality of Fuzzy sets:** Let A, B and C be fuzzy sets defined on X as follows:
Let $A = \{\text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7; \text{Abrahm}/.4\}$
 $B = \{\text{Abrahm}/.4, \text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7\}$.
Then, as degrees of each element in the two sets, equal; we say fuzzy set A equals fuzzy set B, denoted as $A = B$
- However, if $C = \{\text{Abrahm}/.2, \text{Mohan}/.4; \text{Sohan}/1; \text{John}/.7\}$, then $A \neq C$.

Fuzzy Set Representation

- For Crisp sets, we have the operations of Union, intersection & complementation, as illustrated by the example:

$$\text{Let } X = \{x_1, x_2, \dots, x_{10}\}$$

$$A = \{x_2, x_3, x_4, x_5\}$$

$$B = \{x_1, x_3, x_5, x_7, x_9\}$$

$$\text{Then } A \cup B = \{x_1, x_2, x_3, x_4, x_5, x_7, x_9\}$$

$$A \cap B = \{x_3, x_5\}$$

$$A' \text{ or } X \sim A = \{x_1, x_6, x_7, x_8, x_9, x_{10}\}$$

Fuzzy Set Representation

- The concepts of Union, intersection and complementation for crisp sets may be extended to FUZZY sets after observing that for crisp sets A and B , we have
 - (i) $A \cup B$ is the smallest subset of X containing both A and B .
 - (ii) $A \cap B$ is the largest subset of X contained in both A and B .
 - (iii) The complement A' is such that
 - (a) A and A' do not have any element in common and
 - (b) Every element of the universal set is in either A or A' .

Fuzzy Set Representation

- **Associativity**
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - (ii) $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributivity**
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **DeMorgan's Laws**
 - $(A \cup B)' = A' \cap B'$
 - $(A \cap B)' = A' \cup B'$

Fuzzy Reasoning

- The Fuzzy Reasoning is taken care by the following systems in general:
- Non Monotonic reasoning Systems
- Default Reasoning Systems
- Closed World Assumption Systems
- **NON-MONOTONIC REASONING SYSTEMS**
- Monotonic Reasoning: The conclusion drawn in PL and FOPL are only through (valid) deductive methods. When some axiom is added to a PL or an FOPL system, then, through deduction, we can draw more conclusions.

Fuzzy Reasoning

- However, in everyday life, many times in the light of new facts that become available, we may have to revise our earlier knowledge. For example, we consider a sort of deductive argument in FOPL:
 - (i) Every bird can fly long distances
 - (ii) Every pigeon is a bird. (iii) Tweety is a pigeon.
 - Therefore, Tweety can fly long distances
- However, later on, we come to know that Tweety is actually a hen and a hen cannot fly long distances. Therefore, we have to revise our belief that Tweety can fly over long distances.
- This type of situation is not handled by any monotonic reasoning system including PL and FOPL

Fuzzy Reasoning

- A non-monotomic reasoning system is one which allows retracting of old knowledge due to discovery of new facts which contradict or invalidate a part of the current knowledge base. Such systems also take care that retracting of a fact may necessitate a chain of retractions from the knowledge base or even reintroduction of earlier retracted ones from K.B(Knowledge base).
- **Default Reasoning** : One method of handling uncertainty due to incomplete KB is through default reasoning which is also a form of non-monotomic reasoning and is based on the following mechanism:
- Whenever, for any entity relevant to the application, information is not in the KB, then a default value for that type of entity, is assumed and is assigned to the entity. The default assignment is not arbitrary but is based on experiments, observations or some other rational grounds.

Fuzzy Reasoning

- The advantage of this type of a reasoning system is that we need not store all facts regarding a situation. Reiter has given one theory of default reasoning, which is expressed as

$$\frac{a(x) : M b_1(x), \dots, M b_k(x)}{C(x)}$$

- where M is a consistency operator. The inference rule (A) states that if a(x) is true and none of the conditions $b_k(x)$ is in conflict or contradiction with the K.B, then you can deduce the statement C(x) The idea of default reasoning is explained through the following example: Suppose we have

$$\frac{\text{Bird}(x) : M \text{fly}(x)}{\text{Fly}(x)}$$

Bird (twitty)

Fuzzy Reasoning

- $M \text{ fly } (x)$ stands for a statement of the form 'KB does not have any statement of the form that says x does not have wings etc, because of which x may not be able to fly'.
- In other words, $Bird(x) : M \text{ fly } (x)$ may be taken to stand for the statement 'if x is a normal bird and if the normality of x is not contradicted by other facts and rules in the KB.' then we can assume that x can fly. Combining with $Bird(Twitty)$, we conclude that if KB does not have any facts and rules from which, it can be inferred that Twitty can not fly, then, we can conclude that twitty can fly.
- Further, suppose, KB also contains
- (i) $Ostrich(twitty)$
- (ii) $Ostrich(x) \rightarrow \sim FLY(x)$.

Fuzzy Reasoning

- **CLOSED WORLD ASSUMPTION** : Another mechanism of handling incompleteness of a KB is called 'Closed World Assumption' (CWA). This mechanism is useful in applications where most of the facts are known and therefore it is reasonable to assume that if a proposition cannot be proved, then it is FALSE. This is called CWA with failure as negation. This means if a ground atom $P(a)$ is not provable, then assume $\sim P(a)$. A predicate like LESS (x, y) becomes a ground atom when the variables x and y are replaced by constants say x by 2 and y by 3, so that we get the ground atom LESS (2, 3).
- Example of an application where CWA is reasonable is that of Airline reservation where city-to-city flight not explicitly entered in the flight schedule or time table, are assumed not to exist.

Fuzzy Inference

- PL and FOPL are deductive inferencing systems: i.e., the conclusions drawn are invariably true whenever the premises are true. However, due to limitations of these systems for making inferences, as discussed earlier, we must have other systems inferences. In addition to Default Reasoning systems and Closed World Assumption systems, we have the following useful reasoning systems:
- 1) **Abductive inference System**, which is based on the use of causal knowledge to explain and justify a (possibly invalid) conclusion.
- Abduction Rule
$$(P \rightarrow Q, Q) / P$$
- Note that abductive inference rule is different from Modus Ponens inference rule in that in abductive inference rule, the consequent of $P \rightarrow Q$, i.e., Q is assumed to be given as True and the antecedent of $P \rightarrow Q$, i.e., P is inferred

Fuzzy Inference

- The abductive inference is useful in diagnostic applications. For example while diagnosing a disease (say P), the doctor asks for the symptoms (say Q). Also, the doctor knows that for given the disease, say, Malaria (P); the symptoms include high fever starting with feeling of cold etc. (Q)
- i.e., doctor knows $P \rightarrow Q$
- The doctor then attempts to diagnose the disease (i.e., P) from symptoms. However, it should be noted that the conclusion of the disease from the symptoms may not always be correct. In general, abductive reasoning leads to correct conclusions, but the conclusions may be incorrect also. In other words, Abductive reasoning is not a valid form of reasoning.

Rough Set Theory

- Rough set theory can be regarded as a new mathematical tool for imperfect data analysis. The theory has found applications in many domains, such as decision support, engineering, environment, banking, medicine and others. It is a mechanism to deal with imprecise/imprecise knowledge, dealing with such a kind of knowledge is particularly area of research for the scientists, working in the field of Artificial Intelligence.
- Rough set theory is quite comprehensive because of the following reasons :
- It requires no preliminary/additional information about the data as if it is the requirement of probability in statistics, or membership grades in the fuzzy set theory.
- Facilitates the user with efficient tools and techniques to detect the hidden patterns

Rough Set Theory

- Promotes data reductionality i.e. it reduces the original data and, find minimal
- datasets from the data with the similar knowledge as it is in the original dataset.
- Helps to evaluate the data significance.
- Supports the mechanism to Sets the decision rules from the data, automatically
- It is easy to understand, best suited for concurrent or parallel or distributed processing , and offers straightforward interpretation of obtained results.

Important Topics

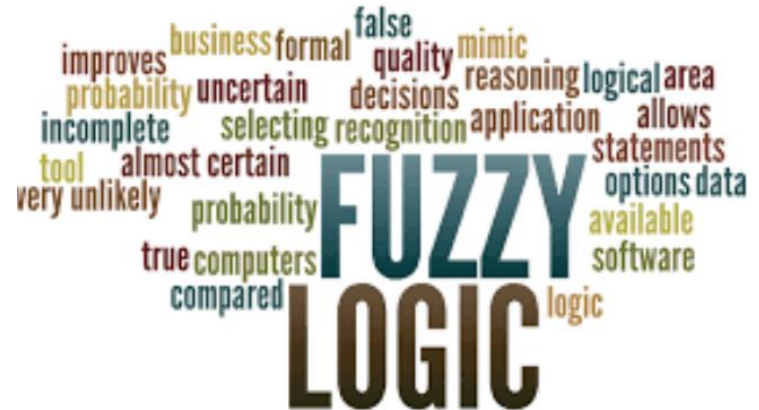
- Fuzzy Systems
- Fuzzy Sets
- Fuzzy Set Representation
- Fuzzy Reasoning
- Fuzzy Inference
- Rough Set Theory



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Summary

- In this session the Fuzzy Systems are discussed along with the Introduction to Fuzzy Sets and their Representation. Later the conceptual understanding of Fuzzy Reasoning is build, and the same is used to perform the Fuzzy Inference. The session finally discussed the concept of Rough Set Theory, also.



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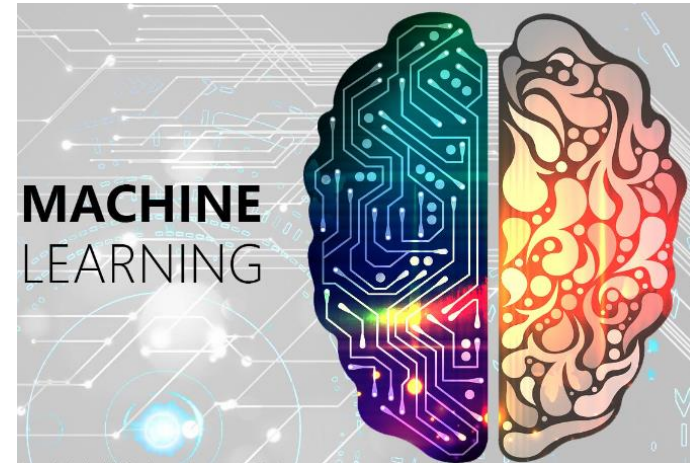
Artificial Intelligence and Machine Learning

Block-3 Unit-1

Introduction to Machine Learning Methods

Topics to be Covered

- Introduction
- Introduction to Machine Learning
- Techniques of Machine Learning
- Reinforcement Learning and Algorithms
- Deep Learning and Algorithms
- Ensemble Methods
- Important Topics
- Summary



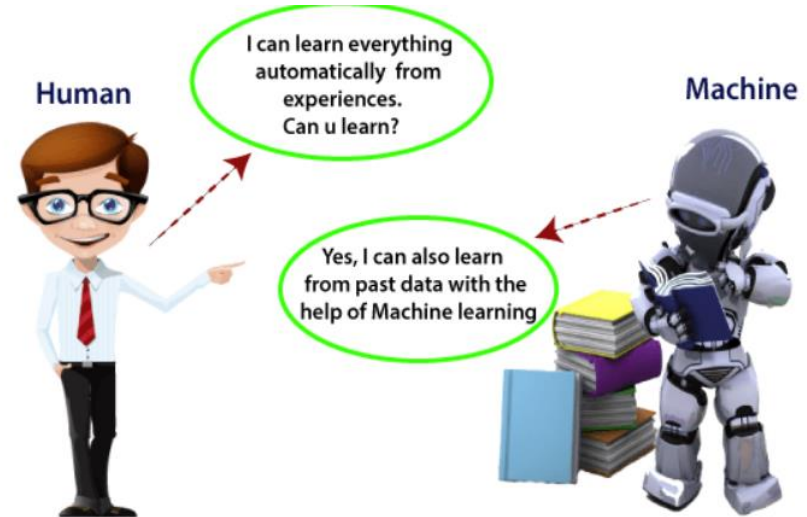
<https://towardsdatascience.com/introduction-to-machine-learning-for-beginners-eed6024fdb08>

Introduction

- After Artificial Intelligence was introduced, in Computing World. There was a need for a machine that would automatically make things better. This needs to be kept in check, so there should be some rules that apply to all learning processes.
- The main goal of machine learning, even at its most basic level, is to be able to analyze and adapt data on its own and make decisions based on calculations and analyses. Machine learning is a way to try to improve computers by imitating how the human brain learns.

Introduction to Machine Learning

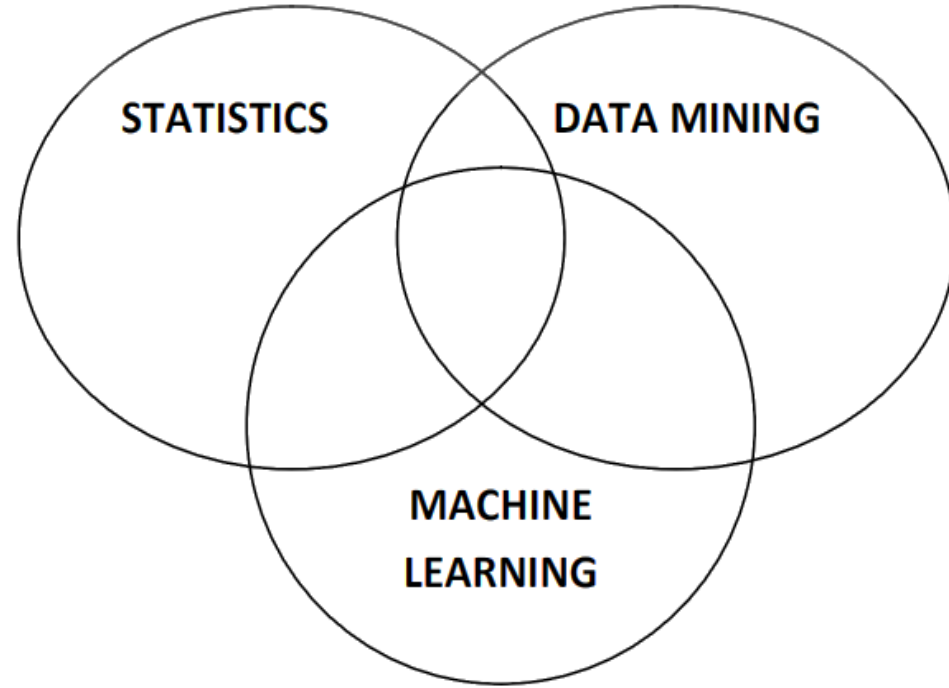
- Machine learning is a growing technology which enables computers to learn automatically from past data. Machine learning uses various algorithms for building mathematical models and making predictions using historical data or information. Currently, it is being used for various tasks such as image recognition, speech recognition, email filtering, Facebook auto-tagging, recommender system, and many more.



<https://www.javatpoint.com/machine-learning>

Introduction to Machine Learning

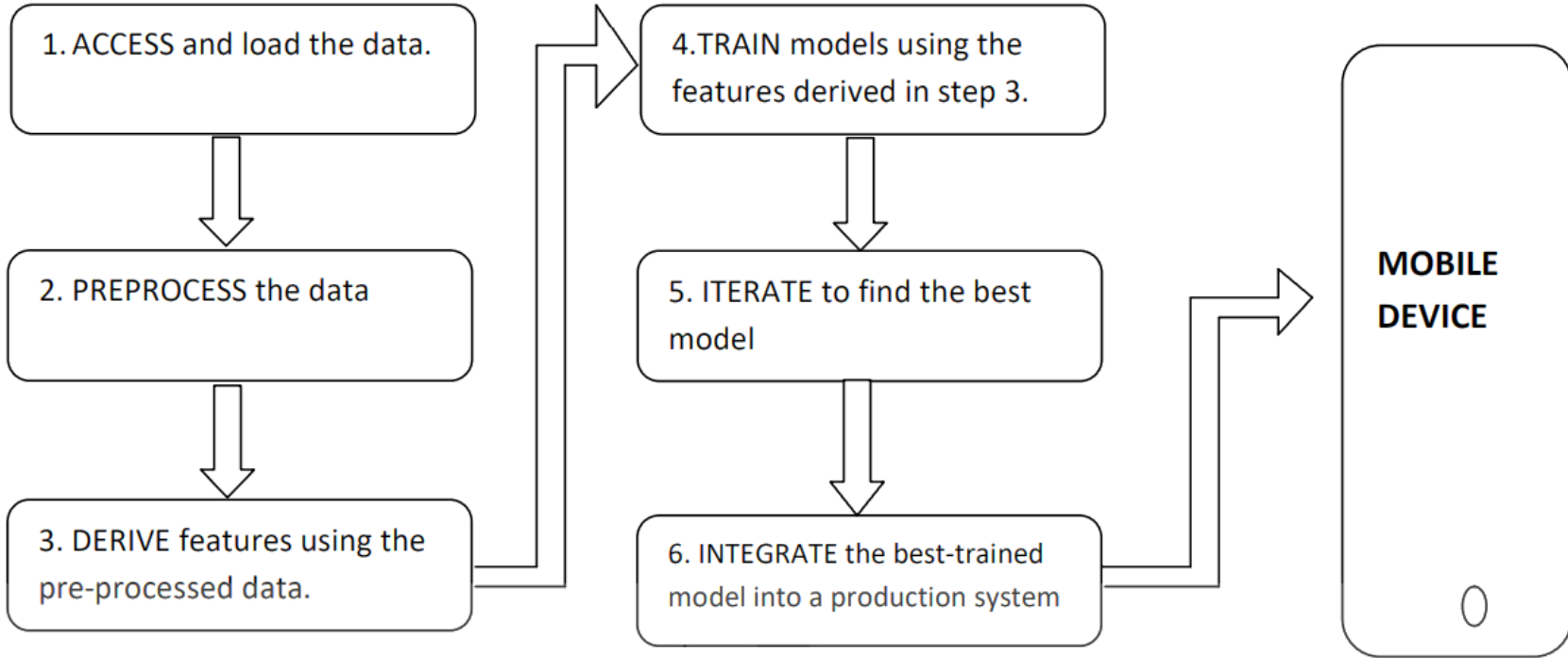
- Machine learning is the core of data mining's technical infrastructure. It is used to extract information from the raw data that is stored in databases; this information is then expressed in a form that is understandable and can be applied to a range of situations.



Techniques of Machine Learning

- Machine learning uses various algorithms to improve, describe, and predict outcomes by repeated learning from data. It is possible to make models that are more accurate as the algorithms learn from the training data. A machine learning model is what you get when you use data to train your machine learning algorithm.
- Machine learning uses various algorithms to improve, describe, and predict outcomes by repeated learning from data. It is possible to make models that are more accurate as the algorithms learn from the training data. A machine learning model is what you get when you use data to train your machine learning algorithm.

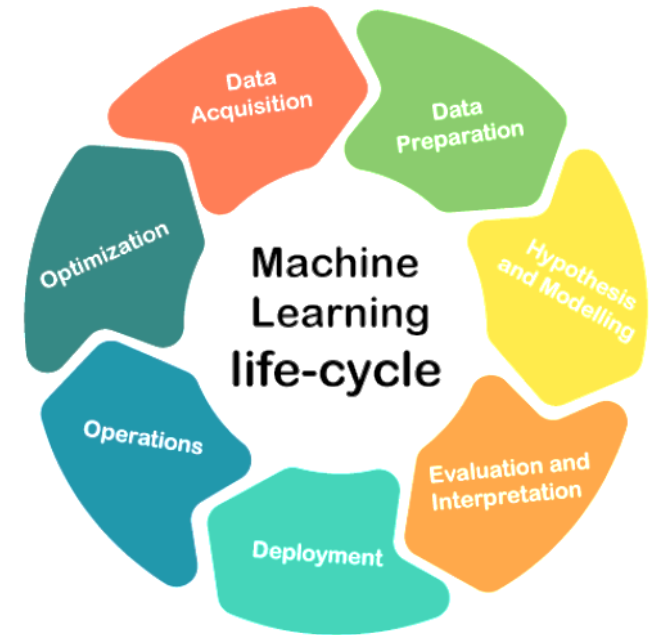
Techniques of Machine Learning



Machine Learning Cycle at a Glance

Techniques of Machine Learning

- The steps in the machine learning cycle are as follows:
- Data Identification
- Data Preparation
- Selection of machine learning algorithm:
- Training the algorithm to develop a model
- Evaluating the model
- Deploying the model
- Performing Prediction
- Assess the predictions



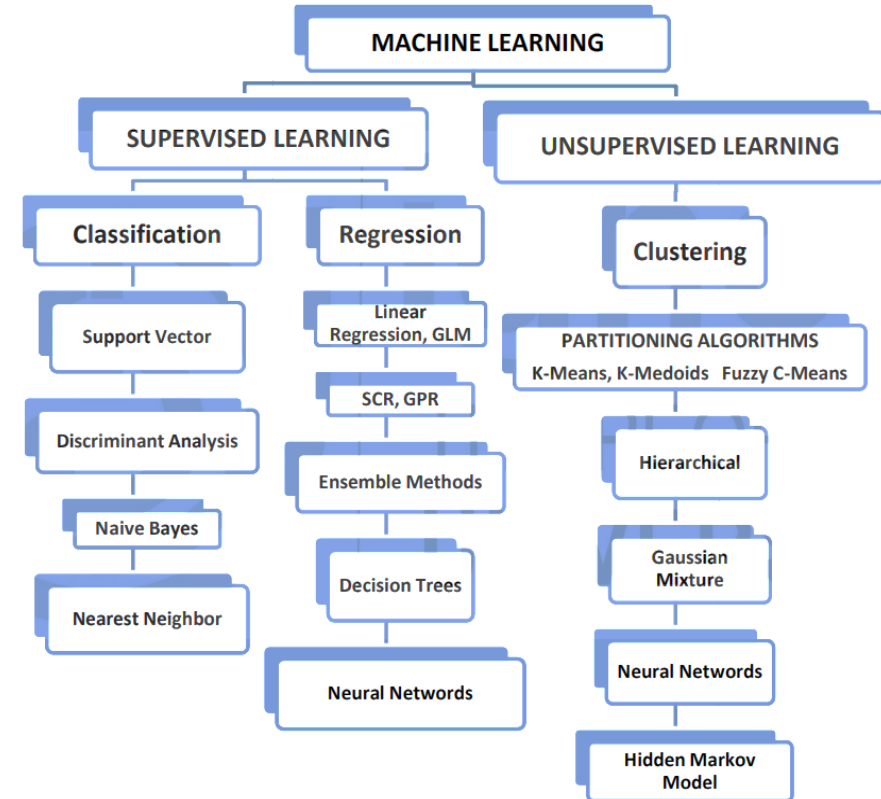
<https://www.tutorialandexample.com/machine-learning-life-cycle>

Techniques of Machine Learning

- Every machine learning workflow begins with three questions:
- What kind of information do you have to work with?
- How do you want to learn something from it?
- How and where will these new ideas come from?
- Fundamentally Machine learning involves two classes of Learning algorithms:
- **Supervised learning**, which requires training a model with data whose inputs and outputs are already known in order for the model to be able to predict future outputs, such as whether or not an email is authentic or spam or whether or not a tumor is cancerous. Classification models classify given data into categories. Imaging for medical purposes, speech recognition, and rating credit are a few examples of typical applications.

Techniques of Machine Learning

- **Unsupervised learning** analyses data to uncover previously unknown used to infer conclusions from sets of data that contain inputs but no tagged answers. The most prevalent method of learning without being observed is clustering.
- Exploratory data analysis is used to uncover hidden patterns analysis, market research, and object recognition.

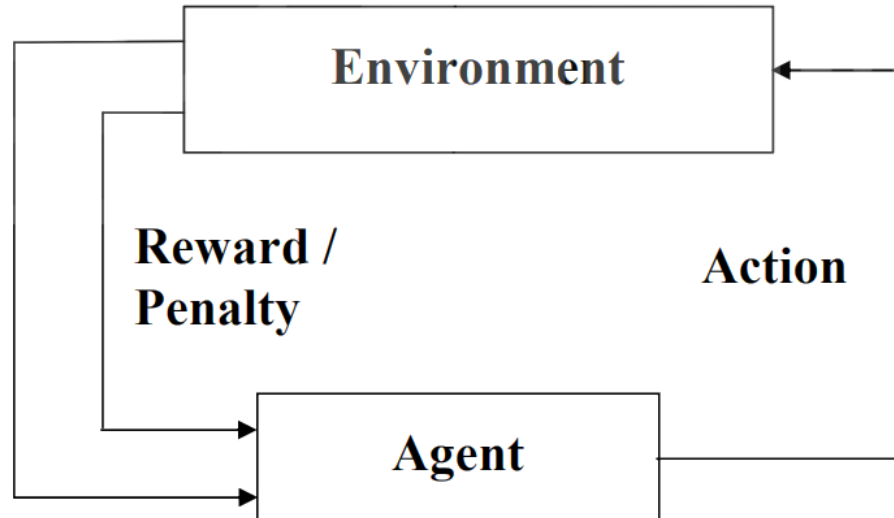


Reinforcement Learning and Algorithms

- In Reinforcement Learning (RL), algorithms get a set of instructions and rules and then figure out how to handle a task by trying things out and seeing what works and what doesn't. As a way to help the AI find the best way to solve a problem, decisions are either rewarded or punished.
- Reinforcement Learning (RL) is a type of Machine Learning in which the agent gets a delayed reward in the next time step to evaluate how well it did in the previous time step. It was mostly used in games, like Atari and Mario, where it could do as well as or better than a person. Since Neural Networks have been added to the algorithm, it has been able to do more complicated tasks.

Reinforcement Learning and Algorithms

- Reinforcement Learning (RL) refers to a kind of Machine Learning method in which the agent receives a delayed reward in the next time step to evaluate its previous action. It was mostly used in games. Typically, a RL setup is composed of two components, an agent and an environment.



Reinforcement Learning and Algorithms

- The following are the meanings of the different parts of reinforcement learning:
- AGENT : The agent is the person who learns and makes decisions.
- ENVIRONMENT : The agent's environment is where it learns and decides what to do.
- ACTION : A group of things that the agent can do.
- STATE : How the agent is doing in its environment.
- REWARD : The environment gives the agent a reward for each action they choose. Usually a scalar value.
- POLICY : Policy is the agent's way of deciding what to do (its control strategy), which is a mapping from situations to actions.

Reinforcement Learning and Algorithms

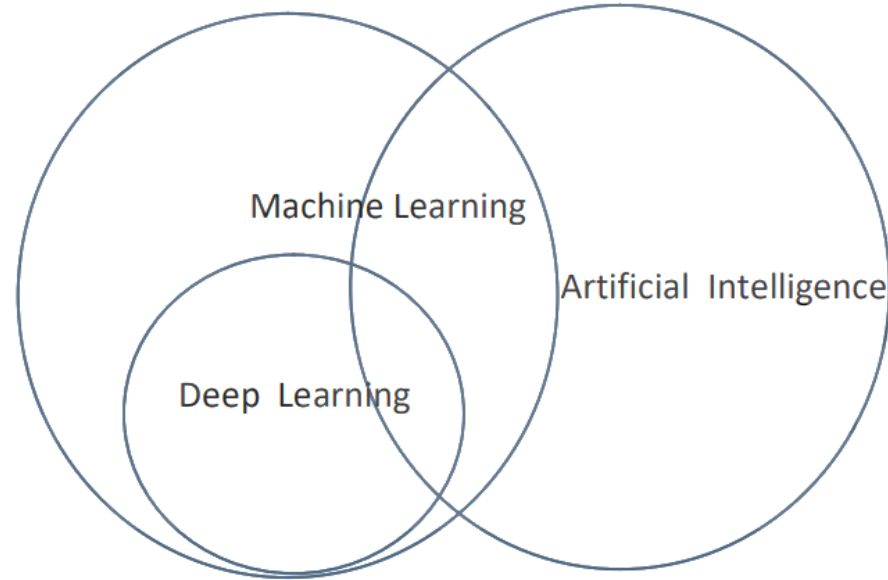
- **VALUE FUNCTION** : A way to map states to real numbers, where the value of a state is the long reward that can be earned by starting in that state and following a certain policy.
- **FUNCTION APPROXIMATOR** : is a term for the problem of figuring out what a function is by looking at training examples. Decision trees, neural networks, and nearest examples of standard approximators.
- **MODEL** : The agent's view of the environment, which maps state-action pairs to probability distributions over states. Note that not every agent that learns from its environment uses a model of its environment.

Deep Learning and Algorithms

- Deep learning is a type of machine learning that uses artificial neural networks and representation learning. It is also called deep structured learning or differential programming.
- Deep learning is a way for machines to learn through deep neural networks. It is used a lot to solve practical problems in fields like computer vision (image), natural language processing (text), and automated speech recognition (audio).
- Machine learning is often thought of as a tool with several algorithms. However, deep learning is actually just a subset of approaches that mostly use neural networks, which are a type of algorithm loosely based on the human brain.
- A deep learning model learns to solve classification tasks directly from images, text, or sound.

Deep Learning and Algorithms

- As seen in the diagram, machine learning (ML), deep learning (DL), and artificial intelligence (AI) are all related.
- Deep Learning is a collection of algorithms inspired by the human brain's workings in processing the data and creating patterns for use in decision making, which are expanding and improving or refining the idea of a single model architecture termed Artificial Neural Network (ANN).



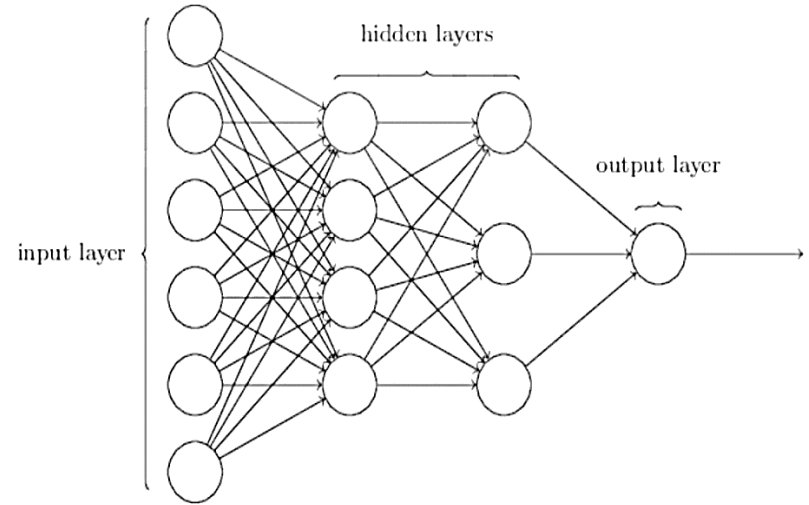
Relation Between Machine learning ,
Deep Learning and Artificial Intelligence

Deep Learning and Algorithms

- **Neural Networks:** Just like the human brain, Neural Networks consist of Neurons. Each Neuron takes in signals as input, multiplies them by weights, adds them together, and then applies a nonlinear function. These neurons are arranged in layers and stacked close to each other.
- **Back Propagation Neural Networks :** Propagation in reverse Back propagation is an iterative process that allows neural networks to learn the desired function by utilising large quantities of data and learning from their previous mistakes. We feed the network data, and in return, it provides us with an output. We start by comparing the output to what we want using a loss function.

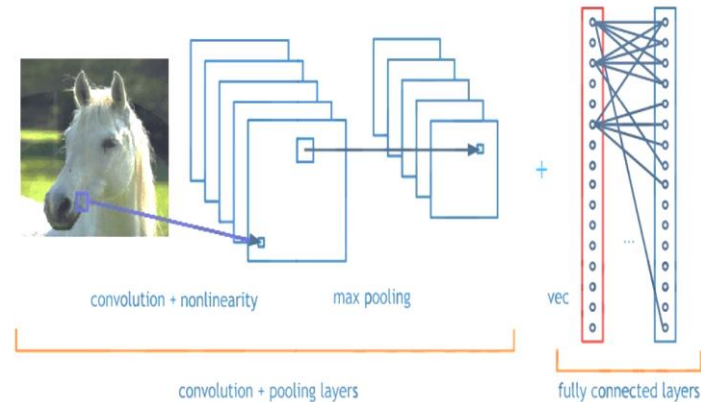
Deep Learning and Algorithms

- **Feed-forward neural networks (FNN) :**
Typically, feed-forward neural networks, also known as FNN, are completely connected, this implies that each neuron in one layer is connected to each neuron in the layer next to it. A "Multilayer Perceptron" is the name given to the structure shown below and that is the topic of discussion here. A multilayer perceptron, has the ability to learn associations between the data that are not linear, in contrast to a single-layer perceptron, which can only learn patterns that can be separated in a linear manner.



Deep Learning and Algorithms

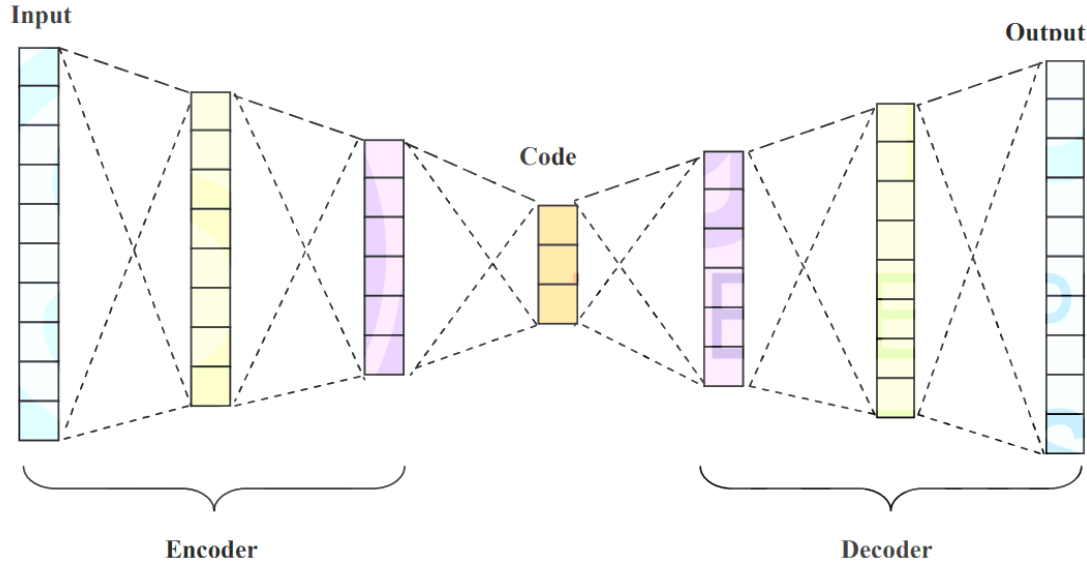
- **Convolutional Neural Networks (CNN)** The term "convolution" refers to the function that is utilized by convolutional neural networks (CNN). The idea that underlies them is that rather than linking each neuron with all of the ones that come after it, we just connect it with a select few of those that come after it (the receptive field). They strive to regularize feed-forward networks in order to avoid overfitting, which is when the model is unable to generalize its findings since it can only learn from the data it has already seen.



Deep Learning and Algorithms

- **Recurrent Neural Networks (RNN)** are utilised in time series forecasting because they are ideal for time-related data. They employ some type of feedback, in which the output is fed back into the input. You can think of it as a loop that passes data back to the network from the output to the input. As a result, they are able to recall previous data and use it to make predictions.
- **Recursive Neural Networks** : Another type of recurrent network is the recursive neural network, which is set up in a tree-like manner. As a result, they can simulate the hierarchical structures training dataset's.
- **Restricted Boltzmann Machines (RBM)** are stochastic neural networks that can learn a probability distribution over their inputs and so have generative capabilities. They differ from other networks in that they only have input and hidden layers (no outputs).

Deep Learning and Algorithms



- An encoder and a decoder are included in Auto-Encoders. The encoder receives the input and encodes it in a lower-dimensional latent space. Whereas, the decoder is used to decode that vector back to the original input.

Deep Learning and Algorithms

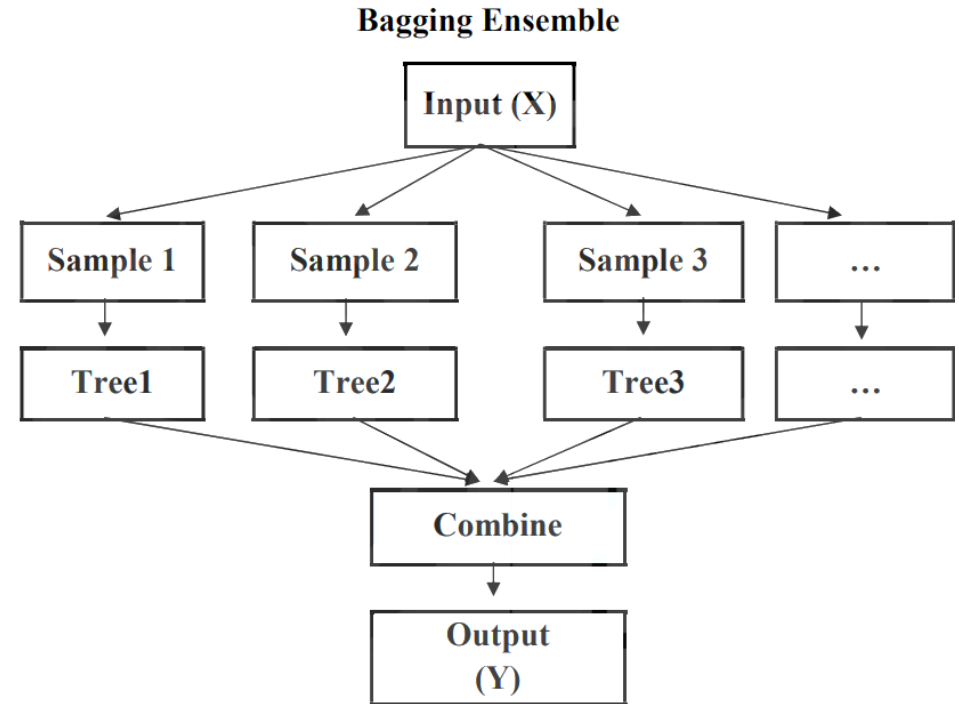
- **Transformers** are also very new, and they are mostly employed in language applications because recurrent networks are becoming obsolete. They are based on the concept of "attention," which instructs the network to focus on a certain data piece.
- **Graph Neural Networks:** Deep Learning does not operate well with unstructured data in general. And there are many circumstances in which unstructured data is organised as a graph in the actual world. Consider social networks, chemical molecules, knowledge graphs, and location information.

Ensemble Methods

- Ensemble learning is a general meta approach to machine learning that combines predictions from different models to improve predictive performance.
- The three primary classes of ensemble learning methods, and it's essential to understand each one thoroughly :
- **Bagging Ensemble learning** is the process of fitting multiple decision trees to various samples of the same dataset and averaging the results. Bagging Ensemble learning is the process of fitting multiple decision trees to various samples of the same dataset and averaging the results. Stacking Ensemble learning is fitting multiple types of models to the same data and then using another model to learn how to combine the predictions in the best way possible.
- Boosting Ensemble Learning entails successively adding ensemble members that correct prior model predictions and produce a weighted average of the predictions

Ensemble Methods

- It is a comprehensive strategy that is simple to expand upon. For instance, additional alterations can be made to the dataset that was used for training, the method that was used to fit the training data can be modified, and the manner in which predictions are constructed can be altered.

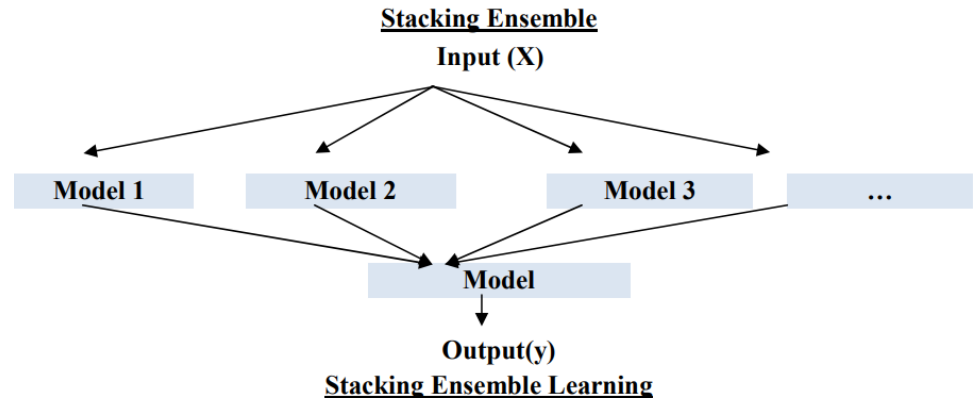


Ensemble Methods

- **Stacking Ensemble learning:** Stacked Generalization, sometimes known as "stacking" due to its abbreviated form, is an ensemble strategy that searches for a diverse group of members by varying the types of models that are fitted to the training data and utilising a model to aggregate predictions. It requires fitting of various kinds of models, applied to the same data, and then using another model to find out how to integrate the predictions in the best way possible. This process is known as model fitting. The key elements of stacking are summarized below:
- Unchanged training dataset.
- Separate machine learning algorithms for respective ensemble member.
- Machine learning model to learn how to combine predictions in the best way.

Ensemble Methods

- Many popular ensemble algorithms are based on this approach, including:
- Stacked Models (canonical stacking)
- Blending
- Super Ensemble



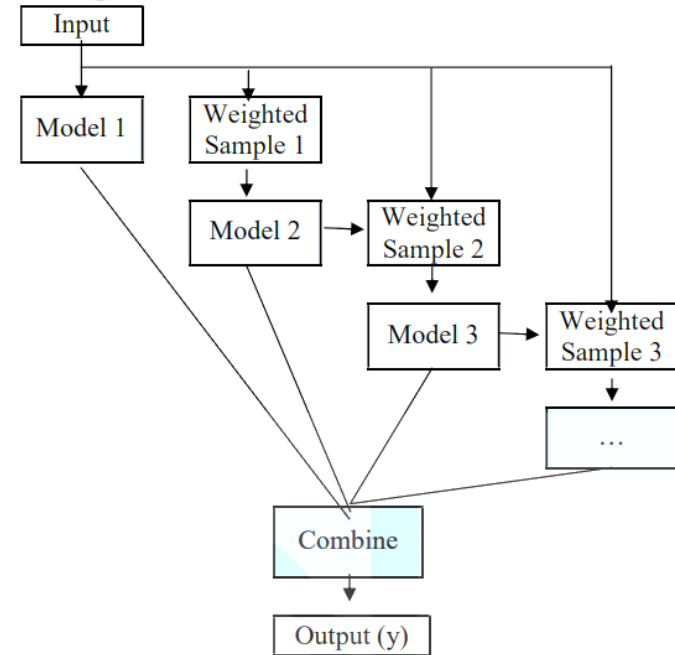
Ensemble Methods

- **Boosting Ensemble learning:** Boosting is an ensemble strategy that aims to alter the training data so that it focuses on examples that earlier models that fit the training data got wrong. Boosting tries to do this by focusing on examples that prior models got wrong. In order for it to function, members are added to the ensemble one at a time, and when each new member is added, the predictions that were produced by the model that came before it are refined. A weighted average of the forecasts is what we get as a result. The key elements of boosting are summarized below:
- Give more weight to examples that are hard to guess when training.
- Add members of the ensemble one at a time to correct the predictions of earlier models.
- Use a weighted average of models to combine their predictions.

Ensemble Methods

- To summarize, many popular ensemble algorithms are based on this approach, including:
- AdaBoost (canonical boosting)
- Gradient Boosting Machines
- Stochastic Gradient Boosting (XGBoost and similar)

Boosting Ensemble



Boosting Ensemble Learning

Important Topics

- Machine Learning
- Techniques of Machine Learning
- Reinforcement Learning
- Deep Learning
- Ensemble Methods



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Summary

- In this session we discussed about the basic concepts of machine learning and also about the various Machine
- learning algorithms. The session also covers the understanding of reinforcement learning and its related algorithms.
- There after we discussed the concept of Deep Learning and various techniques involved in Deep Learning.
- The session finally discussed about the Ensemble Learning and its related methods.

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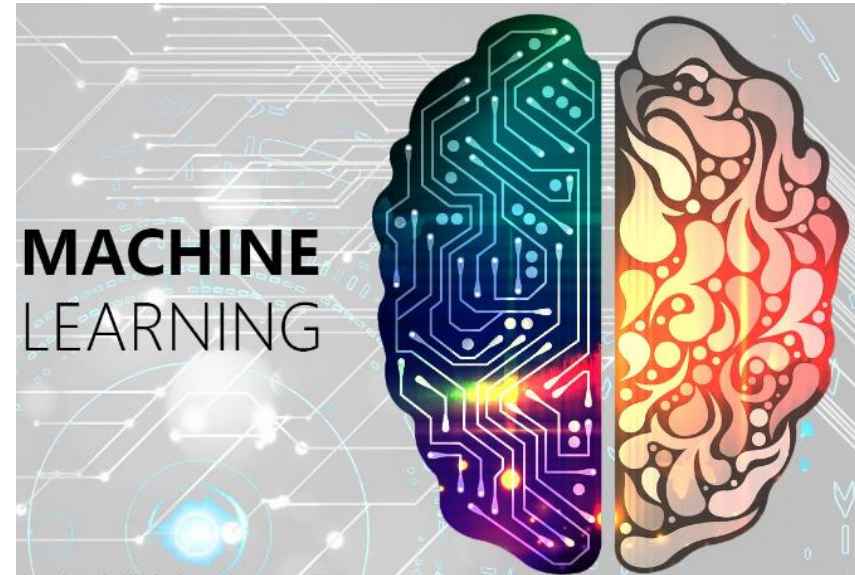
Artificial Intelligence and Machine Learning

Block-3 Unit-2

Classification

Topics to be Covered

- Introduction
- Understanding of Supervised Learning
- Introduction to Classification
- Classification Algorithms
- Important Topics
- Summary

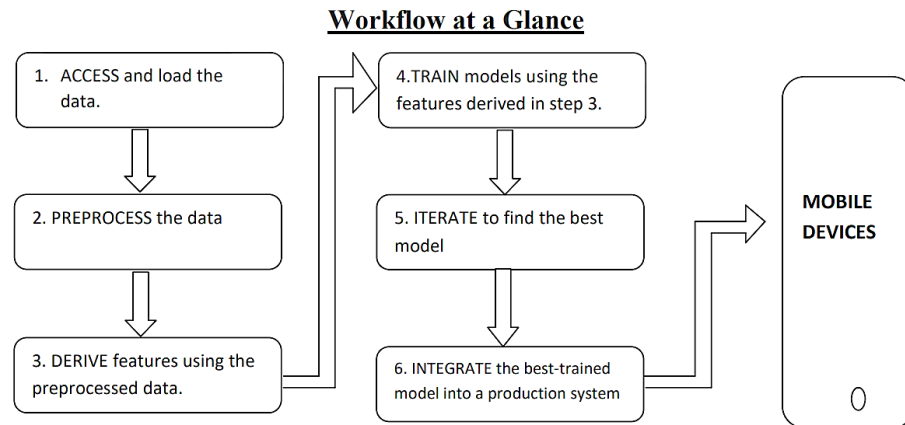


Introduction

- The purpose of the clustering process is to create distinct classes from groups of tuples that share characteristic values. Clustering is the process of defining a mapping, using as input a database containing tuples and an integer value k , in such a way that the tuples are mapped to various clusters.
- Classical Machine Learning Algorithms can be put into two groups: Supervised Learning Algorithms, which use data that has been labelled, and Un-Supervised Learning Algorithms, which use data that has not been labelled and are used for Clustering.
- In this session we will be discussing about the Supervised Learning Algorithms, which are mainly used for the classification purpose.

Understanding of Supervised Learning

- Every workflow for machine learning starts with the following three questions:
- What kind of data do you have available to work with ?
- What kinds of realisations are you hoping to arrive at as a result of it ?
- In what ways and contexts will those realizations be utilized ?

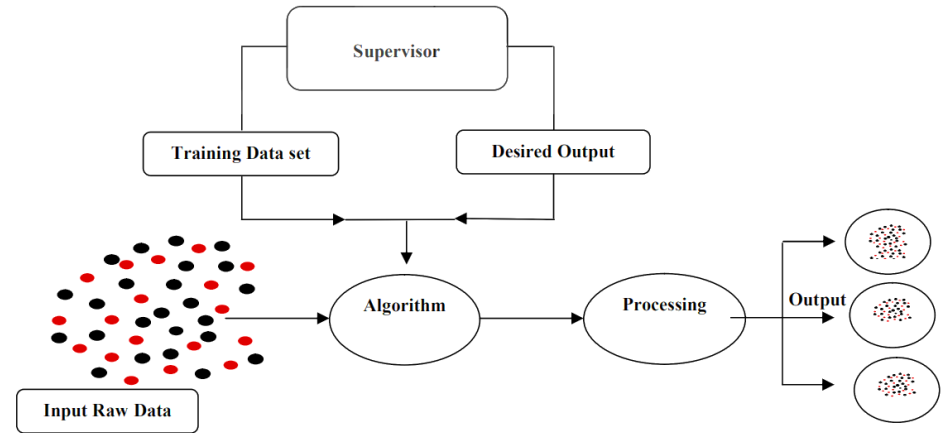


Understanding of Supervised Learning

- In an interesting way, supervised machine learning is like how humans and animals learn "concepts" or "categories." This is defined as "the search for and listing of attributes that can be used to tell exemplars from non-exemplars of different categories."
- The goal of supervised machine learning is to build a model that can make predictions based on evidence even when there is uncertainty. A supervised learning algorithm uses a known set of input data and known responses to the data (output) to train a model to make reasonable predictions about the response to new data.

Understanding of Supervised Learning

- Example: Predicting heart attacks with the help of supervised learning: Let's say doctors want to
- know if someone will have a heart attack in the next year. They have information about the age, weight, height, and blood pressure of past patients. They know if the patients who were there before had heart attacks within a year. So the problem is making a model out of the existing data that can tell if a new person will have a heart attack in the next year.

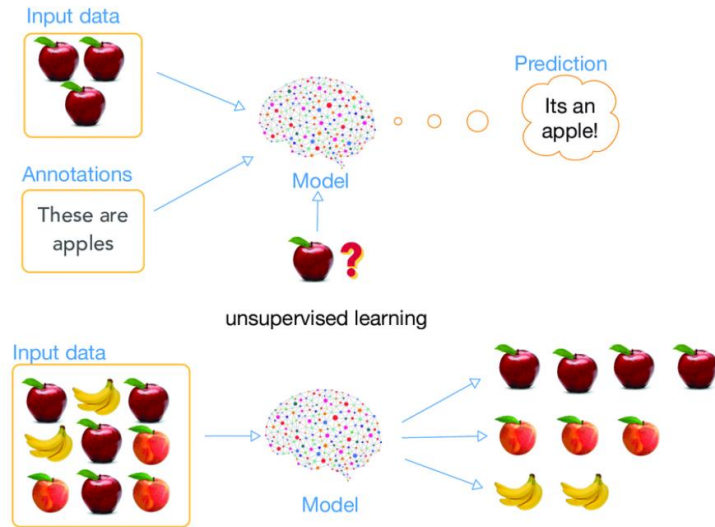


Understanding of Supervised Learning

- Following Steps are Involved in Supervised Learning, and they are self explanatory:
- 1. Determine the Type of Training Examples
- 2. Prepare/Gather the Training Data
- 3. Determine Relation Between Input Feature & Representing Learned Function
- 4. Select a Learning Algorithm
- 5. Run the Selected Algorithm on Training Data
- 6. Evaluate the Accuracy of the Learned Function Using Values from Test Set

Understanding of Supervised Learning

- A very large number of practical applications of the method can be outlined, but the following are some of the common ones
- Detection of spam
- Detection of fraudulent banking or other activities
- Medical Diagnosis
- Image recognition
- Predictive maintenance



Introduction to Classification

- **Classification techniques** : Classification methods make predictions about specific outcomes, such as whether an email is legitimate or spam or whether a tumour is malignant or benign. Classification models classify incoming data into categories. Applications such as medical imaging, speech recognition, and credit scoring are typical examples.
- **Regression techniques** : The techniques of regression can accurately forecast continuous reactions, such as shifts in temperature or variations in the amount of power required. Examples of typical applications are as follows: A few examples of applications are the predicting of stock prices, the recognition of handwriting, the forecasting of power load, and acoustic signal processing.

Introduction to Classification

- **Classification** The process of organizing data into a predetermined number of categories is referred to as classification. Finding out which category or class a new collection of data falls under is the primary objective of a classification problem, which can be stated as follows, Data that is structured as well as data that is not structured can be utilized for classification:
- **Structured data** (data that is in a fixed field in a file or record is called "structured data"). A relational database (RDBMS) is where most structured data is kept.
- **Unstructured data** (unstructured data may have a natural structure, but it isn't set up in a way that can be predicted). There is no data model, and the data is stored in the format in which it was created. Rich media, text, social media activity, surveillance images, and so on, are all types of unstructured data. Following are some of the terminologies frequently encountered in machine learning – classification:

Introduction to Classification

- **Classifier:** A classifier is an algorithm that puts the data you give it into a certain category. A classifier is an algorithm that does classification on a dataset.
- **Classification model:** A classification model looks at the output values to try to figure out what the values used for training mean. It will guess the class labels or categories of the new data.
- **Feature:** property of any object (real or virtual) that can be measured on its own is called a feature.
- **Classification predictive modeling** involves assigning a class label to input examples.

Classification Algorithms

- The Classification algorithm is a type of Supervised Learning that uses the training data to figure out the category of new observations. This method is used to figure out what kind of thing a new observation is. Classification is the process by which a computer programme learns from a set of data or observations and then sorts new observations into different classes or groups. "Yes" or "No," "0" or "1," "Spam" or "Not Spam," "Cat or Dog," and so on are all good examples. Classes are the same thing that have different names, like categories, objectives, and labels.
- In classification, the output variable is not a value but a category, such as "Green or Blue," "Fruit or Animal," etc. This is different from regression, where the output variable is a value. Since the classification method is a supervised learning method, it needs data that has been labelled in order to work

Classification Algorithms

- **Choose supervised learning** if you need to train a model to make a prediction, for instance, the future value of a continuous variable, such as TEMP. or a stock price; use regression techniques and use classification techniques in situations such as identifying makes of cars from webcam video footage or identifying spams from emails; etc.
- **Choose unsupervised learning** if you need to investigate your data and want to train a model to find a decent internal representation, such as by dividing the data into clusters. This type of learning allows for more freedom in exploring and representing the data.

Classification Algorithms

- **AUC-ROC curve:** The letters AUC and ROC stand for "area under the curve" and "receiver operating characteristics curve," respectively.
- This graph shows how well the classification model works at several different thresholds.
- We use the AUC-ROC Curve to see how well the multi-class classification model is doing.
- The (TPR) True Positive Rate and the (FPR) False Positive Rate are used to plot the ROC curve, with TPR on the Y-axis and FPR on the X-axis.

Classification Algorithms

- **NAÏVE BAYES**
- Naive Bayes classifiers are a collection of classification algorithms based on **Bayes' Theorem**. It is not a single algorithm but a family of algorithms where all of them share a common principle, i.e. every pair of features being classified is independent of each other.
- Bayes Theorem: In order to understand this theorem firstly lets understand the meaning of the following symbols or assumptions, they are as follows :
- X is an example of a data set whose class needs to be determined.
- H refers to the hypothesis which states that the data sample X falls into the class C.

Classification Algorithms

- $P(H | X)$: The probability that the data sample X belongs to the class C is expressed by the formula $P(H | X)$, where H is the hypothesis and X is the data sample. This formula represents the likelihood that the data sample X belongs to the class C . The likelihood that the condition H is true for the sample X is often referred to as the posterior probability.
- The prior probability of the H condition is denoted by the notation $P(H)$, which is based on the training data.
- The posterior probability of the X sample is denoted by the symbol $P(X | H)$, which assumes that H is correct.
- The prior probability on the sample X is denoted by the letter $P(X)$.

Classification Algorithms

- **Naiive Bayes** : Steps to perform naiive bayes algorithm
- Step 1: Handling Data : Data is loaded from the CSV File and spread into training and tested assets.
- Step 2: Summarizing the Data : Summarise the properties in the training data set to calculate the
- probabilities and make predictions.
- Step 3: Making a Prediction : A particular prediction is made using a summarise of the data set to make a single prediction.
- Step 4: Making all the Predictions : Generate prediction given a test data set and a summarise data set.

Classification Algorithms

- Step 5: Evaluate Accuracy : Accuracy of the prediction model for the test data set as a percentage correct out of them all the predictions made.
- Step 6: Tying all Together : Finally, we tie to all steps together and form our own model of Naive Bayes Classifier.

Classification Algorithms

- **K-NEAREST NEIGHBOURS (K-NN)** : This approach, places items in the class to which they are “closest” to their neighbour. It must determine distance between an item and a class. Classes are represented by centroid (Central value) and the individual points. One of the algorithms that is used is K-Nearest Neighbors.
Example of classification :
- Teachers classify students’ marks data into a set of grades as A, B, C, D, or F.
- Classification of the height of a set of persons into the classes tall, medium or short.
- The basic approaches to classification are: To create specific models by, evaluating training data, which is basically the old data, that has already been classified by using the domain of the experts’ knowledge.
- Now applying the model developed to the new data.

Classification Algorithms

- Classification Using Distance (K-Nearest Neighbours) - Some of the basic points to be noted about this algorithm are:
- The training set includes classes along with other attributes. (Please refer to the training data given in the Table given below).
- The value of the K defines the number of near items (items that have less distance to the attributes of concern) that should be used from the given set of training data (just to remind you again, training data is already classified data). This is explained in point (2) of the following example.
- A new item is placed in the class in which the most number of close items are placed. (Please refer to point (3) in the following example).
- The value of K should be $\leq \sqrt{\text{Number_of_training_items}}$

Classification Algorithms

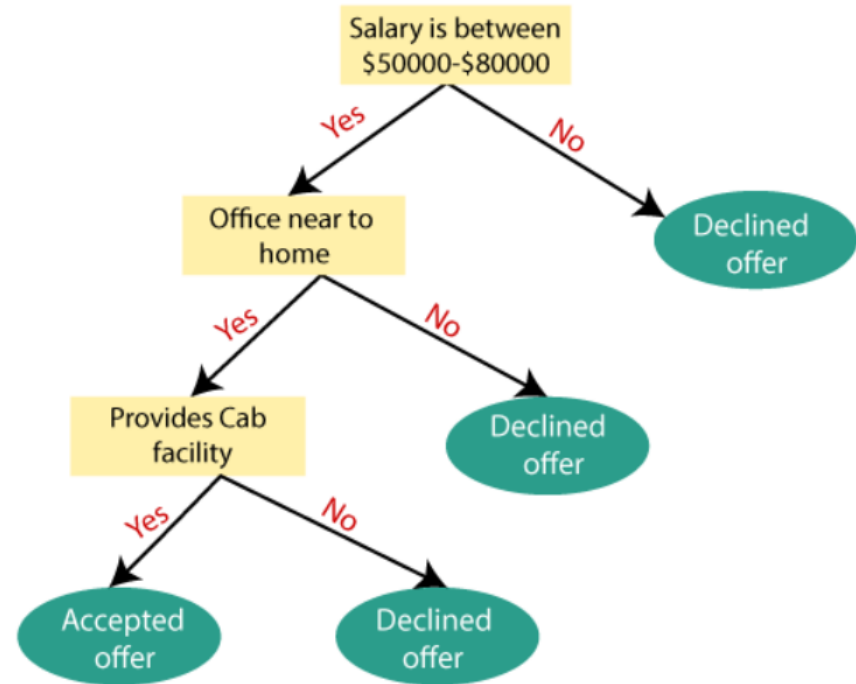
- **Decision Tree** Induction is the process of learning about the classification using the inductive approach. During this process, we create a decision tree from the training data. This decision tree can, then be used, for making classifications.
- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems. It is a tree-structured classifier, where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome.
- In a Decision tree, there are two nodes, which are the Decision Node and Leaf Node. Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.

Classification Algorithms

- Steps of algorithm of decision tree
 1. Data Pre-processing step
 2. Fitting a Decision-Tree algorithm to the Training set
 3. Predicting the test result
 4. Test accuracy of the result(Creation of Confusion matrix)
 5. Visualizing the test set result.
- **Example:** Suppose there is a candidate who has a job offer and wants to decide whether he should accept the offer or Not. So, to solve this problem, the decision tree starts with the root node (Salary attribute by ASM).

Classification Algorithms

- The root node splits further into the next decision node (distance from the office) and one leaf node based on the corresponding labels. The next decision node further gets split into one decision node (Cab facility) and one leaf node. Finally, the decision node splits into two leaf nodes (Accepted offers and Declined offer). Consider the below diagram:



Classification Algorithms

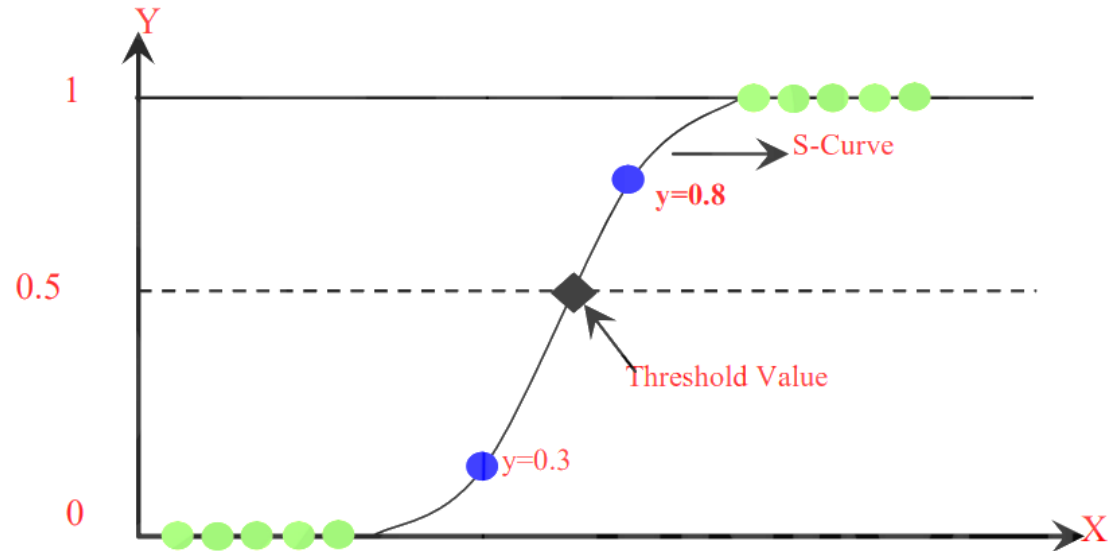
- Some advantages of Decision tree are :-
- It is simple to understand as it follows the same process which a human follow while making any decision in real-life.
- It can be very useful for solving decision-related problems.
- It helps to think about all the possible outcomes for a problem.
- There is less requirement of data cleaning compared to other algorithms.

Classification Algorithms

- **LOGISTIC REGRESSION** : Logistic regression is the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables.
- Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.
- Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas Logistic regression is used for solving the classification problems.

Classification Algorithms

- Logistic regression can be used to classify observations based on different types of data, and it is easy to figure out which variables are the most useful for classifying. The logistic function is shown in the picture below:



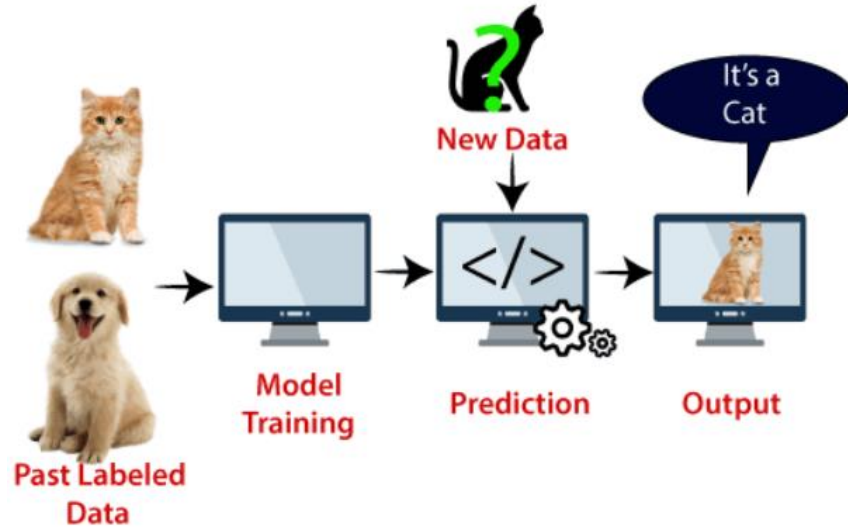
Classification Algorithms

- Types of Logistic Regression: Based on the categories, Logistic Regression can be divided into three types:
- **Binomial**: In binomial logistic regression, the dependent variables can only be either 0 or 1, Pass or Fail, etc.
- **Multinomial**: In multinomial Logistic regression, the dependent variable can be one of three or more types that are not in order, such as "cats," "dogs," or "sheep."
- **Ordinal**: In ordinal Logistic regression, the dependent variables can be ranked, such as "low," "medium," or "high."

Classification Algorithms

- **SUPPORT VECTOR MACHINES** : Support Vector Machine, also called Support Vector Classification, is a supervised and linear Machine Learning technique that is most often used to solve classification problems.
- The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future.
- **Example:** SVM can be understood with the example that we have used in the KNN classifier. Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature.

Classification Algorithms



- SVM algorithm can be used for Face detection, image classification, text categorization, etc.

Classification Algorithms

- There are two main categories that can be applied to SVM:
- **Linear Support Vector Machine:** You can only use Linear SVM if the data can be completely separated into linear categories. The ability to separate a set of data points into two classes using just one straight line is what is meant when we talk about something being "completely linearly separable" (if 2D).
- **Non-Linear Support Vector Machine:** When the data isn't linearly separable, we can use Non-Linear SVM, which means that we apply advanced techniques like kernel tricks to categorise the data points that can't be divided into two classes using a straight line. This allows us to use Non-Linear SVM to classify the data points that aren't linearly separable (in 2D). We don't discover datapoints that are linearly separable in the majority of real-world applications, so we use the kernel approach to solve them instead.

Important Topics

- Supervised Learning
- Naïve Bayes
- K-Nearest Neighbour (K-NN)
- Decision Trees
- Logistic Regression
- Support Vector Machines



<https://gyaanarth.com/web-stories/cbse-12th-chemistry-exam-paper-analysis/>

Summary

- In this session we discussed about the basic concepts of supervised learning.
- This session also covers the various classifications algorithms like:-
 - Naïve Bayes
 - K-Nearest Neighbour (K-NN)
 - Decision Trees
 - Logistic Regression
 - Support Vector Machines

*Thank
You*



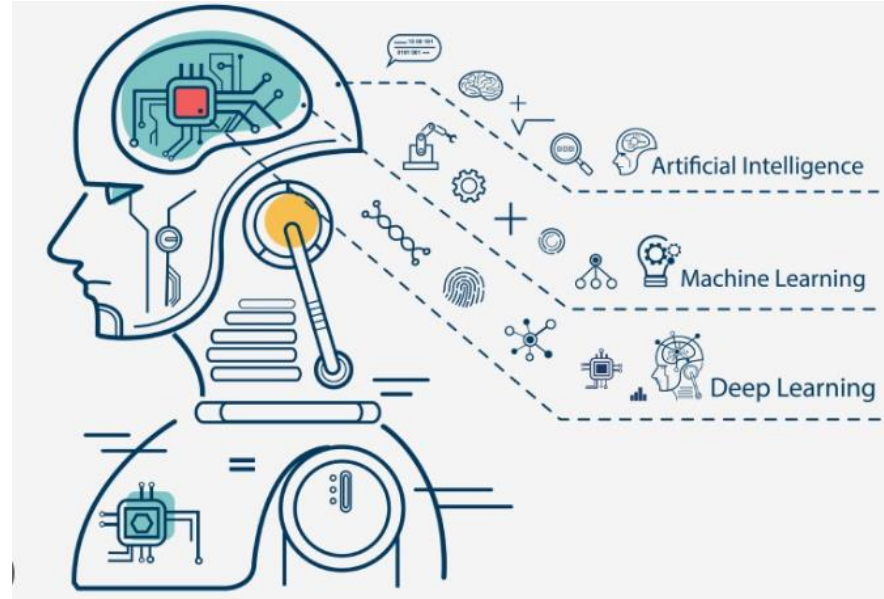
Artificial Intelligence and Machine Learning

Block-3 Unit-3

Regression

Topics to be Covered

- Introduction
- Regression Algorithm
- Linear Regression
- Polynomial Regression
- Support Vector Regression
- Important Topics
- Summary



<https://www.insiris.com/blog/machine-learning-in-the-real-world>

Introduction

- In this session we will
 - Understand the Regression Algorithm
 - Understand and apply Linear Regression
 - Understand and apply Polynomial Regression
 - Understand and apply Support Vector Regression
- Some Examples of Regression
 - Relationship between uploading a picture on Facebook page and number of likes by the friends.
 - Relationship between the height of the child and their parents' heights.
 - Relationship between the average food intake and weight gain.

Regression Algorithm

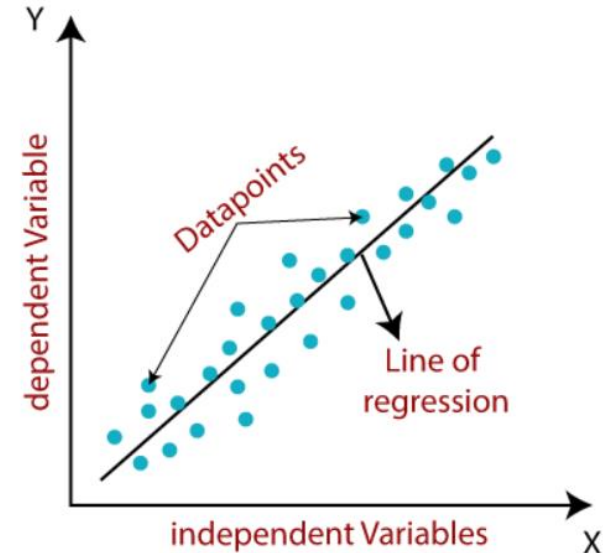
- Following are various types of regression algorithms.
- Linear regression
- Multiple Linear Regression
- Logistic regression
- Polynomial regression
- Ecologic regression
- Ridge regression
- Lasso regression
- Logic regression
- Bayesian regression
- Quantile regression
- Cox regression

Linear Regression

- Linear regression is one of the most popular Machine Learning algorithms. It is a statistical method that is used for predictive analysis. Linear regression makes predictions for continuous/real or numeric variables such as sales, salary, age, product price, etc.
- Linear regression algorithm shows a linear relationship between a dependent (y) and one or more independent (x) variables, hence called as linear regression. Since linear regression shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.

Linear Regression

- The linear regression model provides a sloped straight line representing the relationship between the variables. Consider the below image:
- Mathematically, we can represent a linear regression as:
 - $y = a_0 + a_1x + \epsilon$
 - Y = Dependent Variable (Target Variable)
 - X = Independent Variable (predictor Variable)
 - a_0 = intercept of the line (Gives an additional degree of freedom)
 - a_1 = Linear regression coefficient (scale factor to each input value).
 - ϵ = random error



Linear Regression

Consider the following set of data points (x,y), find the regression line for the given data points

X	1	2	3	4	5
Y	3	4	2	4	5

Solution:

X	y	$(x - \bar{x})$	$(y - \bar{y})$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
3	3.6	0	0	10	4

Linear Regression

$$\text{where } m = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$$

$$m = 4/10 = 0.4$$

$$\bar{x} = \text{mean of } x = 3$$

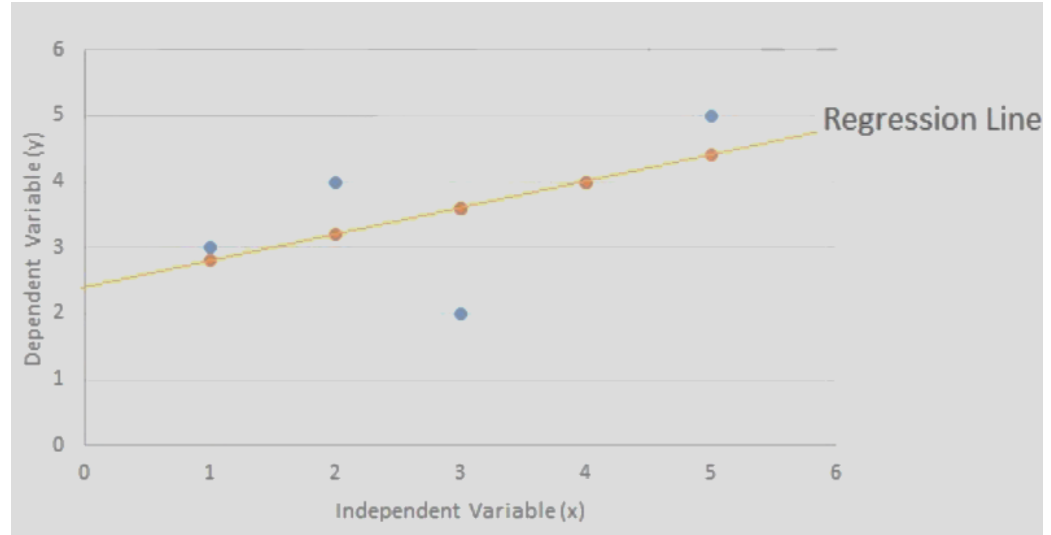
$$\bar{y} = \text{mean of } y = 3.6$$

$$y = mx + c$$

$$m = 0.4$$

$$c = 2.4$$

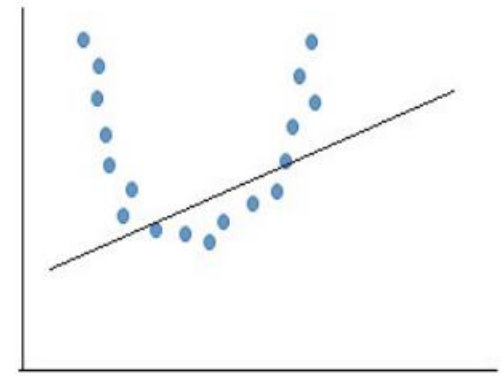
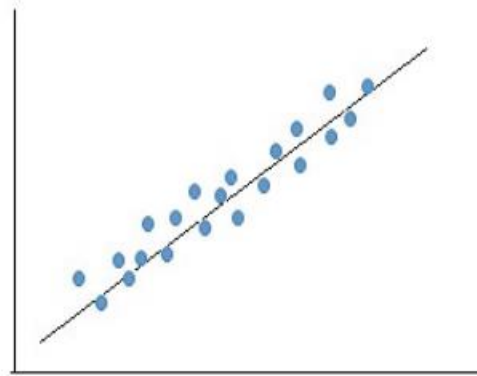
$$y = .4x + 2.4$$



In the above figure blue points are the actual points and yellow points are the predicted points using least square method. Some points represented by blue color lie above the line while some other blue color points lie below the line. However some points represented by the yellow color lie on the line. All other points not lying on the line are the far away from the line with some distance.

Polynomial Regression Algorithm

- It is also called the special case of Multiple Linear Regression in ML. Because we add some polynomial terms to the Multiple Linear regression equation to convert it into Polynomial Regression. It is a linear model with some modification in order to increase the accuracy. The dataset used in Polynomial regression for training is of non-linear nature.



Polynomial Regression Algorithm

Example Consider the following set of data points (x,y). Find the 2nd order polynomial $y = a_0 + a_1 x_i + a_2 x_i^2$, and using polynomial regression determine the value of y when x is 40.

X	40	10	-20	-88	-150	-170
Y	5.89	5.99	5.98	5.54	4.3	3.33

Solution. From the given data points (x,y):

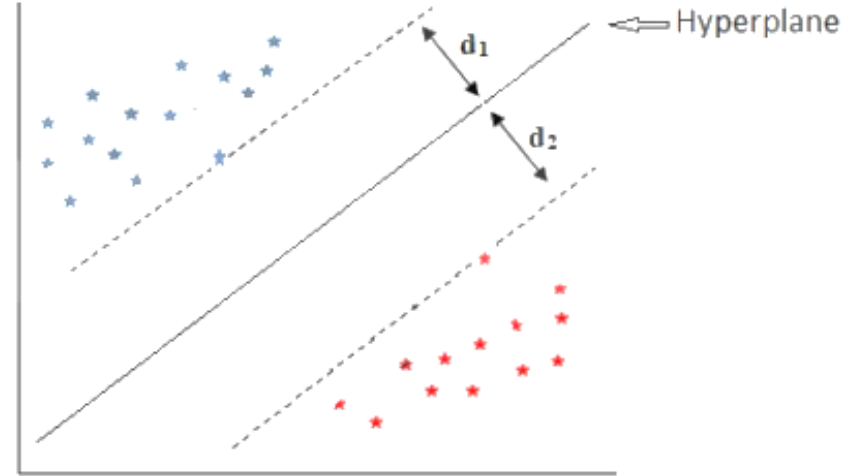
x_i	y_i	x_i^2	x_i^3	x_i^4	$x_i y_i$	$x_i^2 y_i$
40	5.89	1600	64000	2560000	235.6	9424
10	5.99	100	1000	10000	59.9	599
-20	5.98	400	-8000	160000	-119.6	2392
-88	5.54	7744	-681472	59969536	-487.52	42901.8
-150	4.3	22500	-3375000	506250000	-645	96750
-170	3.33	28900	-4913000	835210000	-566.1	96237
$\sum_{i=1}^n x_i$ = -378	$\sum_{i=1}^n y_i$ = 31.03	$\sum_{i=1}^n x_i^2$ = 61244	$\sum_{i=1}^n x_i^3$ = -8912472	$\sum_{i=1}^n x_i^4$ = 1404159536	$\sum_{i=1}^n x_i y_i$ = -1522.7	$\sum_{i=1}^n x_i^2 y_i$ = 248304

Polynomial Regression Algorithm

- By solving above matrix the value a_0, a_1 and a_2 will be
- $a_0 = 6.07647$
- $a_1 = -0.00253987$
- $a_2 = -0.000104319$
- $y = 6.07647 - 0.00253987 x - 0.000104319 x^2$
- $y(50) = 6.07647 - 0.00253987 * 50 - 0.000104319 * 2500 = 5.68$

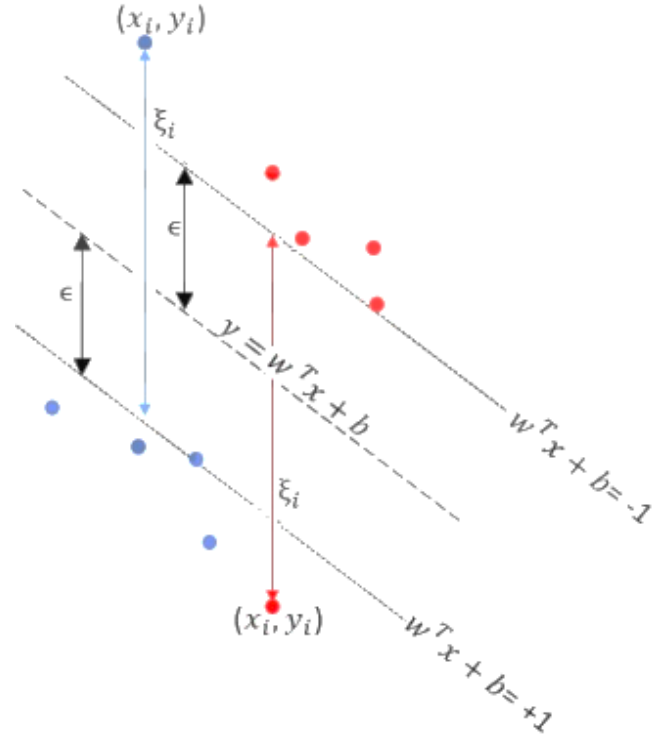
Support Vector Regression

- Support vector machine is used in solving both classification and regression problem. Consider a classification problem having two different categories as shown in figure.
- It is easy to separate these two categories by using a line between the two. There is a hyperplane between these two categories which will separate these two from each other.
- This hyperplane is used to divide the points into different categories lying opposite to the line.



Support Vector Regression

- Consider the following graph As shown in figure equation of hyperplane is $y = w^T x + b$, where b is constant having value zero since line is passes through the coordinate $(0,0)$ and the slope of line m is -1 . $y = w^T x$ (since $b=0$)
- Any point that lies below the hyperplane ($w^T x$) contribute to the positive value of x , and so is an example of the blue data points, while any data point that lie above the hyperplane ($w^T x$) contribute to the negative value of x , and so is an example of the red data points.



Support Vector Regression

- Applications of Support Vector Regression are :-
- Used to solve supervised regression problems.
- Can be used in both linear and non linear type of data.
- Prediction of fire in forest during weather changes.
- Prediction of electric power demand.

Important Topics

- Regression Algorithm
- Linear Regression
- Polynomial Regression
- Support Vector Regression



<https://www.cachindia.com/articles/important-topics-for-ca-fund-new-group-2-46332.asp>

Summary

- In this session, we discussed about the concepts of regression – linear regression and nonlinear regression. We discussed about how to find relationship between response variable and predictor variable.
- Various terminologies used in regression are discussed with an example. Concept of dependent variable, independent variables and how to find the relationships are defined in this unit. Different types of regression also discussed in this session.
- This session also focused on polynomial regression and how to plot a polynomial curve is also discussed.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-3 Unit-4

Neural Networks and Deep Learning

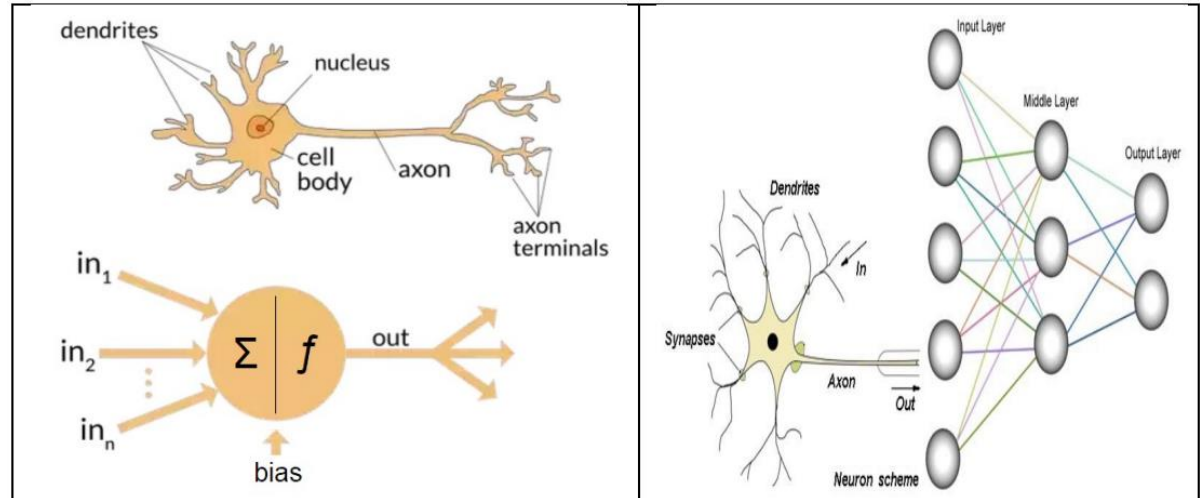
Topics to be Covered

- Introduction
- Overview of Neural Network
- Multilayer Feedforward Neural networks with Sigmoid activation functions
- Sigmoid Neurons: An Introduction
- Back propagation Algorithm
- Feed forward networks for Classification and Regression
- Deep Learning
- Important Topics
- Summary

Introduction

- Following Figure describes a biological neuron's structure, consisting of a cell body and tree-like branches called axons and dendrites. The working of neurons is based on receiving the signals from other neurons through their dendrites, processing the alerts through their body, and finally passing the signals to other neurons via its axon.

Biological Neuron and Artificial Neuron

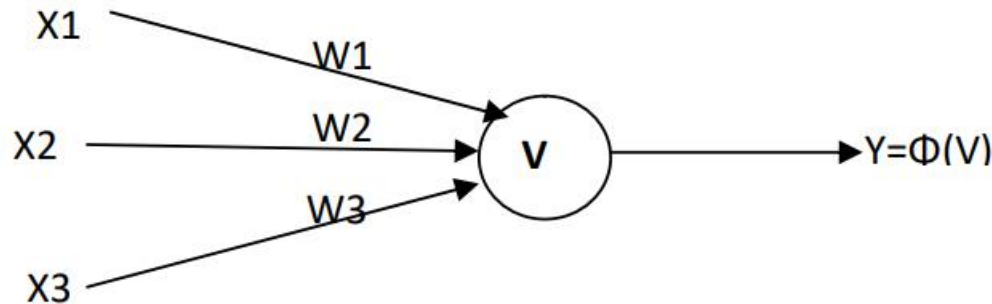


Overview of Artificial Neural Networks

- An artificial neural network (ANN) is like a computing system that simulates how the human brain analyzes information and processes it. It is the branch of artificial intelligence (AI) and solves problems that may be difficult or impossible to solve such issues for humans. In addition, ANNs have the potential for self-learning that provide better results if more data becomes available. Artificial neurons consist of the following things:
- **Interconnecting model of neurons** The neuron is the elementary component connected with other neurons to form the network.
- **Learning algorithm to train the network** Various learning algorithms are available in the literature to train the model. Each layer consists of neurons, and these neurons are connected to other layers.

Overview of Artificial Neural Networks

- Example : Below is a diagram of a single artificial neuron (unit):
- The node has three inputs $x = (x_1, x_2, x_3)$ that receive only binary signals (either 0 or 1). How many different input patterns this node can receive? What if the node had four inputs? Or Five inputs? Can you give a formula that computes the number of binary input patterns for a given number of inputs?



Overview of Artificial Neural Networks

- Answer: For three inputs the number of combinations of 0 and 1 is 8:

x1 :	0	1	0	1	0	1	0	1
x2 :	0	0	1	1	0	0	1	1
x3 :	0	0	0	0	1	1	1	1

and for four inputs the number of combinations is 16:

x1 :	0	1	0	1	0	1	0	1	0	1	0	1	0	1
x2 :	0	0	1	1	0	0	1	1	0	0	1	1	0	0
x3 :	0	0	0	0	1	1	1	1	0	0	0	0	1	1
x4 :	0	0	0	0	0	0	0	0	1	1	1	1	1	1

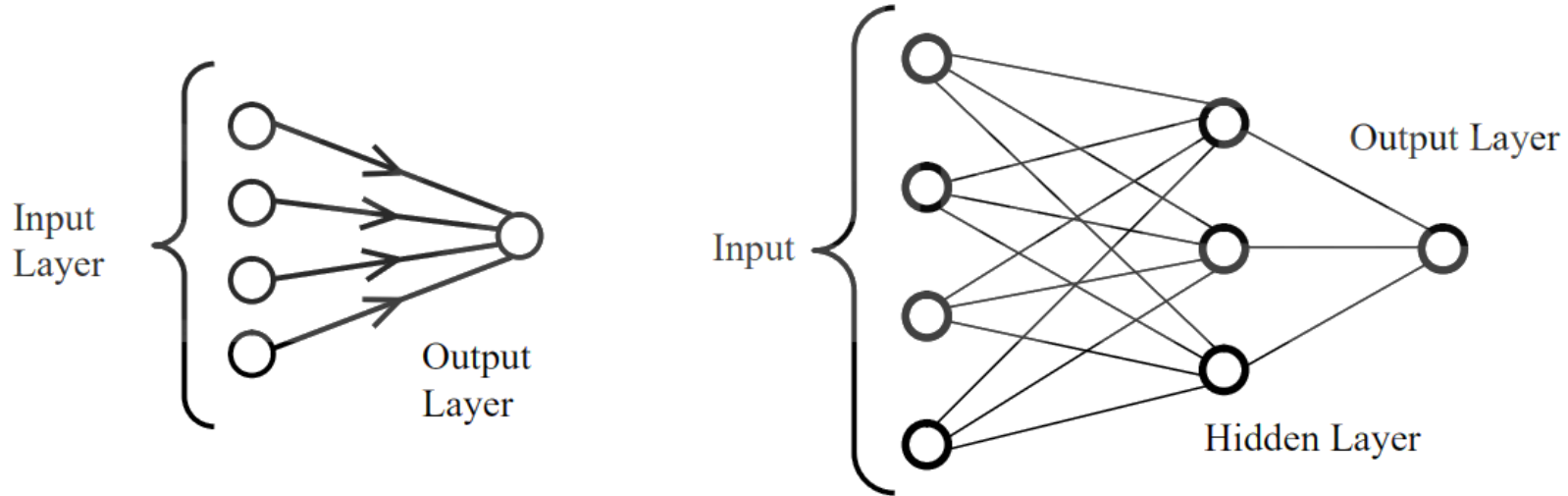
- You may check that for five inputs the number of combinations will be 32. Note that $8 = 2^3$, $16 = 2^4$ and $32 = 2^5$ (for three, four and five inputs).
- Thus, the formula for the number of binary input patterns is: 2^n , where n is the number of inputs.

MultiLayer Feed Forward Neural networks with Sigmoid activation functions

- A multilayer feed forward neural network consists of the interconnection of various layers, named input, hidden layer, and output layer.
- The number of hidden layers is not fixed. It depends upon the requirements and complexity of the problem.
- The simple neural network is one with a single input layer and an output layer is known as perceptrons.
- A Perceptron accepts inputs, moderates them with certain weight values, then applies the transformation function to output the final result.

MultiLayer Feed Forward Neural networks with Sigmoid activation functions

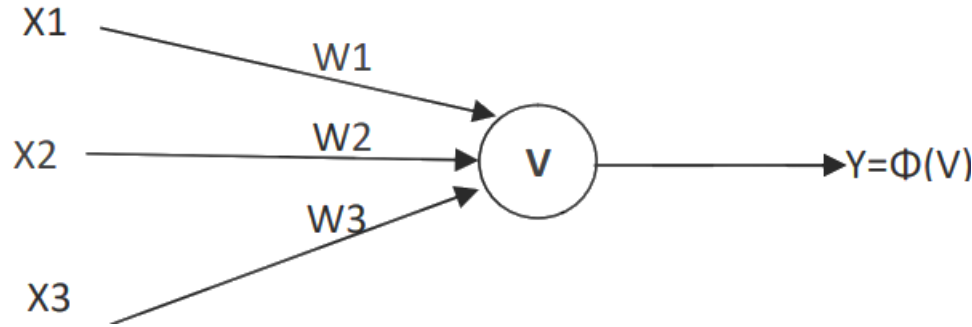
- **Neural Networks with Hidden Layers :** Figure below describes the hidden layers of a neural network by adding more neurons in between the input and output layers. There may be a single hidden layer or multiple hidden layers.



Neural network with a hidden layer

MultiLayer Feed Forward Neural networks with Sigmoid activation functions

- Example – Consider the unit shown below.



- Suppose that the weights corresponding to the three inputs have the following values:
- $w_1 = 2$; $w_2 = -4$; $w_3 = 1$ and the activation of the unit is given by the step-function:
- $\Phi(V) = 1$ for $V \geq 0$ and $\Phi(V) = 0$ Otherwise
- Calculate what will be the output value y of the unit for each of the following input patterns:

MultiLayer Feed Forward Neural networks with Sigmoid activation functions

Pattern	P ₁	P ₂	P ₃	P ₄
X ₁	1	0	1	1
X ₂	0	1	0	1
X ₃	0	1	1	1

- **Answer:** To find the output value y for each pattern we have to: a) Calculate the weighted sum: $v = \sum w_i x_i = w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3$ b) Apply the activation function to v , the calculations for each input pattern are:
- P1 : $v = 2 \cdot 1 - 4 \cdot 0 + 1 \cdot 0 = 2$, $(2 > 0)$, $y = \phi(2) = 1$
- P2 : $v = 2 \cdot 0 - 4 \cdot 1 + 1 \cdot 1 = -3$, $(-3 < 0)$, $y = \phi(-3) = 0$
- P3 : $v = 2 \cdot 1 - 4 \cdot 0 + 1 \cdot 1 = 3$, $(3 > 0)$, $y = \phi(3) = 1$
- P4 : $v = 2 \cdot 1 - 4 \cdot 1 + 1 \cdot 1 = -1$, $(-1 < 0)$, $y = \phi(-1) = 0$

Sigmoid Neurons: An Introduction

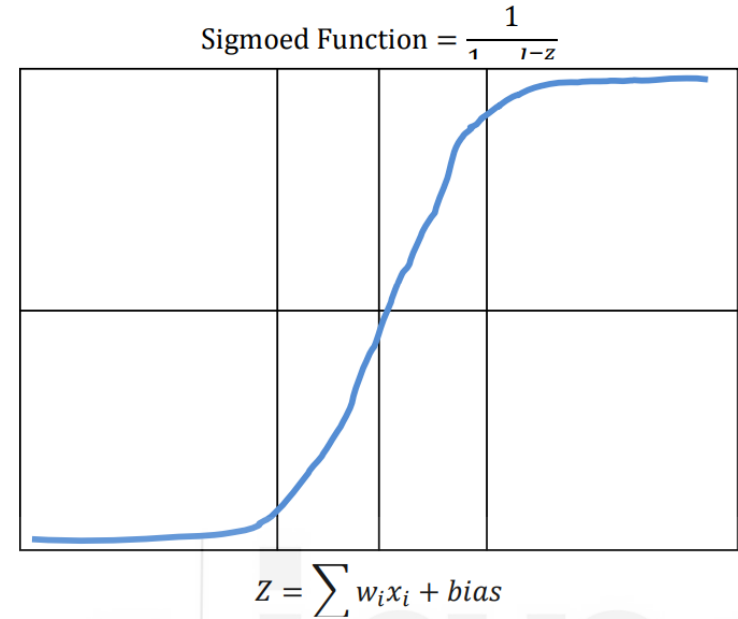
- The activation function is a mathematical function that decides the threshold value for a neuron, it may be linear or non-linear. The purpose of an activation function is to add non-linearity to the neural network.
- **Binary Step Function** : Binary step function depends on a threshold value that decides whether a neuron should be activated or not. The input fed to the activation function is compared to a certain threshold; if the input is greater than it, then the neuron is activated, else it is deactivated, meaning that its output is not passed on to the next hidden layer.

Sigmoid Neurons: An Introduction

- A **Sigmoid function** is a mathematical function which has a characteristic S-shaped curve. There are a number of common sigmoid functions, such as the logistic function, the hyperbolic tangent, and the arctangent. In machine learning, the term sigmoid function is normally used to refer specifically to the logistic function, also called the logistic sigmoid function.
- All sigmoid functions have the property that they map the entire number line into a small range such as between 0 and 1, or -1 and 1, so one use of a sigmoid function is to convert a real value into one that can be interpreted as a probability.

Sigmoid Neurons: An Introduction

- The function is called the sigmoid function. First, the value, Z , is computed then the sigmoid function is applied to Z .
- However, it looks very abstract or strange to you how it works.
- Those who don't have good knowledge of mathematics need not worry. Figure explains its curve and its derivative.



Back Propagation Algorithm

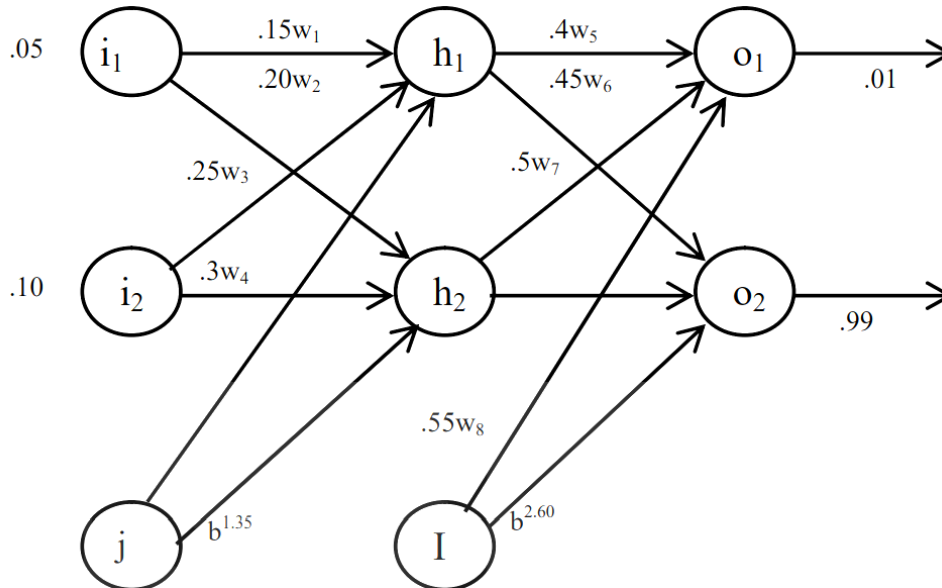
- The Backpropagation algorithm is a supervised learning algorithm for training the neural network model.
- It is one of the most fundamental building blocks of any neural network., if you have multiple layers in the neural network. Then it is used to adjust the weight in the backward direction.
- This algorithm trains the neural network model based on chain rule method. In simple terms, we can say that after every forward pass through a network, the backpropagation algorithm works to perform a backward pass to adjust the weights and biased parameters of the model.
- In other words, we can conclude that Backpropagation is used to minimize the error/ cost function by repeatedly adjusting the network's weights and biases.

Back Propagation Algorithm

- The steps of backpropagation model are given below:
- 1. First, some random value 'W' is initialized as the weights and propagated forward accordingly.
- 2. Then, find out the Error after reducing that Error by propagating backward and increasing the value weight 'W.'
- 3. After that, observe the Error and whether it has been increased. If supplementing, then don't increase the value of 'W.'
- 4. Once again propagated backward and, at this time, decreased the value of 'W'.
- 5. Now, notice the Error, and check it has been reduced or not.

Back Propagation Algorithm

- **How does Backpropagation work?**
- consider below Neural Network for a better understanding:

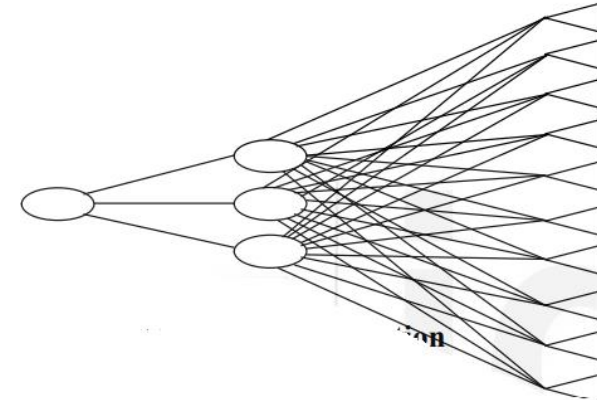


Back Propagation Algorithm

- This network contains:
- 1. Three input layers
- 2. Two layers of hidden neurons
- 3. Two neurons at the output layer.
- The following steps are used in the Backpropagation:
- Step1: We need to use forward propagation
- Step 2: After that, we have to follow backward propagation
- Step 3: We put all the values to calculate the updated weight

Feed Forward Networks For Classification And Regression

- Feed forward neural network is used for various problems, including classification, regression, and pattern encoding. In the first case, the web returns a value called $z=f(w,x)$, which is very close to the target value y . While in the second case, the target becomes the input itself $v(x, y, f(w, x))$. To deal with multiclassification, we can use either of the techniques.
- The above-mentioned left-hand side network is a modular architecture. Here, every class connects with three distinct hidden neurons. While mentioned right-hand side network defines a fully connected network, which is used for a richer classification process. The left-side network is advantageous as it is modular and supports the classifiers' gradual construction.



Deep Learning

- Deep learning is a subset of artificial intelligence, commonly called AI, that tells us the workings of the human brain to process data and patterns defining for decision making.
- Deep learning has capable of learning unsupervised from unstructured data or unlabeled data.
- Deep learning is further classified as an AI function that is used to simulate the workings of the human brain in processing data to detect objects, recognize speech, translate languages, and make decisions.

Deep Learning

- **How Deep Learning Works ?**
- Initially, there was a limitation of computing resources, and the concept of deep learning was not so popular. Once these resources were available, deep learning took the attention of the researchers.
- Deep Learning can handle all forms of data from all world regions. This data is available in massive amounts, termed big data, and is taken from various sources, including social media, search engines, different e-platforms, and others multimedia sources.
- Big data is accessible through multiple fintech applications such as cloud computing.

Deep Learning

- **Deep Learning vs. Machine Learning**
- One of the popular AI techniques available for processing big data is machine learning.
- For example, if a digital payments company wants to find out the occurrence of fraud in their system, such a company might use machine learning tools.
- The used algorithm, built on the machine learning technique, will process all transactions on the digital platform, try to find out the patterns, and mention the detection of anomalies by the pattern.
- The program built for machine learning linearly uses data, while deep learning systems enable a nonlinear approach to process the data.

Deep Learning

- **Deep Learning Example**
- The fraud detection system mentioned above can be solved through deep learning. However, as a machine learning system starts working with parameters, such as transactions of dollars an account holder sends or receives, the deep-learning method can also use the same parameters, but it works on a neural network.
- As we have mentioned earlier, each layer of the neural network takes the inputs from the previous layer; for example, the input layer has the parameters like sender information, data from social media, a credit score of the customer, using IP address, and others and passed the output to next layer for decision making.
- The final layer, the output layer, decides whether fraud has been detected or not.

Important Topics

Summary

- In this session we learned about the fundamental concepts of Neural networks, and various concepts related to area of neural networks and deep learning, this includes the understanding of the activation function, back propagation algorithm, feed forward networks and many more.
- In this session the concepts are simplified with the help of the numerical, which will help you map the theoretical concepts of neural networks with that of their implementation part.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-4 Unit-1

Feature selection and Extraction

Topics to be Covered

- Introduction
- Dimensionality Reduction
- Principal Component Analysis
- Linear Discriminant Analysis
- Singular Value Decomposition.
- Important Topics
- Summary



<https://mitsloan.mit.edu/ideas-made-to-matter/7-lessons-to-ensure-successful-machine-learning-projects>

Introduction

- Data sets are made up of numerous data columns, which are also referred to as data attributes.
- These data columns can be interpreted as dimensions on an n-dimensional feature space, and data rows can be interpreted as points inside that space.
- If your data is given in the form of rows and columns, such as in a spreadsheet, then features is another term that can be used interchangeably with input variables.
- The points (data rows) in that space reflect a small and non-representative sample of the space's contents.

Dimensionality Reduction

- The Data mining and Machine Learning methodologies both have processing challenges when working with big amounts of data (many attributes).
- The dimensions of the feature space utilised by the approach, often referred to as the model attributes, play the most important function.
- These elements, also known as the model attributes, are the fundamental qualities, and they can either be variables or features. When there are more features, it is more difficult to see them all, and as a result, the work on the training set becomes more complex as well.
- The process of making a set of major variables from a huge number of random variables is what is referred to as dimension reduction. Dimension reduction can be accomplished in two ways :

Dimensionality Reduction

- **Feature selection:** It is the process of selecting some attributes from a given collection of prospective features, and then discarding the rest of the attributes that were considered. The use of feature selection can be done for one of two reasons: either to get a limited number of characteristics in order to prevent overfitting or to avoid having features that are redundant or irrelevant.
- **Feature Selection Methods:** Feature selection methods can be divided into two categories: supervised, which are appropriate for use with labelled data, and unsupervised, which are appropriate for use with unlabeled data.

Dimensionality Reduction

- **Filter methods:** Filter methods choose features based on statistics instead of how well they perform in feature selection cross-validation. Using a chosen metric, irrelevant attributes are found and recursive feature selection is done.
- **Wrapper methods:** Wrapper feature selection methods look at the choice of a set of features as a search problem. Their quality is judged by preparing, evaluating, and comparing a set of features to other sets of features. This method makes it easier to find possible interactions between variables.
- **Embedded methods:** Embedded feature selection approaches incorporate the feature selection machine learning algorithm as an integral component of the learning process. This allows for simultaneous classification and feature selection to take place within the method.

Dimensionality Reduction

- **Forward feature selection:** The first step in the process of feature selection is to evaluate each individual feature and choose the one that results in the most effective algorithm model. This is referred to as "forward feature selection." After that step, each possible combination of the feature that was selected and a subsequent feature is analysed, and then a second feature is selected, and so on, until the required specified number of features is chosen.

Dimensionality Reduction

- **Feature extraction:** The process of reducing the amount of resources needed to describe a large amount of data is called "feature extraction." One of the main problems with doing complicated data analysis is that there are a lot of variables to keep track of. A large number of variables requires a lot of memory and processing power, and it can also cause a classification algorithm to overfit to training examples and fail to generalise to new samples.
 - **Feature extraction fills the following requirements:** It takes raw data, called features, and turns them into useful information by reformatting, combining, and changing the primary features into new ones. This process continues until a new set of data is created that the Machine Learning models can use to reach their goals.
- Methods of Dimensionality Reduction .

Principal Component Analysis

- Principal Component Analysis is an unsupervised learning algorithm that is used for the dimensionality reduction in machine learning.
- It is a statistical process that converts the observations of correlated features into a set of linearly uncorrelated features with the help of orthogonal transformation. These new transformed features are called the Principal Components.
- It is one of the popular tools that is used for exploratory data analysis and predictive modeling. It is a technique to draw strong patterns from the given dataset by reducing the variances.

Principal Component Analysis

- **Merits of Dimensionality Reduction**

- It helps to compress data, which reduces the amount of space needed to store it and the amount of time it takes to process it.
- If there are any redundant features, it also helps to get rid of them.

- **Limitations of Dimensionality Reduction**

- You might lose some data.
- You might lose some data.
- PCA fails when the mean and covariance are not enough to describe a dataset.
- We don't know how many major parts we need to keep track of, but in practice, we follow some rules.

Linear Discriminant Analysis

- The application of logistic regression has been restricted to problems involving two classes of subjects. On the other hand, the Linear Discriminant Analysis is the linear classification method that is recommended to use when there are more than two classes.
- The algorithm for linear classification known as logistic regression is known for being both straightforward and robust. On the other hand, there are a few restrictions or faults in the system that highlight the requirement for more complex linear classification algorithms. The following is a list of some of the problems:
- **Binary class Problems.** Concerns regarding the binary class is that the Logistic regression is utilised for issues that involve binary classification or two classes. It is possible to enhance it such that it can manage multiple-class categorization, but in practise, this is not very common.

Linear Discriminant Analysis

- **Unstable, but with well-defined classes.** When the classes are extremely distinct from one another, logistic regression may become unstable.
- It is prone to instability when there are only a few occurrences. When there are not enough examples from which to draw conclusions about the parameters, the logistic regression model may become unstable.
- **Understanding LDA Models :** In order to simplify the analysis of your data and make it more accessible, LDA will make the following assumptions about it:
 - The distribution of your data is Gaussian, and when plotted, each variable appears to be a bell curve.
 - Each feature has the same variance, which indicates that the values of each feature vary by the same amount on average in relation to the mean.

Linear Discriminant Analysis

- **Problems with Classification:** LDA is used to solve classification problems where the output variable is a categorical one. This may seem obvious, as LDA works with both two and more than two classes.
- **Gaussian Distribution:** The standard way to use the model assumes that the input variables have a Gaussian distribution. Think about looking at the univariate distributions of each attribute and using transformations to make them look more like Gaussian distributions.

Linear Discriminant Analysis

- **Remove Outliers:** Think about removing outliers from your data. These things can mess up the basic statistics like the mean and the standard deviation that LDA uses to divide classes.
- **Same Variance:** LDA assumes that the variance of each input variable is the same. Before using LDA, you should almost always normalise your data so that it has a mean of 0 and a standard deviation of 1.

Single Value Decomposition

- The Singular Value Decomposition (SVD) method is a well-known technique for decomposing a matrix into a large number of component matrices.
- This method is valuable since it reveals many of the interesting and helpful characteristics of the initial matrix.
- We can use SVD to discover the optimal lower-rank approximation to the matrix, determine the rank of the matrix, or test a linear system's sensitivity to numerical error.

Single Value Decomposition

- Singular value decomposition is a method of decomposing a matrix into three smaller matrices.
- $A = USV^T$
- Where:
 - A : is an $m \times n$ matrix
 - U : is an $m \times m$ orthogonal matrix
 - S : is an $n \times n$ diagonal matrix
 - V : is an $n \times n$ orthogonal matrix

Important Topics

- Feature Selection
- Feature extraction
- Principal Component Analysis
- Linear Discriminant Analysis
- Singular Value Decomposition



<https://www.careerindia.com/tips/nest-important-topics-to-read-during-the-last-week-022723.html>

Summary

- In this session we learned about the concept of Dimension Reductionality, wherein we understood basics of Feature Selection and Feature extraction techniques.
- Thereafter an explicit discussion of Principal Component Analysis(PCA), Linear Discriminant Analysis(LDA) and Singular Value Decomposition(SVD) was given.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-4 Unit-2

Association Rules

Topics to be Covered

- Introduction
- What are Association Rules?
- Apriori Algorithm
- FP Tree Growth
- Pincer Search
- Important Topics
- Summary



<https://mitsloan.mit.edu/ideas-made-to-matter/7-lessons-to-ensure-successful-machine-learning-projects>

Introduction

- In this session we will understand the purpose of association rules, pattern search and various algorithms for pattern search.
- Imagine a salesperson at a shop. If you buy a product from the shop, the sales person recommends you more items related to the product you have purchased. He may also suggest you about the items that are frequently bought with the product you have purchased.
- The salesperson may also try to figure out your
- choices about the product you observe at the shop and may recommend you further.
- If salesman notices a male customer, then he may suggest adding beer too from his historical analysis of male customer preferring to buy beer as well.

What are Association Rules ?

- Association rules are one of the important concepts that is widely applied in problems like market basket analysis. Consider a supermarket, where all the related items such as grocery items, dairy items, cosmetics, stationary items etc are kept together in same aisle.
- This helps the customers to find their required items timely. This further helps them to remember the items to purchase they might have forgotten or to they may like to purchase if suggested. Association rules thus enable one to correlate among various products from a huge set of available items.
- Analyzing the items customer buy together also helps the retailers to identify the items they can offer on discount.

What are Association Rules ?

- **Basic Concepts** There are few terms that one should understand before understanding the algorithm.
- **k-Itemset**: It is a set of k items. For example, 2-itemset can be {pencil, eraser} or {bread, butter} etc., 3-itemset can be {bread, butter, milk}.
- **Support**: Frequency of appearance of an item appears in all the considered transactions is called as the support of an item. Mathematically, support of an item x is defined as:

$$\text{support}(x) = \frac{\text{Number of transactions containing } x}{\text{Total number of considered transactions}}$$

What are Association Rules ?

- Confidence: Confidence is defined as the likelihood of obtaining item y along with an item x. Mathematically, it is defined as the ratio of frequency of transactions containing items x and y to the frequency of transactions that contained item x.

$$\text{confidence}(y) = \frac{\text{Number of transactions containing } x \text{ and } y}{\text{Number of considered containing } x}$$

- Confidence can also be defined as probability of occurrence of y, given probability of occurrence of x.

$$\text{confidence}(x \Rightarrow y) = P(y/x)$$

What are Association Rules ?

- where x is antecedent, and y is a consequent. In terms of support, confidence can be described as:

$$confidence(y) = \frac{support(x \cup y)}{support(x)}$$

- Frequent Itemset: An item whose support is at least the minimum support threshold is known as a frequent itemset.
- For example, let minimum support threshold is 10, then an item set with support score 11 is a frequent itemset but an item set with support score 9 is not.

What are Association Rules ?

- **Association Rules: Binary Representation** Let $I = \{I_1, I_2, I_3, \dots, I_n\}$ be a set of n items and $T = \{T_1, T_2, T_3, \dots, T_t\}$ be a set of t transactions, where each transaction T_i contains a not null set of items purchased by a customer such that $T_i \subseteq I$. Let each item I_i in the store be represented by a binary variable, B . The variable takes up the value 0 or 1, representing the absence or presence of item at the store.
- $B(i) = 1$, if an item I_i is available at the store 0, otherwise

What are Association Rules ?

- For example, consider a set of four transactions T1, T2, T3 and T4 with the following items:
- T1 = {milk, cookies, bread},
- T2 = {milk, bread, egg, butter},
- T3 = {milk, cookies, bread, butter},
- T4 = {cookies, bread, butter}.

Transaction id	Milk	Cookies	Bread	Butter	Egg
T1	1	1	1	0	0
T2	1	0	1	1	1
T3	1	1	1	1	0
T4	0	1	1	1	0

Binary representation of the transactions

What are Association Rules ?

- The binary variable can also be used to analyze the purchasing patterns of the customers. One can analyze a basket in terms of binary values of the items that customer has purchased. Such items are said to frequently bought together or associated to each other.
- These binary patterns are represented in the form of association rules. For example, the customer who buys milk, also tend to buy bread at the same time. This can be represented using support and confidence as: Milk => Bread with support as 75% (or value as 0.75) and confidence as 100% (or value as 1).

$$\text{Support(milk)} = \frac{\text{transactions containing milk}}{\text{total number of transactions}} = \frac{3}{4} = 0.75$$

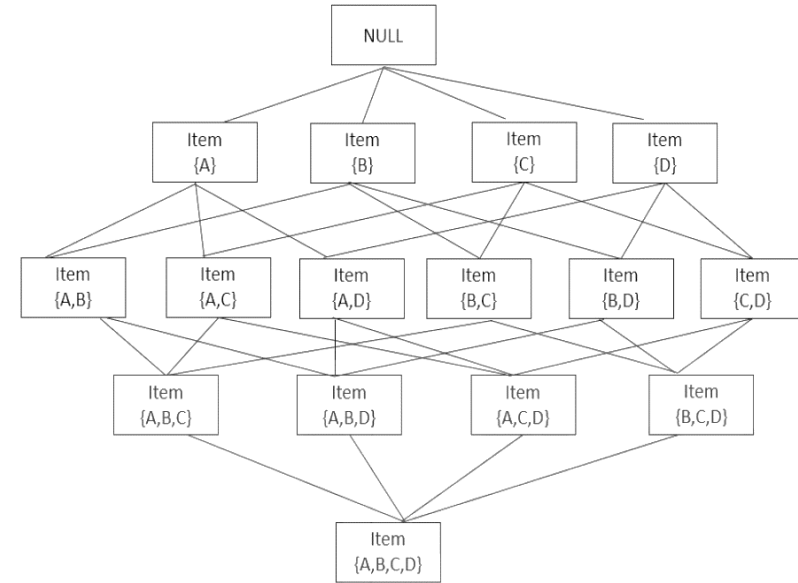
$$\text{Confidence (bread)} = \frac{\text{support(milk,bread)}}{\text{support(milk)}} = \frac{\frac{3}{4}}{0.75} = \frac{0.75}{0.75} = 1$$

What are Association Rules ?

- **Association Rules Discovery** The problem of discovery of association rules can be stated as: Given a set of transactions T , find the rules whose support and confidence are greater than equal to the minimum support and confidence threshold. `
- Traditional approach to generate association rules is to compute the support and confidence for every possible combination of items. But this approach is computationally not possible as the number of combinations of items can be exponentially large.
- To avoid such large number of computations, basic approach should be to ignore the needless computations without computing their support and confidence scores.

Apriori Algorithm

- This algorithm is used to obtain the frequent item sets for association rules. The algorithm is named so, as it needs the prior knowledge of the frequent item sets.
- Frequent Item sets Generation For a given set of n items, there are $2^n - 1$ possible combinations of items. Consider an itemset $I = \{A, B, C, D\}$ with four items, there are 15 combinations of items such as $\{\{A\}, \{B\}, \{C\}, \{D\}, \{A, B\}, \{A, C\}, \{A, D\}, \{B, C\}, \{B, D\}, \{C, D\}, \{A, B, C\}, \{A, B, D\}, \{A, C, D\}, \{B, C, D\}, \{A, B, C, D\}\}$. This can be represented by a lattice diagram as shown in Figure.



Apriori Algorithm

- Apriori algorithm searches the items level by level; to find the $(k+1)$ item sets, it uses the k item sets. To determine the frequent item sets, the algorithm first finds the candidate item sets from the lattice representation.
- Apriori Principle: If an itemset is frequent, then all of its subsets must be frequent. To understand the principle, let's consider the itemset $\{A, B, C\}$ is the frequent itemset, then its subsets $\{A, B\}$, $\{A, C\}$, $\{B, C\}$, $\{A\}$, $\{B\}$ and $\{C\}$ are also the frequent item sets (marked in yellow) as shown in Figure. This is because if a transaction contains the itemset $\{A, B, C\}$, then the transaction will also contain all the subsets of this itemset.

Apriori Algorithm

Input: T : a set of transactions; $minsup$: minimum support threshold
Output: $F(k)$: set of k -frequent itemsets (result)

```
1.  $F_1 \leftarrow \{\text{large 1 - itemsets}\}$ 
2.  $k \leftarrow 2$ 
3. while  $F(k-1)$  is not empty
4.    $C(k) \leftarrow \text{Apriori\_gen}(F(k-1), k)$ 
5.   for transactions  $t$  in  $T$ 
6.      $D_t \leftarrow \{c \text{ in } C(k) : c \subseteq t\}$ 
7.     for candidates  $c$  in  $D_t$ 
8.        $\text{count}[c] \leftarrow \text{count}[c] + 1$ 
9.
10.     $F(k) \leftarrow \{c \text{ in } C(k) : \text{count}[c] \geq minsup\}$ 
11.     $k \leftarrow k + 1$ 
12.
13. return  $\text{Union}(F(k))$ 
14.
15. Apriori_gen( $F, k$ )
16. for all  $X \subseteq F, Y \subseteq F$  where  $X_1 = Y_1, X_2 = Y_2, \dots, X_{k-2} = Y_{k-2}$  and  $X_{k-1},$   

    $Y_{k-1}$  are in lexicographic order
17.    $c = X \cup \{Y_{k-1}\}$ 
18.   if  $u \subseteq c$  for all  $u$  in  $C$ 
19.      $\text{result} \leftarrow \text{append}(\text{result}, c)$ 
20. return  $\text{result}$ 
```

Apriori Algorithm

- **Generating Association Rules using Frequent Itemset** Once the frequent itemsets $F(k)$ are generated from set of transactions T , next step is to generate strong association rules from them. Recall that strong association rules satisfy both minimum support as well as minimum confidence. Theoretically, confidence is defined as the ratio of frequency of transactions containing items x and y to the frequency of transactions that contained item x .

FP Tree Growth

- In Data Mining, finding frequent patterns in large databases is very important and has been studied on a large scale in the past few years. Unfortunately, this task is computationally expensive, especially when many patterns exist.
- **Working of FP growth algorithm:**
- Create Frequent Pattern Tree, or FP-tree by compressing the transaction database. Along with preserving the information about the itemsets, the tree structure also retains the association among the itemsets.
- Divide the transaction database into a set of conditional databases. where each associated with one frequent item or “pattern fragment,” and examines each database separately.
- For each “pattern fragment,” examine its associated item sets only.

FP Tree Growth

- Advantages of FP growth over Apriori algorithm:
- Efficient than Apriori algorithm
- No candidate itemset generation
- Only two passes over the transaction set
- **FP Tree Construction**
- Obtain the 1-itemsets and their support score as computed in the first step of Apriori algorithm.
- Sort the item sets in descending order of their support score.
- Create the root of tree as NULL.

FP Tree Growth

- Examine the first transaction in the set T , create the nodes of tree as the itemset with the maximum support score at the top, the next itemset with lower count and so on. At the end, the leaves of the tree will have the itemsets with the least support score.
- Examine every transaction in the set, whose itemsets are also sorted in descending order of their support score. If any itemset of this transaction is already existing in the tree, then they share the same node, but count is incremented by 1. Else, a new node and branch will be created for a new itemset.

FP Tree Growth

- Divide the FP tree obtained at the end of step 5 into conditional FP trees. For this, examine the leaf (lowest) node of FP tree. The lowest node represents the frequency pattern of length 1. For each item, traverse the path in the FP Tree to create conditional pattern base. It is the path labels of all paths to any node of the given item in FP tree.
- Conditional pattern base generates the frequent patterns. Thus, one conditional tree is generated for one frequent pattern. Perform this step recursively to generate all the conditional FP trees.

Pincer Search

- In Apriori algorithm, the computation begins from the smallest set of frequent item sets and proceeds till it reaches the maximum frequent itemset. But the algorithm's efficiency decreases as the algorithm passes through many iterations. Solution to the problem is to generate the frequent item sets by using bottom-up and top-down approach together.
- Thus, the Pincer algorithm works on bidirectional approach. It attempts to find the frequent item sets in a bottom – up manner but, at the same time, it maintains a list of maximal frequent item sets. While iterating the transactions set, it computes the support score of the candidate maximal frequent item sets to record the frequent itemsets.
- This way, the algorithm outputs the subsets of the frequent item sets and, hence, they need not maintain support score in the next iteration. Thus, Pincer–search is advantageous over Apriori algorithm when the frequent item set is long.

Important Topics

- Association rules Discovery
- Apriori Algorithm
- FP Tree Growth
- FP Tree Construction
- Pincer Search



<https://www.careerindia.com/tips/nest-important-topics-to-read-during-the-last-week-022723.html>

Summary

- This session presented the concepts of association rules. In this unit, we briefly described about what are these association rules and how various models help to analyse data for patterns, or cooccurrences, in a transaction database. These rules are created by finding the frequent item sets in the database using the support and the confidence. Support indicates the frequency of occurrence of an item given the entire transaction database.
- High support score indicates a popular item among the users. Confidence of a rule indicates the rule's reliability. Higher confidence indicates more occurrence of item(s) in a set. A typical example that has been used to study association rules is market basket analysis.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-4 Unit-3

Clustering

Topics to be Covered

- Introduction
- Types of clustering
- Partition Based
- Hierarchical Based
- Density Based Clustering techniques
- Clustering algorithms
- Important Topics
- Summary



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Introduction

- Clustering or cluster analysis is a method for dividing a group of data objects into subgroups based on a single observation.
- Here, each cluster is a subset of the data, where objects with resemblance or similar properties are grouped.
- Also, they are similar to each other in one cluster but differ from those objects which are in different clusters.
- In simple terms, the effort of dividing a population into multiple groups so that data points of one group can easily be compared with data points of another group.

Introduction

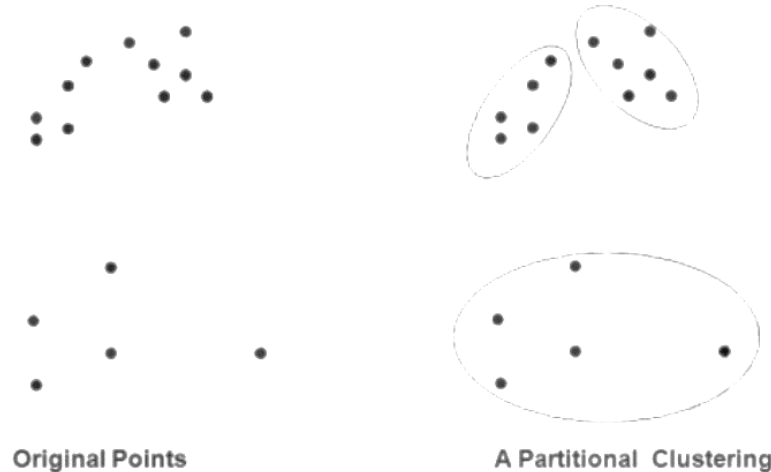
- Clustering is a widely used technique in the industry. It is being used in almost every domain, ranging from banking to recommendation engines, document clustering to image segmentation.
- Customer Segmentation Customer segmentation is one of the most common applications of clustering. And it isn't just limited to banking.
- This strategy is across functions, including telecom, e-commerce, sports, advertising, sales, etc.
- Document Clustering Document Clustering is another common application of clustering. Let's assume that you have multiple documents, and you need to cluster similar documents together.
- Clustering helps us group these documents such that similar documents are in the same clusters.

Types of Clustering

- There are many clustering algorithms in the literature. Traditionally, documents are grouped based on how similar they are to other documents. Similarity-based algorithms define a function for computing document similarity and use it as the basis for assigning documents to clusters.
- Each cluster should have data that are comparable to one another but different from those in other clusters.
- Clustering algorithms fall into different categories based on the underlying methodology of the algorithm (agglomerative or partition), the structure of the final solution (flat or hierarchical), or the density based. All the above-mentioned clustering types are discussed in detail in the rest of this session.

Partition Based Clustering

- This clustering method classifies the information into multiple groups based on the characteristics and similarities of the data. In the partitioning method when database(D) contains multiple(N) objects then the partitioning approach divides the data into user-specified(K) partitions, each of which represents a cluster and a specific region.
- That is, the data is divided into K groups. That is, it divides the data into K groups or partitions in such a manner that each group should have at least one object from the existing data.

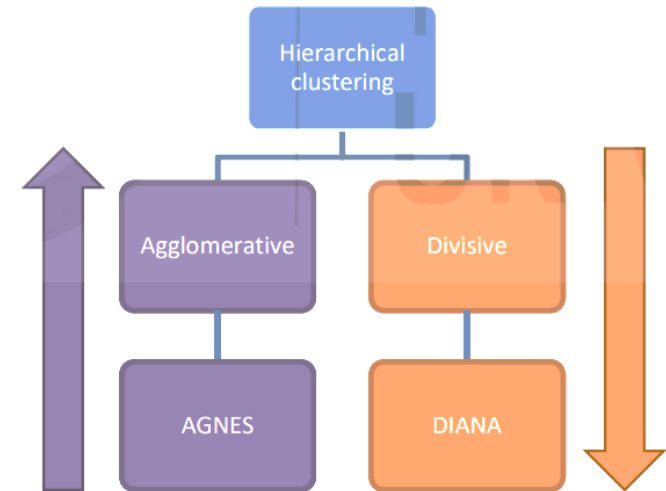


Hierarchical Based Clustering

- Hierarchical Clustering analysis is an algorithm that groups the data points with similar properties and these groups are termed “clusters”. As a result of hierarchical clustering, we get a set of clusters, and these clusters are always different from each other. Clustering of this data into clusters is classified as:
- Agglomerative Clustering (involving decomposition of cluster using bottom-up strategy)
- Divisive Clustering (involving decomposition of cluster using top-down strategy)
- Hierarchical clustering helps in creating clusters in the proper order (or hierarchy).
- Example: The most common everyday example we see is how we order our files and folders in our computer by proper hierarchy.

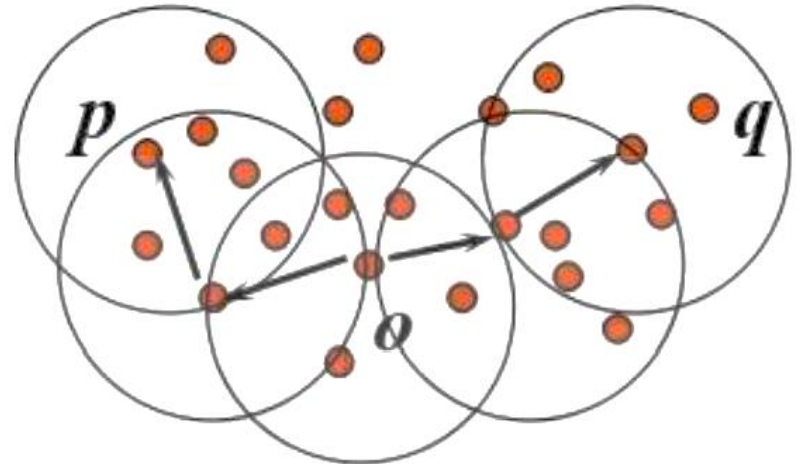
Hierarchical Based Clustering

- As mentioned, Hierarchical clustering is classified into two types i.e., Agglomerative clustering (AGNES) and Divisive Clustering (DIANA)
- Hierarchical clustering Technique in terms of space and time complexity:
- Space complexity = $O(n^2)$ where n is the number of data points.
- Time complexity = $O(n^3)$ where n is the number of data points.



Density Based Clustering Technique

- Clusters are formed in the Density Depending Clustering Technique based on the density of the data points represented in the data space. Those locations that become dense as a result of the large amount of data points that reside there are termed as clusters.



Clustering Algorithms

- **K-Means** Among clustering algorithms, is an algorithm that tries to minimize the distance of the points in a cluster with their centroid – the k -means clustering technique.
- K-means is a centroid-based algorithm. It can also be distances between points are calculated to allocate a point to a cluster. Each cluster in K is paired with a centroid. The K-Means algorithm's main goal is to reduce the sum of distances between points and th corresponding cluster centroid.

Clustering Algorithms

- Let's have a look at an example. Assume you went to a bookstore to purchase some books. There are several types of books that can be found there. One thing you'll notice is that the books are sorted into groups based on their category. All of the literature books will be kept in one location, while science books will be organized by type.
- Working of K-means clustering algorithm**
- Selection of k values.
- Initialization of the centroids.
- Selection of the cluster and finding the average.

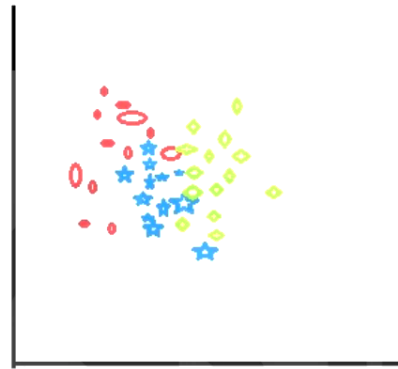


fig 1: before applying
k-means clustering

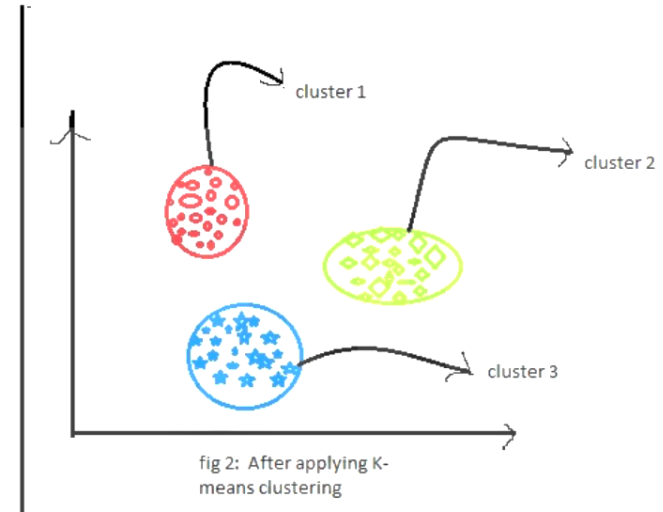
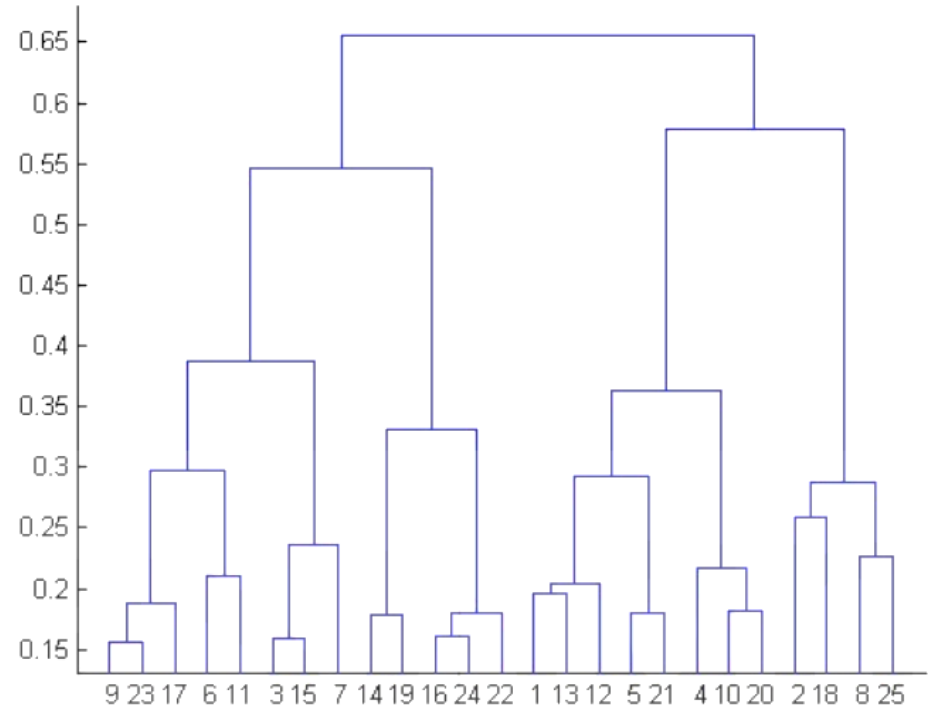


fig 2: After applying K-
means clustering

Clustering Algorithms

- **Agglomerative clustering**
- In this case of clustering, the hierarchical decomposition is done with the help of bottom-up strategy where it starts by creating atomic (small) clusters by adding one data object at a time and then merges them together to form a big cluster at the end, where this cluster meets all the termination conditions. This procedure is iterative until all the data points are brought under one single big cluster.

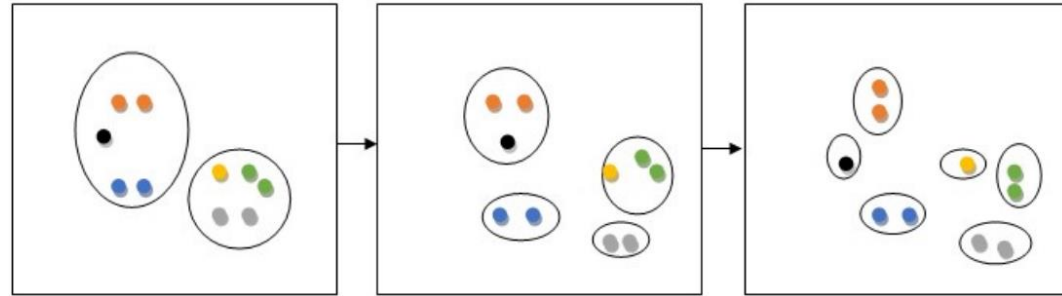


Clustering Algorithms

- **Basic algorithm of agglomerative clustering**
- 1. Determine the proximity matrix.
- 2. Assume that each data point belongs to a cluster.
- 3. Do it again.
- 4. Combine the two groups that are the closest together.
- 5. Make changes to the proximity matrix.
- 6. Continue until just one cluster remains.

Clustering Algorithms

- **Divisive Clustering (DIANA)**
- Diana basically stands for Divisive Analysis; this is another type of hierarchical clustering where basically it works on the principle of top-down approach (inverse of AGNES) where the algorithm begins by forming a big cluster, and it recursively divides the most dissimilar cluster into two, and it goes on until we're all the similar data points belong in their respective clusters.
- These divisive algorithms result in highly accurate hierarchies than the agglomerative approach, but they are computationally expensive.



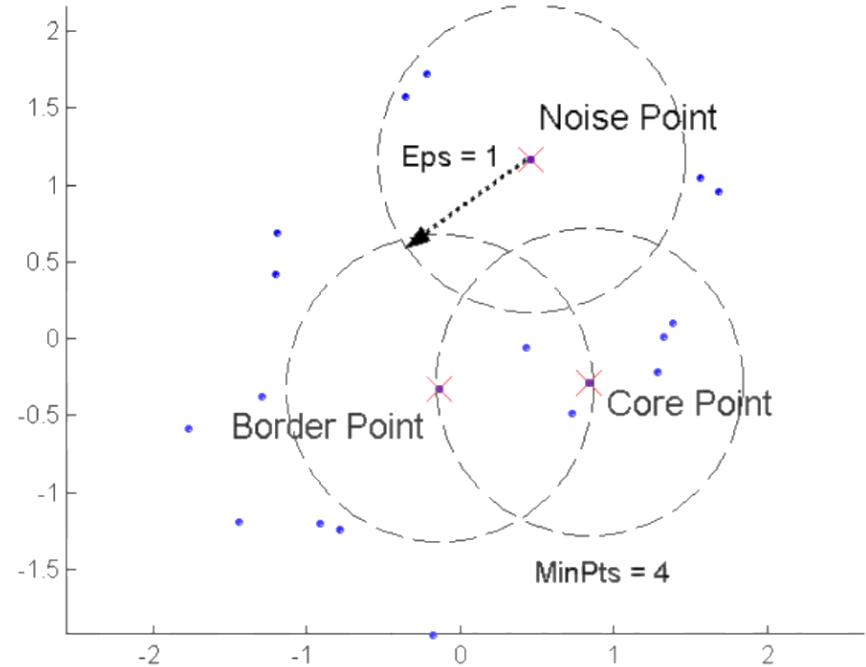
Divisive clustering step by step process

Clustering Algorithms

- **Density-Based Spatial Scan (DBSCAN)** : The distance between objects is used to cluster things in most partitioning methods. Such algorithms can only locate spherical-shaped clusters and have trouble finding clusters of other forms. DBSCAN can form clusters in different shapes; this type of algorithm is most suitable when the dataset contains noise or outliers. Also, it depends on a density-based concept of cluster: A cluster is defined as the most densely connected set of points.
- DBSCAN uses noise to find clusters of any shape in spatial databases.
- If a point has more than a given number of points (MinPts) within Eps, it is considered a core point.
- ▪ These points belong in a dense region and are at the interior of a cluster.

Clustering Algorithms

- DBSCAN Algorithm
- 1. Label points as core, border and noise
- 2. Eliminate noise points for every core point p that has not been assigned to a cluster
- 3. Create a new cluster with the point p and all the points that are density-connected to p .
- 4. Assign border points to the cluster of the closest core point.



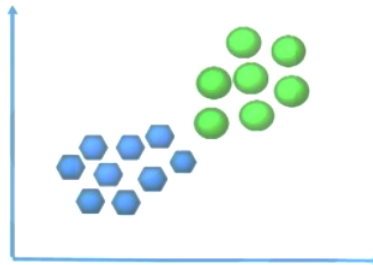
Clustering Algorithms

- **Fuzzy Clustering** is the most used clustering algorithm present in real world and is a sort of clustering in where each data point is assigned to multiple clusters. The algorithm suggests that the data points can belongs to more than one cluster unlike hard clustering where data points can actually belong to only one cluster.
- **Example** : A Guava can either be Yellow and Green (hard clustering), but a Guava can also be Yellow and Green (this is fuzzy clustering).

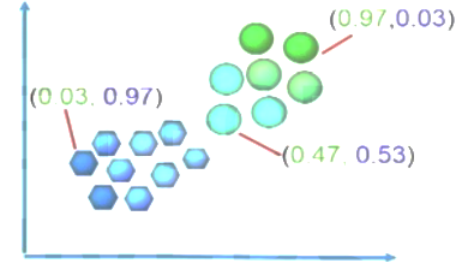
Clustering Algorithms

- Here, the Guava can be Yellow to a certain degree as well as Green to a certain degree. Instead of the Guava belonging to Green [green = 1] and not Yellow [Yellow = 0], the Guava can belong to Green [green = 0.5] and Yellow [Yellow = 0.5].
- These values are normalized between 0 and 1; however, they do not represent probabilities, so the two values do not need to add up to 1.

Hard Clustering



Soft Clustering



Important Topics

- Partition Based
- Hierarchical Based
- K-Means
- Agglomerative and Divisive
- DBSCAN
- Fuzzy Clustering



<https://www.careerindia.com/tips/nest-important-topics-to-read-during-the-last-week-022723.html>

Summary

- The technique of separating a set of data objects into subgroups based on some observation is known as clustering or cluster analysis.
- Each subset is referred to as a 'cluster,' with objects that are related to one another but different from those in other clusters.
- In simple terms, the process of separating a population or set of data points into several groups so that data points from the same group can be compared to data points from different groups.
- Clustering is often confused with classification in data analysis, where separation of data happens on the basis of class labels, while in clustering partitioning of large data sets occurs in groups occurs on the basis of similarity.

Summary

- In Density Based Clustering Technique, the clusters are created based upon the density of the data points which are represented in the data space.
- K-Means: Among clustering algorithms, is an algorithm that tries to minimize the distance of the points in a cluster with their centroid the k-means clustering technique. The main objective of the K-Means algorithm is to minimize the sum of distances between the points and their respective cluster centroid.
- AGNES is a type of agglomerative clustering that combines the data objects into a cluster based on similarity.
- DBSCAN uses noise to find clusters of any shape in spatial databases.

*Thank
You*



Artificial Intelligence and Machine Learning

Block-4 Unit-4

Machine Learning-Programming Using Python

Topics to be Covered

- Introduction
- Classification Algorithms
- Regression Algorithms
- Feature Selection and Extraction
- Association Rules
- Clustering Algorithms
- Important Topics
- Summary



<https://www.livemint.com/brand-stories/why-use-python-for-ai-and-machine-learning-11686566054332.html>

Introduction

- In this session we will see the implementation of learned in this course. To understand the codes you need to have understanding of respective Machine learning algorithms along with that understanding of programming is must.
- The codes are readily using various libraries of Python programming language viz. Scikit Learn, Matplotlib, numpy etc., you can execute these codes through any of the Python programming tools.
- Most of the machines learning algorithms, you learned in this course, are implemented here, just try to execute them and analyze the results.

Classification Algorithms

- **NAIVE BAYES** : It is a method of classification that is founded on Bayes makes the assumption that predictors are free to act independently of one another. A Naive Bayes classifier, to put it in layman's words, makes the assumption that the existence of one particular characteristic in a class is unrelated to the presence of any other feature.
- The following procedures need to be carried out in order to classify data using the Naive Bayes method.
- In the first step, we will begin by importing the dataset as well as any necessary dependencies.
- The second step is to get the prior probability of each class using the formula $P(y)$.

Classification Algorithms

- The Third Step is to Determine the likelihood of each
- characteristic using the table you just created.
- Final and the Fourth Step is to Calculate the Posterior Probability for each class by applying the Naive Bayesian equation.
- **K-Nearest Neighbour (K-NN)**
- We learned that Suppose the value of K is 3. The KNN calculating the distance of point X from all the points. It then finds the 3 nearest points with least distance to point X

Classification Algorithms

- In the example shown below following steps are performed:
- In Step 1, the scikit-learn package is used to import the k-nearest neighbour algorithm.
- Step 2. is to create the feature variables and the target variables.
- Step 3. Separate the data into the test data and the training data.
- Step 4. Generate a k-NN model using neighbours value.
- Step 5. Train the model using the data or adjust the model based on the data.
- Proceed to Step 6, which is to make a forecast.

Classification Algorithms

- **Decision Tree Implementation** : A decision tree is a type of supervised machine learning algorithm that may be used for both regression and classification tasks. It is one of the most popular and widely used machine learning techniques.
- The decision tree method creates a node for each attribute present in the dataset, with the attribute that is considered to be the most significant being placed at the top of the tree. When we first get started, we think of the entire training set as the root. There must be a categorical breakdown of the feature values.

Classification Algorithms

- **Logistic Regression** : LR is a classification algorithm that is used in Machine Learning to predict the likelihood of a categorical dependent variable. It is also known as "logistic regression." The dependent variable in logistic regression is a binary variable, which means that it comprises data that is either recorded as 1 (yes, success, etc.) or 0. (no, failure, etc.)
- **Support Vector Machine** : Support Vector Machine, more usually referred to as SVM, is a technique for supervised and linear machine learning that is most frequently utilized purpose of addressing classification issues. Support Vector Classification is another name for SVM. In addition, there is a subset of SVM known as SVR, which stands for Support Vector Regression

Classification Algorithms

- The following are the steps involved in implementation:
- Import the Libraries
- Make sure the Dataset is loaded.
- Dataset will be divided into X and Y.
- Create a Training set and a Test set from the X and Y Datasets.
- Scaling the features should be done.
- Ensure that the SVM is adjusted to the Training set.
- Make a guess about the results of the test set.
- Construct the Matrix of Confusion.

Regression Algorithms

- **Linear Regression** The purpose of a linear regression model is to determine whether or not there is a connection between one or more characteristics (also known as independent variables) and a target variable that is continuous (dependent variable). Linear Regression is referred to as Uni-variate Linear Regression when there is only one feature, and it is referred to as Several Linear Regression when there are multiple features. Following are the stages involved in the implementation of a linear regression :
- Firstly, initialize the parameters.
- Given the value of an independent variable, predict what the value of a dependent variable will be.
- Determine the amount of error that each forecast has for each data point.

Regression Algorithms

- Using a_0 and a_1 , perform the calculation for the partial derivative.
- Add up the individual costs that you have determined for each of the numbers.
- **Polynomial Regression** is a type of linear regression in which the relationship between the independent variable x and the dependent variable y is described as an n th degree polynomial.
- This type of regression is also known as "extended" linear regression. Polynomial regression is used to model a nonlinear relationship between the value of an independent variable x and the conditional mean of a dependent variable y .
- This relationship is represented by the notation $E(y | x)$. solely as a result of the non exists between the dependent and the independent variables When we want to transform linear regression into polynomial regression, we just add some polynomial terms

Feature Selection and Extraction

- Feature selection and extraction are one of the most important steps that must be performed in order for machine learning to be successful.
- While we covered the theoretical aspects of this process in the earlier sessions of this course, it is now time to understand the implementation part of the mechanisms that we have learned for Feature selection and extraction.
- Dimensionality reduction may be seen of as a way to streamline the analysis process.

Feature Selection and Extraction

- **Principal Component Analysis (PCA)** : Among the various techniques the Principal Component Analysis (PCA) most frequently used.
- PCA works by considering the variance of each attribute because the high attribute shows the good split between the classes, and hence it reduces the dimensionality. Some real-world applications of PCA are image processing, movie recommendation system, optimizing the power allocation in various communication channels. It is a feature extraction technique, so it contains the important variables and drops the least important variable.

Association Rules

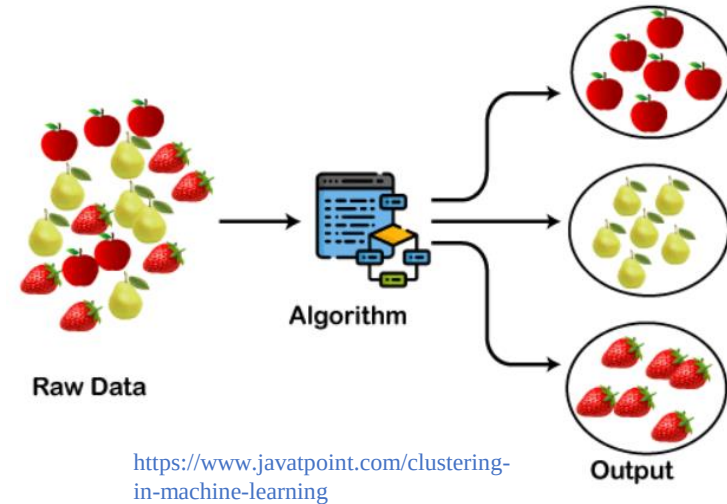
- **Apriori Algorithm :**
- Apriori algorithm refers to the algorithm which is used to calculate the association rules between objects. It means how two or more objects are related to one another. In other words, we can say that the apriori algorithm is an association rule learning that analyzes that people who bought product A also bought product B.
- The Apriori algorithm is a data mining technique that is used for mining frequent item sets and appropriate association rules. It does this by using a mathematical formula. We focused on the definitions of association rule mining and Apriori algorithms, as well as the application of an Apriori algorithm, in the area of this class that was most pertinent to the topic.

Clustering Algorithms

- Clustering or cluster analysis is a machine learning technique, which groups the unlabeled dataset. It can be defined as **"A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group."**
- It does it by finding some similar patterns in the unlabelled dataset such as shape, size, color, behavior, etc., and divides them as per the presence and absence of those similar patterns.
- It is an unsupervised learning method, hence no supervision is provided to the algorithm, and it deals with the unlabeled dataset.

Clustering Algorithms

- **Example:** Let's understand the clustering technique with the real-world example of Mall:
- When we visit any shopping mall, we can observe that the things with similar usage are grouped together. Such as the t-shirts are grouped in one section, and trousers are at other sections, similarly, at vegetable sections, apples, bananas, Mangoes, etc., are grouped in separate sections, so that we can easily find out the things. The clustering technique also works in the same way.



Important Topics

- Classification Algorithms
- Regression Algorithms
- Apriori Algorithm
- Clustering Algorithms



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Summary

- In this session we understood the implementation of various machine learning algorithms for Classification, Regression, Dimension Reductionality and clustering.
- The theoretical aspects of the respective algorithm were already discussed in the respective units of this course.

*Thank
You*